

Causal Wasserstein Distribution Boosting (C-WDB): Method, Simulation Evidence, and What May Be Claimed

A consolidated technical report through Phase 5 and its repair track

Prepared for the Wasserstein Causal Forests project

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Abstract

This report consolidates everything the project has established about Causal Wasserstein Distribution Boosting (C-WDB), the proposed contribution, before the applied study begins. It states the estimands and the identification argument, develops the proper scoring rule the method optimizes and the gradient the implementation actually uses, describes the boosting algorithm in both its independent-arm and shared-partition forms, documents the ten simulation regimes and the oracle truth construction, and reports the preregistered tournament together with the repair track that followed it. The tournament returned **NOT-GO** on one of six rules, the null-regime correctness rule, and the repair track returned **GO** for two variants of the contrast regularizer. The final section states precisely which claims the evidence supports, which it does not, and what an applied study may therefore assert.

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1 Purpose and scope of this document

The project builds a treatment-aware learner for the *conditional law* of a distribution-valued potential outcome. Its selling point is that a method holding a whole conditional distribution can answer questions about functionals that were not named at training time, whereas a method holding a fixed vector of target coordinates cannot. Everything below exists to test whether that selling point survives contact with a preregistered simulation tournament that includes the published incumbent.

The document is deliberately self-contained. Sections 2 through 5 are the mathematics and the algorithm, Section 6 the data generating processes, Section 7 the evaluation protocol and the gate rules, Sections 8 through 10 the results, and Sections 11 and 12 the limitations and the consequences for the applied study. Every number reported here comes from a frozen artefact in the repository; the artefact list and checksums are in Section 13.

2 Setup, estimands, identification

2.1 Observation regime

The observational unit is $O = (X, A, Y)$ with $X \in \mathcal{X}$ pre-treatment covariates, $A \in \{0, 1\}$ a binary treatment, and $Y = Y^A$ an outcome that is itself a probability measure, $Y^0, Y^1 \in \mathcal{P}_2(\mathbb{R})$. Each unit therefore carries a whole distribution as its response: a firm’s distribution of daily returns, a hospital’s distribution of patient waiting times, a country’s distribution of household incomes. A pre-specified low-dimensional moderator $V = g(X)$ is fixed before any outcome analysis.

The current observation regime is called **ORACLE-V1**: each Y_i is treated as fully observed and represented exactly by its quantile vector on a declared grid. Inner-sample uncertainty, histogram-count noise, and measurement error in the inner distribution are deliberately deferred, and the regime where they matter (D10 in the original template) is dormant. This is assumption A6 in the ledger, and it is the single largest simplification in the work so far.

2.2 Finite-grid representation and geometry

Fix levels $0 < u_1 < \dots < u_K < 1$ and positive quadrature weights w_k normalized so that $\sum_k w_k = 1$. Writing $Q_Y(u) = \inf\{t : F_Y(t) \geq u\}$ for the left-continuous generalized inverse, the stored representation of an outcome is

$$q(Y) = (Q_Y(u_1), \dots, Q_Y(u_K)) \in \mathcal{Q}_K, \quad \mathcal{Q}_K = \{q \in \mathbb{R}^K : q_1 \leq \dots \leq q_K\},$$

the closed monotone cone. With $W = \text{diag}(w_1, \dots, w_K)$ the declared metric is

$$d_W(q, q') = \left\{ \sum_k w_k (q_k - q'_k)^2 \right\}^{1/2}, \quad z = W^{1/2}q, \quad d_W(q, q') = \|z - z'\|_2.$$

This is the finite-grid, quadrature version of the 2-Wasserstein distance on $\mathcal{P}_2(\mathbb{R})$, and the contract writes it $W_{2,K}$ rather than W_2 so that the discretization is never silently identified with the continuum object. The rescaling $L(q) = W^{1/2}q$ is injective because every $w_k > 0$; this fact is used in the propriety proof.

2.3 Estimands

The frozen estimand dictionary separates four levels that are routinely conflated in this literature. Table 1 lists the identifiers that appear in every result row.

Table 1: Frozen estimand dictionary (contract G0-WP0-A-v1). Grid and continuum versions carry different identifiers and are never pooled.

Identifier	Level	Definition
LAW-A-K	grid law	$P_a^K(x) = \mathcal{L}\{q(Y^a) \mid X = x\}$
MEANQ-A-K	grid barycenter coordinate	$\bar{q}_a(x) = \mathbb{E}\{q(Y^a) \mid X = x\}$
BARY-A	continuum barycenter	$m_a(x) \in \arg \min_{\nu} \mathbb{E}\{W_2^2(Y^a, \nu) \mid X = x\}$
TATE-K- j	grid outcome level	$\mathbb{E}\{h_j(q(Y^1))\} - \mathbb{E}\{h_j(q(Y^0))\}$
TCATE-K- j	grid outcome level	$\mathbb{E}[h_j(q(Y^1)) - h_j(q(Y^0)) \mid V = v]$
REF-A-K	grid outcome level	$r_a^K(x) = \mathbb{E}\{d_W(q(Y^a), q(\nu_*)) \mid X = x\}$
REF-ATE-K, REF-TCATE-K	grid outcome level	marginal and moderator-conditional contrasts of r_a^K
LAW-A-M-K	finite-particle law	the fixed- M score projection of Section 3.4

The mandatory common target, the one every method in the tournament can produce without any substitution, is the grid causal mean

$$\tau_q^K(x) = \mathbb{E}\{q(Y^1) - q(Y^0) \mid X = x\} \in \mathbb{R}^K.$$

2.4 Identification, and what is not identified

Under A1 (binary treatment), A2 (consistency, $Y = Y^A$), A3 (conditional ignorability, $(Y^0, Y^1) \perp A \mid X$), and A4 (overlap, $0 < e(X) < 1$), the usual g-formula holds at the level of whole laws: for every Borel $B \subseteq \mathcal{Q}_K$,

$$P_a^K(x)(B) = \mathbb{P}\{q(Y^a) \in B \mid X = x\} = \mathbb{P}\{q(Y) \in B \mid A = a, X = x\},$$

and consequently, for every integrable measurable h , $\mathbb{E}\{h(q(Y)) \mid A = a, X = x\} = \mathbb{E}\{h(q(Y^a)) \mid X = x\}$. Every target in Table 1 is a functional of the two arm-marginal conditional laws and is therefore identified.

Two negative statements matter as much as the positive one, and both are enforced in code rather than left to discipline.

First, the joint law $\mathcal{L}(Y^0, Y^1 \mid X = x)$ is *not* identified. The witness in the contract takes X constant, randomized A , and $Y^a = \delta_{B_a}$ with B_0, B_1 marginally Bernoulli(1/2): the coupling $B_1 = B_0$ and the coupling $B_1 = 1 - B_0$ induce the same law of (A, Y) but give $\mathbb{P}(Y^1 = Y^0) = 1$ and $\mathbb{P}(Y^1 = Y^0) = 0$ respectively. Individual treatment-effect distributions, cross-arm particle pairings, and shared ranks across arms are therefore outside the contract. In particular, if the method outputs M particles per arm, the quantity $p_{1m}(x) - p_{0m}(x)$ has no causal meaning, and every reported output must be invariant to permuting particle labels within an arm (assumption A14).

Second, the barycenter level and the outcome level are different. In one dimension the conditional barycenter has quantile function $Q_{m_a(x)}(u) = \mathbb{E}\{Q_{Y^a}(u) \mid X = x\}$, so $q\{m_a(x)\} = \mathbb{E}\{q(Y^a) \mid X = x\}$ holds coordinatewise and MEANQ-A-K is recoverable. But for nonlinear T_j ,

$$T_j\{m_a(x)\} \neq \mathbb{E}\{T_j(Y^a) \mid X = x\}$$

in general. A barycenter draw, or a posterior draw of a mean surface, may not be relabelled as a draw from the outcome law. This prohibition is what forces the two Bayesian baselines to report **not_applicable** on every law-level metric rather than a substitute quantity, and it is the reason the transfer claim in D7 is testable at all.

3 The proper score

3.1 Definition

For $\varepsilon \geq 0$ define the smoothed distance and the loss-oriented energy score

$$\rho_\varepsilon(p, q) = \{d_W^2(p, q) + \varepsilon^2\}^{1/2} - \varepsilon, \quad S_\varepsilon(P, y) = \mathbb{E}_{Z \sim P} \rho_\varepsilon(Z, y) - \frac{1}{2} \mathbb{E}_{Z, Z' \sim P} \rho_\varepsilon(Z, Z').$$

For the equal-weight M -particle law $P_M = M^{-1} \sum_m \delta_{p_m}$ this is

$$S_{M, \varepsilon}(p_{1:M}, y) = \underbrace{\frac{1}{M} \sum_{m=1}^M \rho_\varepsilon(p_m, y)}_{\text{attraction}} - \underbrace{\frac{1}{2M^2} \sum_{m, \ell=1}^M \rho_\varepsilon(p_m, p_\ell)}_{\text{repulsion}}. \quad (1)$$

The subtraction of ε makes $\rho_\varepsilon(p, p) = 0$; additive constants would cancel from the energy divergence anyway. The value $\varepsilon = 10^{-3}$ is frozen in the manifest and the same value is used in the objective, the gradient, the tuning risk, and the definition of the fixed- M target. It is not an undocumented denominator patch: S_ε is exactly the score being optimized, and it is proper in its own right.

3.2 Strict propriety

Proposition 1 (Strict propriety on the weighted monotone cone). *For every $\varepsilon \geq 0$, S_ε is strictly proper on $\mathcal{P}_1(\mathcal{Q}_K, d_W)$: the population risk $R_\varepsilon(P, Q) = \mathbb{E}_{Y \sim Q} S_\varepsilon(P, Y)$ satisfies $R_\varepsilon(P, Q) \geq R_\varepsilon(Q, Q)$ with equality if and only if $P = Q$.*

Direct expansion gives $R_\varepsilon(P, Q) - R_\varepsilon(Q, Q) = \frac{1}{2} D_{\rho_\varepsilon}(P, Q)$ with $D_{\rho_\varepsilon}(P, Q) = 2\mathbb{E} \rho_\varepsilon(X, Y) - \mathbb{E} \rho_\varepsilon(X, X') - \mathbb{E} \rho_\varepsilon(Y, Y')$ for independent $X, X' \sim P$ and $Y, Y' \sim Q$. The identity

$$\sqrt{t + \varepsilon^2} - \varepsilon = \frac{1}{2\sqrt{\pi}} \int_0^\infty \{1 - e^{-st}\} e^{-s\varepsilon^2} s^{-3/2} ds, \quad t \geq 0,$$

applied with $t = \|L(p) - L(q)\|_2^2$ turns the divergence into a positive mixture of squared maximum mean discrepancies,

$$D_{\rho_\varepsilon}(P, Q) = \frac{1}{2\sqrt{\pi}} \int_0^\infty \text{MMD}_{k_s}^2(L_\# P, L_\# Q) e^{-s\varepsilon^2} s^{-3/2} ds, \quad k_s(z, z') = e^{-s\|z - z'\|_2^2}.$$

Every integrand is nonnegative, Gaussian kernels are characteristic on Euclidean space, and the mixing density is strictly positive, so the divergence vanishes only when $L_\# P = L_\# Q$, hence only when $P = Q$ by injectivity of L . Restricting to the closed subset $\mathcal{Q}_K$ does not weaken strictness. At $\varepsilon = 0$ this reduces to the classical energy-score result of Gneiting and Raftery (2007, Theorem 5); the distance-to-RKHS equivalence is that of Sejdinovic et al. (2013) and Székely and Rizzo (2013).

The practical content of this proposition is assumption A13: because the score is strictly proper over the *unrestricted* finite-grid law space, minimizing its conditional risk identifies $P_a^K(x)$ itself, not some summary of it.

3.3 Gradient, and why the repulsion term cannot be dropped

Writing $r_{my, \varepsilon} = \{(p_m - y)^\top W(p_m - y) + \varepsilon^2\}^{1/2}$ and similarly $r_{m\ell, \varepsilon}$ for particle pairs, differentiating (1) and counting both ordered appearances of p_m in the double sum gives

$$\nabla_{p_m} S_{M, \varepsilon} = \frac{1}{M} \frac{W(p_m - y)}{r_{my, \varepsilon}} - \frac{1}{M^2} \sum_{\ell=1}^M \frac{W(p_m - p_\ell)}{r_{m\ell, \varepsilon}}. \quad (2)$$

The second term couples every particle to the entire ensemble. A particlewise attraction gradient that omits it is not the gradient of the certified score, and the consequence of omitting it is not cosmetic.

Proposition 2 (Squared- W_2 collapse). *The improper ablation that removes the repulsion term minimizes $L_{\text{sq}}(p_{1:M}) = M^{-1} \sum_m \mathbb{E}\{d_W^2(p_m, q(Y))\}$. Each summand is independent of the other particles, $\mathcal{Q}_K$ is convex, and $\mathbb{E}\{q(Y)\} \in \mathcal{Q}_K$, so every particle has the same unique minimizer $p_m^* = \mathbb{E}\{q(Y)\}$. The ensemble collapses to the barycenter whatever the outer law.*

This is why the tournament carries a squared- W_2 booster as a preregistered ablation: it is the same tree machinery with (1) replaced by a squared loss, so any difference isolates the repulsion term and nothing else. The certified numerical witness records a final particle spread of 3.5×10^{-18} for the ablation, and the D6 tournament result in Section 8 is the empirical counterpart.

3.4 The fixed- M target

Strict propriety identifies the truth over all laws. It does *not* promise exact recovery inside a finite particle class. The class $\mathcal{P}_M^{\text{emp}}(\mathcal{Q}_K)$ consists of equal-weight empirical measures with M slots and repeated particles allowed, so after merging duplicates the atom masses are integer multiples of $1/M$. The implemented population target is the constrained risk projection

$$P_{a,M,\varepsilon}^{K,*}(x) \in \arg \min_{P \in \mathcal{P}_M^{\text{emp}}(\mathcal{Q}_K)} \mathbb{E}\{S_\varepsilon(P, q(Y)) \mid A = a, X = x\}.$$

Existence follows from coercivity: with $r_m = d_W(p_m, 0)$ and $C = \mathbb{E}\{d_W(q(Y^a), 0) \mid X = x\} < \infty$ under A8, the triangle inequality and the missing diagonal pairs give $R_\varepsilon(p_{1:M}) \geq M^{-2} \sum_m r_m - C - \varepsilon$, so sublevel sets are bounded and closed in finite dimension. Uniqueness of the parameter vector is impossible by permutation symmetry, and uniqueness of the optimal law is not claimed.

Assumption A17 follows: the method reports LAW-A-M-K, not LAW-A-K, and its M -sensitivity must be measured. It is, in Section 8.6.

3.5 Monotone-cone projection

A tree step can leave the cone, so every candidate is projected,

$$\Pi_W(v) = \arg \min_{q_1 \leq \dots \leq q_K} \frac{1}{2} \sum_k w_k (q_k - v_k)^2,$$

which exists and is unique because W is positive definite and $\mathcal{Q}_K$ is nonempty, closed, and convex. Weighted pool-adjacent-violators computes it exactly by replacing each violating adjacent block with its weighted mean; the recorded maximum KKT residual is 1.7×10^{-17} . The implementation skips rows already inside the cone, which is exact (PAVA pools nothing there) and skips essentially all rows during boosting. Projection keeps particles in the space on which propriety was proved. It is not claimed that projecting an arbitrary tree step always decreases the score; that is what the line search in Section 4.2 handles.

4 The algorithm

4.1 Two architectures

C-WDB comes in two forms that differ only in how the two arms share tree structure.

v0, independent arms. One particle booster is fitted per arm on that arm's observations. It is the separate-head limit and serves as the ablation that isolates the value of sharing.

v1, shared partition. One sequence of trees is grown on the pooled sample; each tree has a single covariate partition but arm-specific leaf vectors. This is the architecture the project claims, and the mechanism is that a split found on one arm's data localizes both arms.

4.2 The boosting loop

Let $F^{(t)}(x, a) \in \mathcal{Q}_K^M$ denote the current particle block. The loop is a projected functional gradient descent on the empirical energy risk.

1. *Initialize.* Every unit starts from a base law $F^{(0)}$ built by `compute_init_base`, which lexicographically sorts the observed quantile vectors and takes M evenly spaced order statistics, a deterministic spread-preserving empirical initialization. Whether the base is computed per arm or pooled is the `init_sharing` switch, and Section 9 shows this switch was a genuine defect.
2. *Gradient.* Compute (2) at every training row, then form the preconditioned descent target $-M \nabla/w$, which is the negative gradient expressed in the rescaled coordinates where the geometry is Euclidean.
3. *Fit one weak learner* to that target. In v0 this is an ordinary multi-output regression tree; in v1 it is the arm-shared tree of Section 4.3. Splits use only the gradient of each unit's *realized* arm, so no counterfactual quantity enters the split search.
4. *Line search with projection.* With step η starting at the learning rate, form $\Pi_W(F^{(t)} + \eta \text{direction})$ and accept it only if the empirical risk did not increase; otherwise halve η , up to twelve backtracks. If no step is accepted the boosting stops early. This is what makes the descent guarantee hold despite the projection and the tree approximation.
5. *Record* the step, the loss before and after, the number of backtracks, and the projection magnitude, so the whole path is auditable.

Prediction replays the accepted steps and their step sizes on new X , projecting after each. Outputs are canonicalized so that the particle ordering carries no information, which is A14 enforced in code.

The split criterion is multi-output sum-of-squares reduction on the gradient block. The implementation uses the closed form $\text{gain} = \|S_L\|^2/n_L + \|S_R\|^2/n_R - \|S\|^2/n$ with S the column sums, so one sort plus one cumulative sum per covariate scores every admissible threshold. That reduces the naive $O(n^2 MKp)$ rescan to $O(pn(MK + \log n))$, and a slow reference implementation is retained as the semantic definition and checked against it in the tests.

4.3 The arm-shared tree and its leaf rule

In v1, a candidate split is admissible only if both children contain at least `min_arm_leaf` observations from *both* arms, which is what keeps every leaf able to express a contrast. Given a leaf with index set I , let $\bar{g}$ be the pooled gradient mean over I , and $\bar{g}_0, \bar{g}_1$ the arm-specific means with counts n_0, n_1 .

The frozen G3 rule, `arm_shrinkage` with parameter λ , sets

$$v_a = \frac{n_a \bar{g}_a + \lambda \bar{g}}{n_a + \lambda}, \quad a \in \{0, 1\}.$$

This pulls each arm mean toward the pooled mean. Crucially it shrinks the *contrast* only as a side effect: at a balanced leaf the surviving contrast is the raw gap times $n_a/(n_a + \lambda)$, which at $\lambda = 5$ and a typical main-grid leaf is 0.86. Almost nothing is regularized. This is the defect diagnosed in Section 9.

The repair reparameterizes the leaf so that the pooled component is immovable:

$$v_a = \bar{g} + (a - \pi) \kappa \delta, \quad \delta = \bar{g}_1 - \bar{g}_0, \quad \pi = \frac{n_1}{n_0 + n_1}. \quad (3)$$

Then $v_1 - v_0 = \kappa \delta$ and $\pi v_1 + (1 - \pi) v_0 = \bar{g}$ *identically*, so a contrast rule shrinks the contrast and provably cannot move the pooled fit. This is what makes the three candidate rules comparable: they differ only in the scalar $\kappa \in [0, 1]$.

R1, fixed ridge. $\kappa = n_{\text{eff}}/(n_{\text{eff}} + \lambda_\tau)$ with $n_{\text{eff}} = n_0 n_1 / (n_0 + n_1)$, the effective sample size the arm gap is estimated from. This is the posterior mean under a mean-zero Gaussian prior on the contrast, and is the direct analogue of the half-Cauchy prior that Bayesian causal forests place on the treatment-effect scale. The constant $\lambda_\tau = 50$ was hand-tuned on pilot seeds, which is declared: R1 is the tuned reference in the roster and the two adaptive rules received no equivalent tuning.

R2, adaptive threshold. $\kappa = (1 - c \hat{\sigma}^2 / \|\delta\|^2)_+$, where $\hat{\sigma}^2 = \sum_d (s_{0d}^2/n_0 + s_{1d}^2/n_1)$ is the plug-in variance of the arm gap in that leaf. This is a positive-part James-Stein factor calibrated against the leaf’s own noise: under an exactly null gap $\mathbb{E}\|\delta\|^2 = \hat{\sigma}^2$, so the retained fraction is zero in expectation, while a gap far above the noise passes through undamped. Two calibrations are entered and both reported, $c = 1$ (null-calibrated, the principled default) and $c = 3$ (conservative sensitivity). Within-leaf covariate heterogeneity inflates $\hat{\sigma}^2$, so the rule is conservative in exactly the direction a null regime rewards and costs power where within-leaf structure is strong.

R3, cross-fitted ridge. R1’s functional form with λ_τ chosen on held-out energy risk rather than frozen. Candidates $\{0, 50, 500\}$ over two arm-stratified folds; each held-out unit is scored only against its *own observed arm*, so no counterfactual enters the selection and the score is strictly proper for the law that unit realizes. Ties break toward the stronger regularizer, on the grounds that when the data cannot distinguish two strengths the null-safe one is the right default for a method whose failure mode is manufacturing effects. Assumption A15 is satisfied: every candidate is scored only on folds excluded from the fit that produced it.

The identification argument for R3 is worth stating because it is the reason cross-fitting can tune a *causal* quantity without a counterfactual. Over-shrinking the contrast when the arms genuinely differ pushes both arm laws toward a pooled law that fits neither, so the held-out risk rises; under-shrinking when they do not differ fits sampling noise in the arm gap, and the risk rises too. Because the pooled component is fixed by construction in (3), the minimizer is informative about the contrast rather than about the pooled fit.

4.4 Cost

R3 costs roughly $1 + n_{\text{folds}} \times |\text{candidates}| \times (n_{\text{folds}} - 1)/n_{\text{folds}}$ ordinary fits. Two folds over three candidates is about four fits, which is why that configuration was frozen on cost grounds before any accuracy result existed. The energy computations chunk their pairwise tensors to a 128 MB budget, exactly, because the (rows, M, M, K) temporaries reach several gigabytes at the largest cells.

5 The comparison roster

Three points about the roster carry restrictions into everything below.

Causal-DRF is the published direct-hit incumbent and the comparison that matters scientifically. It is run from the authors’ supplied `code/causal_drf_paper-main/R/simulation_study_setup.R`, using the `causal-clean` branch of `herbbs10/drif`, commit `0a1a508444176b5b1553f13e832be93a374b0af2`. Its kernel bandwidth uses a median-distance rule because the package convention is not scale equivariant.

W-DRF-T is an ordinary two-forest T-learner on the rescaled coordinates, pinned to CRAN `drf` 1.3.1. It is not Causal-DRF, and calling it that would attribute a method it does not implement.

PTA-S and PTA-F are the two surviving endpoints of a line that failed its own preregistered crossover test, so the project returned `RETAIN-STRONGEST-ENDPOINT` for that line: partial pooling did not beat its endpoints, and neither endpoint is a contribution of this project. They enter only as baselines, they estimate conditional *means* of a fixed target vector, and under the contract they

Table 2: Methods in the tournament. Only C-WDB can return `G0`; the others are held to correctness and cost reporting.

Label	Registry	Role and mechanism	Law?
C-WDB-v1	<code>cwdb_v1</code>	claimant: shared partition, arm-specific leaves	yes
C-WDB-v0	<code>cwdb_v0</code>	ablation: independent arms, isolates sharing	yes
no-shrink	<code>cwdb_v1_noshrink</code>	ablation: <code>arm_shrinkage=0</code>	yes
sq- W_2	<code>sqw2_booster</code>	ablation: repulsion term deleted	degenerate
PTA-S	<code>pta_s</code>	baseline: separately tuned Bayesian scalar heads	no
PTA-F	<code>pta_f</code>	baseline: forced-shared multivariate BCF	no
W-DRF-T	<code>wdrft</code>	baseline: two ordinary predictive DRFs, one per arm	yes
Causal-DRF	<code>causal_drf</code>	incumbent: one shared treatment-aware forest with weighted-MMD splitting and signed weights	yes

report `not_applicable` on every law-level metric rather than substituting a posterior draw of a mean surface.

6 Data generating processes

6.1 One generative form

Every regime shares a single form, which is what makes the oracle truth uniform across regimes:

$$q(Y^a)_k = m_a(x) + \xi + e^{s_a(x)+\eta} \psi(z_k; \gamma_a(x)), \quad z_k = \Phi^{-1}(u_k), \quad (4)$$

with the outer latent pair (ξ, η) independent of x and A given the arm. Here ξ shifts the outer location and η the outer log scale: they are what makes the conditional object a genuine *law over distributions* rather than a single distribution per covariate value. The inner reshaping is built from the second and third probabilists’ Hermite polynomials,

$$\psi(z; \gamma) = z + \gamma \frac{\text{He}_2(z)}{2} + \gamma \frac{\text{He}_3(z)}{6} = z + \gamma \frac{z^2 - 1}{2} + \gamma \frac{z^3 - 3z}{6}.$$

Both Hermite terms are orthogonal to the identity under the standard normal, so $\mathbb{E}\{\psi(Z)\} = 0$: the reshaping moves skewness and the upper tail while leaving the mean untouched. Its derivative is $1 + \gamma(z + (z^2 - 1)/2)$, minimized at $z = -1$ with value $1 - \gamma$, so every draw stays strictly inside the monotone cone for $0 \leq \gamma < 1$ at any K . The monotone-cone guarantee therefore does not weaken as the grid refines, which matters because a violated cone would be an artefact of the simulator rather than of any method.

Covariates are $X \sim \text{Uniform}[-1, 1]^5$. The moderator $V = g(X)$ is a four-bin discretization of X_0 at cut points $-0.5, 0, 0.5$; a discrete moderator keeps every `TCATE-K-j` statement a finite collection of conditional means, which is what the contract’s prohibition on pointwise claims at every continuous v requires.

6.2 The ten regimes

Regimes differ only in m_a , s_a , γ_a , the outer law, and the propensity. Table 3 states what each is built to test.

D7 deserves emphasis because it is the regime the contribution is really about. Both arms share the location and scale surfaces, and ψ is mean-free, so `grid_mean` carries almost no signal while `grid_skewness` and `grid_upper_tail_mean` carry all of it. Those two functionals are deliberately *excluded from every training manifest* and evaluated anyway. A method that outputs a full law integrates them at evaluation time; a method holding fixed target coordinates cannot produce them at all.

Table 3: The ten regimes. Propensities are clipped logistic: mild $0.8(X_0 + 0.5X_1)$ clipped to $[0.1, 0.9]$; strong $2.5(X_0 + 0.7X_1 - 0.5X_2)$ clipped to $[0.05, 0.95]$; deteriorating $4.0(X_0 + 0.5X_1)$ clipped to $[0.01, 0.99]$.

ID	Name	What it is for
D0	deterministic	no outer randomness, so the conditional law is a point mass and the conditional mean is exactly recoverable; a correctness floor
D1	smooth stochastic	the reference regime: location-scale surfaces with Gaussian outer location and log-scale
D2	null effect	both arms share m , s , γ exactly, so the true effect is identically zero; the false-effect test
D3	separate-head favorable	the two arms' location surfaces depend on disjoint covariates, so one shared partition must carry the union of both arms' structure
D4	shared-structure favorable	one moderator drives prognosis and effect in both arms, so sharing splits pays
D5	equal barycenters	the two arms match in every mean but differ in outer law, so any mean-based method sees nothing
D6	multimodal outer law	the outer location is an equal-weight two-component mixture at ± 1.5 ; the particle-collapse test
D7	unseen functional transfer	both arms share m and s and differ only in the inner shape γ_a , so the entire effect sits in skewness and the upper tail, which are excluded from every training manifest
D8	strong confounding, weak effect	a large prognostic surface, a small effect, and a strong propensity
D9	overlap deterioration	propensities pushed to the clipping bounds

6.3 Oracle truth

Because (ξ, η) is low dimensional with a known Gaussian or Gaussian-mixture law, the conditional expectation of any grid functional is a Gauss–Hermite quadrature sum rather than a Monte Carlo average. Truth entering an RMSE therefore carries deterministic quadrature error, not sampling noise. Twelve nodes per non-degenerate dimension agree with a twenty-node rule to 7.5×10^{-3} across every regime and target, and an independent check against a 50,000-draw Monte Carlo at 140 regime-arm-target combinations returns PASS with worst discrepancy 0.0179 against Monte Carlo standard errors of 0.001 to 0.007.

One target is not computed this way. The `tail_calibration` truth is the expectation of an indicator, and no Gaussian quadrature rule integrates a step function: the generic path was wrong by 0.069 against a Monte Carlo standard error of 0.002. The outer location is therefore integrated in closed form through the normal survival function and only the smooth log-scale latent goes through quadrature.

7 Evaluation protocol

7.1 Metrics

Law-level metrics require a conditional law and are reported only by the four law methods.

Excess energy risk (`arm_energy_risk`) is $\mathbb{E}_{q \sim P_a(x)} S_\varepsilon(\hat{P}_a(x), q)$ minus the same quantity at the truth, evaluated against the quadrature representation of the true conditional law rather than a single draw. The irreducible regime-dependent term is subtracted so that zero is attainable only at the truth.

Kernel law error (`kernel_law_error`) is the squared maximum mean discrepancy under a Gaussian kernel on the rescaled coordinates z , with a median-distance bandwidth so that the

discrepancy is scale equivariant in the outcome’s own units. This is the *declared primary law metric*. The kernel is characteristic, so it vanishes only at the true law.

Mode coverage counts a true mode as covered when the predicted atoms within a radius of its centre carry at least a floor of predicted mass. A law collapsed onto one mode of a bimodal outer law scores 0.5 however well it fits that mode, which is precisely the collapse signature.

Tail calibration is a predeclared threshold event under the grid law.

Outcome-level metrics are computed by every method that can produce the target: `mean_quantile_rmse` against `MEANQ-A-K` (the mandatory common target), `tcate_functional_rmse` and `tate_functional_rmse` against the named functional, and `reference_tcate_rmse` and `reference_effect_rmse` against the Wasserstein reference effect with $\nu_\star$ the standard normal, frozen before any outcome analysis.

The two forest baselines are scored on their native representation, a weighted empirical law over the whole training sample, and C-WDB on its M particles. Both are valid predictive distributions and the score is proper, so a richer representation is allowed to win; every row records its atom count so a reader can see which is which. This is a deliberate choice not to handicap the baselines.

7.2 Design grids and budgets

Table 4 lists the six grids of the frozen manifest. Every method in a replication receives the same training sample and is evaluated on the same test design of 1000 points drawn at seed $900000 + s$, disjoint from every training seed. PTA-F appears only at $K = 5$ because its cost accelerates in the target dimension $D = K + J + 1$ (measured at 20.4 s for $D = 2$, 27.4 s for $D = 4$, 51.0 s for $D = 8$, with the ratio itself rising), which is a declared cost limitation and never reported as a loss.

Table 4: The frozen design grids, 4110 cells in total.

Grid	Purpose	DGPs	n	K	M	Seeds
main	primary tournament	D0–D9	500, 1000	25	10	20
smallk	full roster including PTA-F	D0–D9	500	5	10	20
particles	finite- M claim	D1, D6	1000	25	2,5,10,25	20
resolution	grid resolution	D1, D5, D6, D7	1000	49	10	10
shrinkage	causal regularization	D2, D8	1000	25	10	20
scaling	runtime and memory	D1, D4, D6	2000	25	10	10

Budgets are frozen and regime-independent, and a validator rejects any registry entry carrying a regime-dependent parameter, which is how comparable tuning effort is enforced rather than asserted. Boosting uses 100 trees, learning rate 0.12, depth 4, minimum leaf 10, minimum arm leaf 5, $\varepsilon = 10^{-3}$, and $\lambda = 5$ for `arm_shrinkage`. W-DRF-T uses 1000 trees with honesty on; Causal-DRF 2500 trees; PTA-S 50 prognostic and 20 treatment trees with 200 draws after 100 burn-in and five cross-fitting folds.

A cell that raises is recorded as a failure with its traceback and is *never* retried under a different seed, because silently replacing a seed a method failed on converts a robustness result into a selection artefact. The merge refuses rather than cleans: a duplicate key, a result absent from the manifest, a manifest cell with no rows, or a row disagreeing with the manifest on its own coordinates makes the audit **FAIL**.

7.3 The six gate rules

A win means a seed-paired difference more negative than twice its paired standard error; methods are paired within a replication because they share a training sample and a test design. All six rules must hold for **G0**.

Rule 4 is evaluated only on rule 3’s winners by design: passing by beating PTA-S somewhere C-WDB lost to Causal-DRF would not be the claimed transfer. The gate flags are recomputed by

Table 5: Preregistered decision rules, frozen before the first decisive seed.

#	Name	Threshold
1	correctness and nulls	D0 <code>mean_quantile_rmse</code> ≤ 0.15 ; D2 ≤ 0.15 <i>and</i> $\leq 1.25\times$ the best baseline’s
2	law advantage	beats Causal-DRF on <code>kernel_law_error</code> in at least two of D1, D5, D6, D7
3	transfer	the advantage reaches at least one functional or reference target in at least one of D5, D6, D7, against Causal-DRF
4	beats the direct learner	on a target rule 3 actually won, also beats PTA-S
5	no collapse	D6 mode coverage ≥ 0.90 and particle participation ratio $\geq 0.60M$
6	cost	median runtime at most 60 times Causal-DRF’s

an independent script that shares only the thresholds with the analysis code and reimplements the statistics, so a bug appears as a disagreement rather than as two copies of one wrong answer.

8 Results of the preregistered tournament

8.1 Execution

The manifest declared 4110 cells; 4110 produced rows, 97,650 in total, with zero failures for every method and a merge audit of **PASS**. Wall time was about 2.5 hours on ten workers, and the independent recomputation agreed with the analysis code on every rule with zero disagreements.

One defect was found and fixed *before* this run was reported: law methods had been handed only the training functionals, so `grid_skewness` and `grid_upper_tail_mean` returned `not_applicable` for C-WDB too, which meant the D7 transfer claim was measuring nothing. That required a full re-run; the defective run is archived rather than deleted.

8.2 The verdict, and the rule that failed

Table 6: Gate outcome for C-WDB-v1, preregistered tournament.

Rule	Statement	Result
1	correctness and nulls	FAIL
2	law advantage over Causal-DRF	PASS (3 wins, 2 required: D1, D5, D7)
3	transfer to a causal functional	PASS (5 winning targets)
4	beats PTA-S on a rule-3 winner	PASS (3 wins, all on capability)
5	no particle collapse	PASS (mode coverage 0.99, participation 10 of 10)
6	cost	PASS (13.3 s against 1.17 s, ratio 11.3, cap 60)
Verdict		NOT-GO , recommendation PIVOT

Rule 1 failed on D2, where the true effect is exactly zero. C-WDB-v1’s `mean_quantile_rmse` there was 0.1466 against a best baseline of 0.0545, a false-effect ratio of 2.69 against a cap of 1.25. In absolute terms 0.147 is inside the 0.15 tolerance, so the method is not wildly wrong, but it manufactures roughly two and a half times as much apparent heterogeneity as the best comparator on data containing none. A distributional causal method whose selling point is detecting effects that scalar methods miss cannot also be the method most likely to report effects that are not there. That is disqualifying on its own terms, not a threshold technicality.

8.3 What the tournament established

Three mechanisms behaved as the theory predicts, and each is isolated by an ablation rather than inferred from a cross-method comparison.

The repulsion works. On D6, whose outer law is a two-component mixture, C-WDB-v1 reaches mode coverage 0.990 with the participation ratio at 10.00 of $M = 10$, meaning no collapse whatever. The squared- W_2 booster, the same machinery with the repulsion term deleted and nothing else changed, reaches 0.230, below the 0.5 that sitting on one mode of two would give. The paired differences are large against their standard errors: mode coverage -0.755 (SE 0.005), kernel law error -0.160 (SE 0.001), excess energy risk -0.569 (SE 0.002). See Figure 1.

Full-law transfer works. On D7, C-WDB-v1’s error on TCATE-K-grid_skewness is 0.0121 against Causal-DRF’s 0.0732, a factor of six on the functional carrying the regime’s entire treatment effect, and PTA-S cannot produce the quantity at all. This is the clearest single result in the tournament and it is the mechanism the contribution claims.

Sharing pays where it should. Against the independent-arm ablation C-WDB-v0, the shared partition wins mean_quantile_rmse by -0.0320 (SE 0.0036) on D3 and by -0.1012 (SE 0.0032) on D4, the two regimes built to discriminate separate-head from shared structure.

Finite-particle cost is controlled. See Section 8.6.

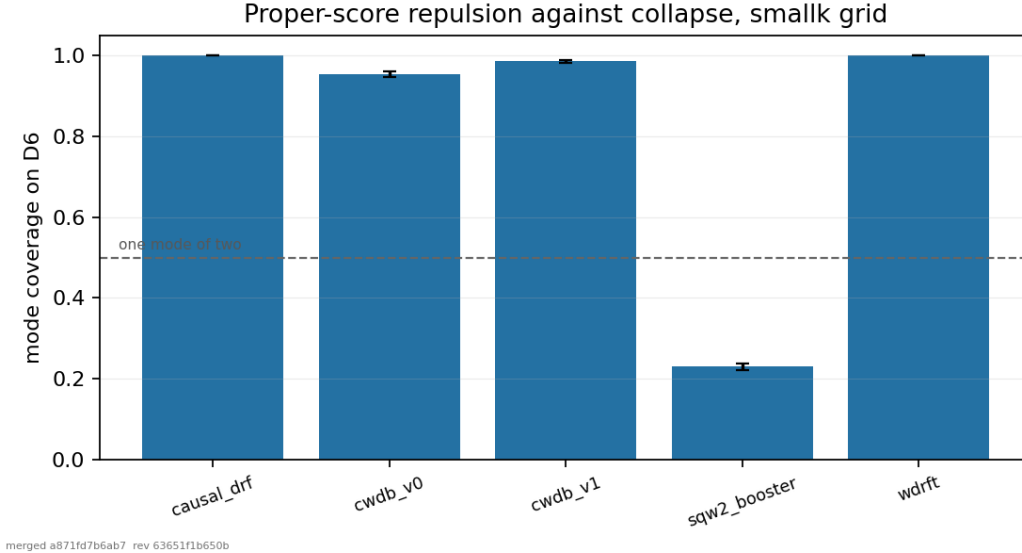


Figure 1: The repulsion ablation on D6. The squared- W_2 booster is the identical tree machinery with the repulsion term of (1) deleted; it collapses below the level a single mode would give, while every law method that keeps the term covers both modes.

8.4 The adversarial reading

The memo states the losses first, on the principle that averaging them into an overall score would hide them. C-WDB-v1 is beaten by more than two paired standard errors on twelve regime-target combinations, the most important of which are collected in Table 7.

Two of these matter for how the contribution is framed.

First, C-WDB loses the primary law metric on D6, the very regime built to showcase its repulsion mechanism, while simultaneously dominating mode coverage there. The two are consistent: the method places particles on both modes and so wins any metric sensitive to missing a mode, but with only ten atoms it represents the shape *within* each mode more coarsely than a forest’s weighted empirical law over a thousand training points. This is a representation-size effect, and the M -sensitivity evidence suggests raising M narrows it slowly. Figure 3 shows D6 as the single red bar against the incumbent.

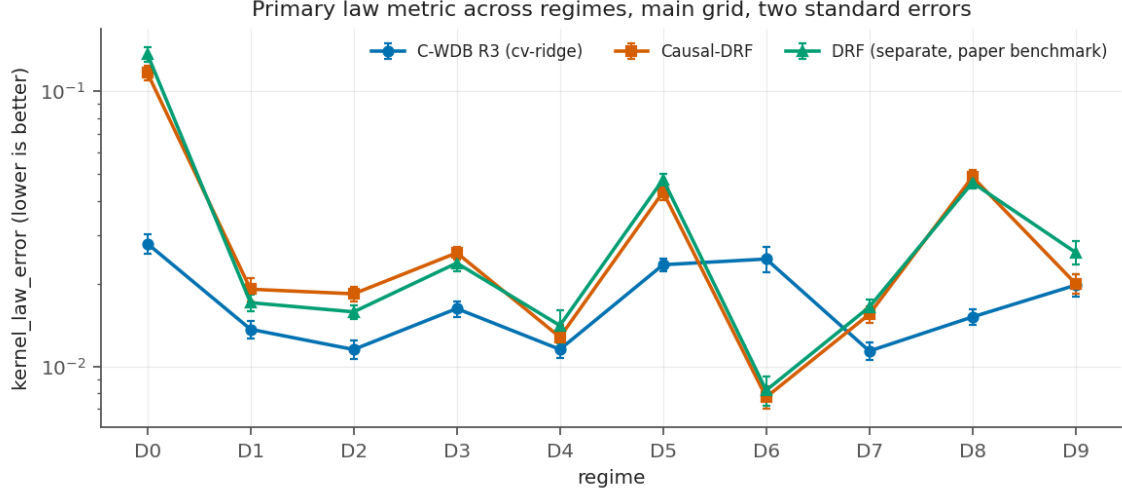


Figure 2: The declared primary law metric across all ten regimes on the main grid, with two standard errors. Lower is better. The frozen claimant C-WDB-v1 and the recommended repaired claimant C-WDB R3 are both shown, together with the two forest baselines and the independent-arm ablation. Both C-WDB variants are at or near the front almost everywhere; the exception is D6, discussed in the text. Numbers behind this figure are in Table 18.

Table 7: Selected cells where the claimant loses by more than two paired standard errors. Positive differences favour the comparator.

Regime	Target	Comparator	Claimant	Comparator	Difference	SE
D6	kernel_law_error	causal.drf	0.0191	0.00945	+0.00965	0.00045
D5	REF-ATE-K	causal.drf	0.0937	0.0179	+0.0758	0.0048
D5	REF-TCATE-K	causal.drf	0.0961	0.0230	+0.0730	0.0046
D6	REF-TCATE-K	causal.drf	0.0746	0.0412	+0.0335	0.0061
D5	TCATE-K-grid_sd	causal.drf	0.0422	0.0275	+0.0146	0.0024
D7	TCATE-K-grid_mean	pta_s	0.0341	0.0223	+0.0118	0.0041

Second, rule 4 passed on **zero accuracy wins and three capability wins**. On every target where both methods produce an estimate, PTA-S is at least as accurate. C-WDB passes only because on three targets PTA-S produces nothing at all. That capability advantage is real and it is what a transfer test is for, but the honest reading weakens it: PTA-S’s remedy is to add the functional to its manifest and fit one more scalar head, measured at 1.14 s. The capability gap is a convenience advantage of seconds, not a barrier. Set against that, the advantage over Causal-DRF, the published incumbent, *is* an accuracy advantage, on three of four eligible regimes for the primary law metric and on five per-target functional comparisons. The incumbent comparison is the stronger of the two, and the paper should be built on it.

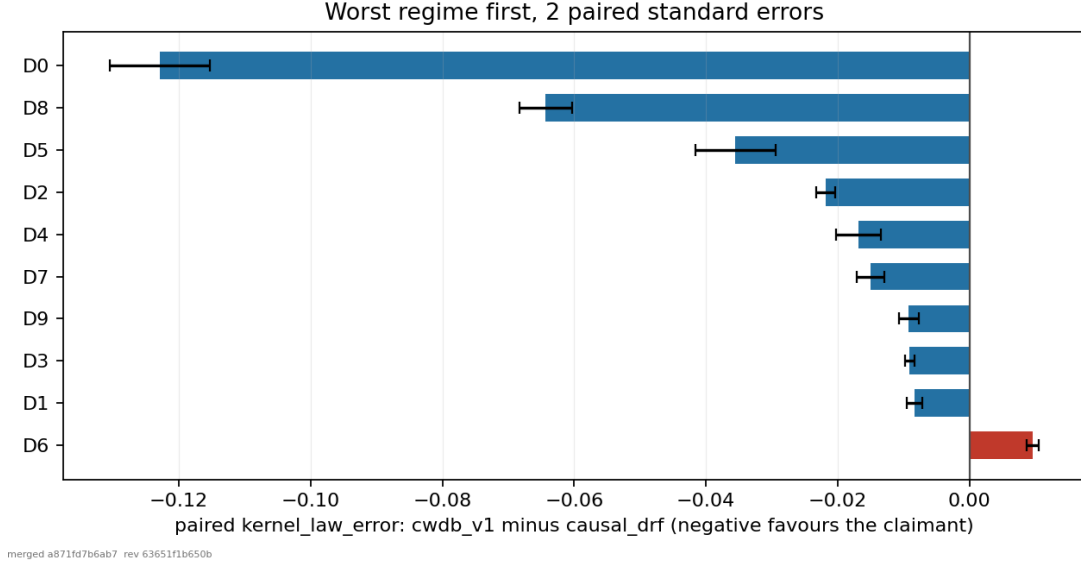


Figure 3: Paired primary-law-metric difference against the published incumbent, worst regime first. Negative favours the claimant. Only D6 favours Causal-DRF.

8.5 The recommended claimant placed in the same comparisons

Everything above is the frozen tournament, in which the only claimant was C-WDB-v1. Because the repair track reuses the frozen coordinates and seeds exactly (Section 10), the recommended variant `cwdb_r3.cvridge` can be dropped into precisely the same comparisons without recomputing a single frozen number, and this subsection does that. The complete metric-by-method tables are collected in Section 11; what follows is the reading of them.

Table 8: Where C-WDB R3 stands against the three comparators that matter, on the frozen main grid. Read with Table 30, which gives every paired difference and its standard error.

Comparator	Outcome for <code>cwdb_r3.cvridge</code>
Causal-DRF (incumbent)	wins the primary law metric in nine of ten regimes, losing only D6; wins the grid causal mean in six regimes, loses D1 and D5, ties D4 and D9
PTA-S (direct learner)	loses the grid causal mean in five regimes and wins two; ties on most functionals both estimate; wins outright only on the two functionals PTA-S cannot produce at all
C-WDB-v1 (frozen claimant)	improves six regimes on the grid causal mean, ties two, regresses on D1 and D9, cuts the null-regime error by two thirds, and <i>loses the pure-shape transfer result</i> discussed below

Four readings deserve stating in words rather than left in the tables, and the fourth is the

most important thing in this report that is not in either memo.

The incumbent comparison strengthens. Figure 4 is the same construction as Figure 3 with the recommended variant in place of the frozen one. The pattern is essentially unchanged: R3 wins nine of ten regimes and D6 remains the only one the incumbent takes, with the margins within a hair of the frozen claimant’s (larger for R3 in four regimes, smaller in five, and the D6 loss itself smaller at +0.0131 against +0.0163). The law-level advantage is therefore preserved rather than improved, which is the point: the contrast rule was not supposed to move the pooled fit and the numbers say it did not. On the grid causal mean, where the contrast *is* the estimand, the D8 advantage is now dramatic, 0.0735 against Causal-DRF’s 0.3769.

The null regime is no longer a liability. On D2 the grid causal mean falls from v1’s 0.1466 to 0.0506, which is *below* the best baseline’s 0.0545 rather than 2.69 times it. Figure 7 shows what each contrast rule pays for that: the cross-fitted rule is the only one that clears the cap on D2 while leaving D0 and D3 essentially where the frozen claimant left them.

The direct target learner is still the one to beat on the common target. On the grid causal mean, PTA-S beats R3 in five regimes (D0, D1, D4, D5, D9) and loses in two (D3, D8). Nothing in the repair changes the parent memo’s judgement that this comparison is the weak part of the case.

The repair damages the pure-shape transfer result, and this is not reported anywhere else in the project’s artefacts. The clearest single result of the frozen tournament was D7’s TCATE-K-grid_skewness, where C-WDB-v1 scored 0.0152 against Causal-DRF’s 0.0840, a factor of five and a half on the functional carrying that regime’s entire treatment effect. Every contrast rule degrades it, in proportion to how hard it shrinks:

Table 9: D7 TCATE-K-grid_skewness, the functional that carries the whole D7 treatment effect and appears in no training manifest. The middle column is the paired difference against the frozen claimant, the right column against the published incumbent. Positive is worse.

Method	Error	vs C-WDB-v1	vs Causal-DRF
cwdb_v1	0.0152	—	−0.0688 (0.0028)
cwdb_v1_pooledinit	0.0253	+0.0101 (0.0015)	−0.0587 (0.0030)
cwdb_r1_ridge	0.0645	+0.0493 (0.0027)	−0.0195 (0.0029)
cwdb_r3_cvridge	0.0904	+0.0752 (0.0069)	+0.0064 (0.0074)
cwdb_r2_threshold3	0.1986	+0.1834 (0.0052)	+0.1146 (0.0046)
causal.drf	0.0840		
wdrft	0.0464		
pta_s	<i>n/a</i>		

The mechanism is not mysterious and it follows directly from Section 4.3. D7 places the entire effect in the arm contrast of the inner shape, and a contrast regulariser shrinks exactly that. Cross-fitting selects a median strength of 50 on D7, so it shrinks a contrast that is real. The result is that R3 ties the incumbent on this functional instead of beating it by a factor of five, and cwdb_r2_threshold3 loses to the incumbent outright. The effect is not an artefact of one grid or one estimand: it reproduces at $K = 5$ (0.0629 for R3 against 0.0095 for v1) and on the marginal TATE-K-grid_skewness (0.0746 against 0.0070), where R3 is also worse than both forest baselines.

Two things keep this from being fatal, and both belong in the same breath as the finding. First, the transfer claim does not rest on skewness alone: on D7’s TCATE-K-grid_upper_tail_mean, the other functional excluded from every training manifest, R3 still beats Causal-DRF (0.0314 against 0.0821) and PTA-S still cannot produce it. Second, the gate rules were satisfied on their own terms, because rules 3 and 4 are evaluated over the declared target set and not over skewness specifically. But a paper whose headline example is the skewness result cannot be written around cwdb_r3_cvridge as if the repair were free, and an applied study that targets a pure shape contrast

should expect the regularised claimant to be conservative about exactly that contrast.

Figure 6 shows all four functionals across the three eligible regimes and makes the trade visible in one picture.

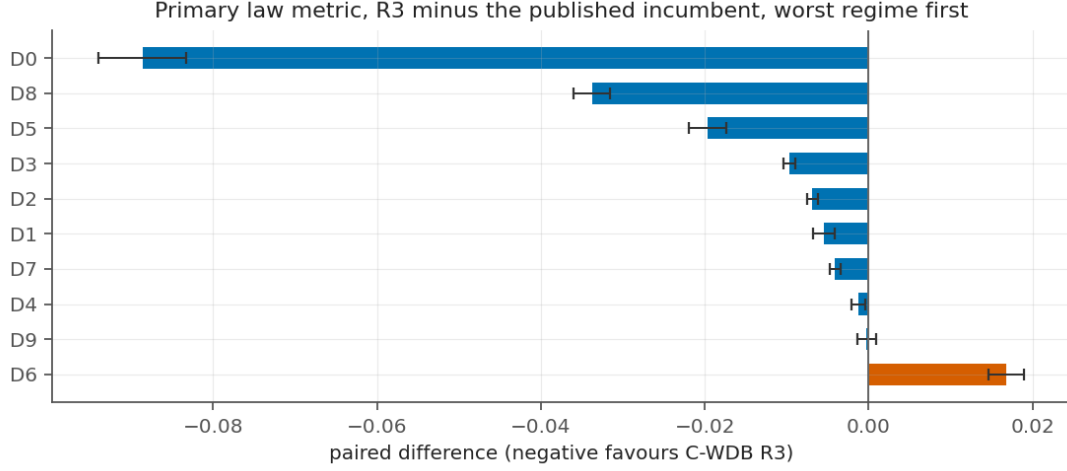


Figure 4: Paired primary-law-metric difference of the recommended claimant against the published incumbent, worst regime first, two paired standard errors. Negative favours C-WDB R3. Compare Figure 3, which is the same construction for the frozen claimant.

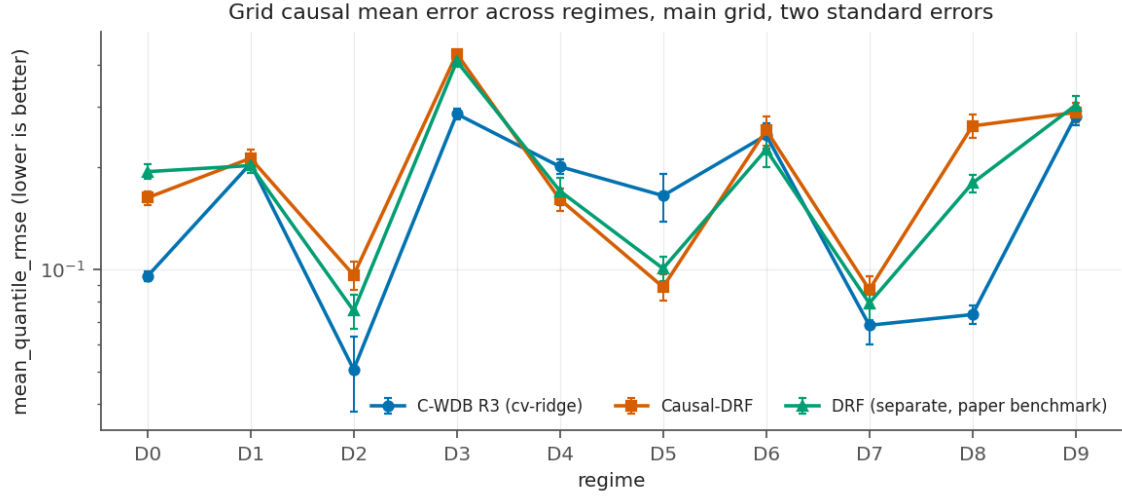


Figure 5: Grid causal mean error, the mandatory common target every method can produce, across all ten regimes. The D2 gap between the two C-WDB curves is the repair. PTA-S is the strongest comparator on this metric and remains so in six regimes, which is the honest statement of where a direct target learner is hard to beat. Numbers in Table 17.

8.6 Finite-particle approximation

Excess risk falls monotonically in M and is nearly flat past $M = 10$ (Figure 8). Ten particles buy most of what twenty-five buy, which is what keeps the method at 11.3 times the incumbent's runtime rather than far more. This is the empirical counterpart to the fixed- M contract of Section 3.4: the approximation is real, it is measured, and it is small at the operating point.

Moderator-conditional functional error; a missing bar means the method cannot estimate that functional

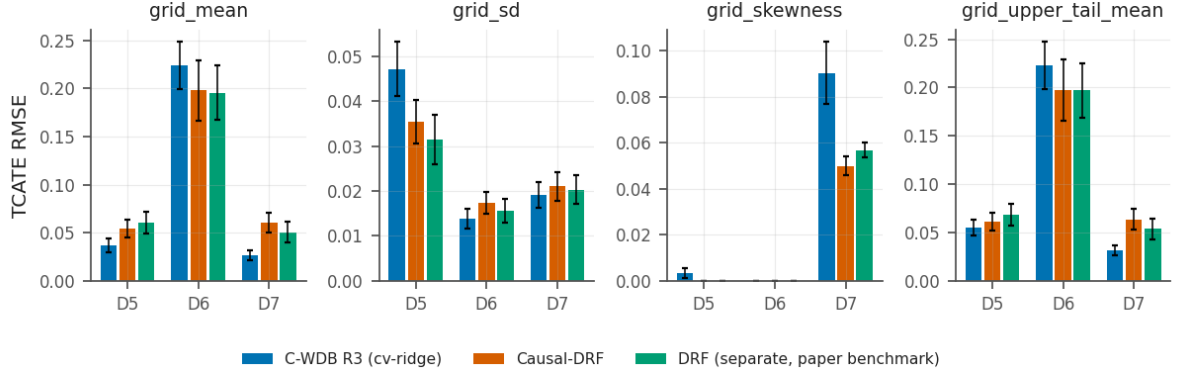


Figure 6: Moderator-conditional functional error on the three regimes eligible for the transfer rules. `grid_skewness` and `grid_upper_tail_mean` are excluded from every training manifest, so PTA-S is marked *n/a* there: that is the transfer test, not a missing measurement. On D6 the true skewness effect is numerically zero, which is why that panel’s bars vanish for every method. Numbers in Table 25.

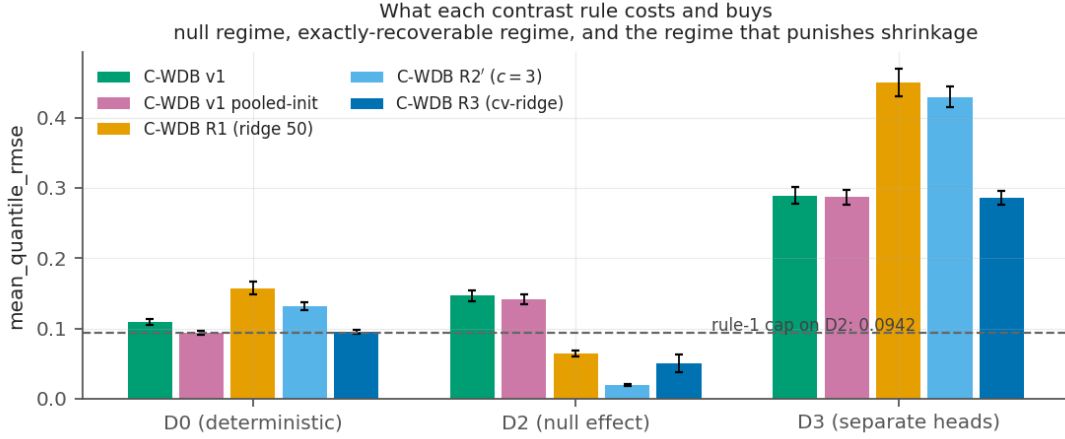


Figure 7: What each contrast rule costs and buys, on the grid causal mean. D2 is the null regime the gate failed on and the dashed line is its preregistered cap. D0 is exactly recoverable, so a rule that cannot switch itself off is penalised there, which is what happens to the hand-tuned fixed ridge. D3 is built to favour separate heads, so a rule that shrinks every contrast is penalised there, which is what happens to both fixed rules. Only the cross-fitted rule is acceptable in all three.

Table 10: Excess energy risk over the true law as a function of the particle count, 20 replications each.

Regime	$M = 2$	$M = 5$	$M = 10$	$M = 25$
D1	0.0897	0.0584	0.0508	0.0468
D6	0.1063	0.0851	0.0696	0.0651

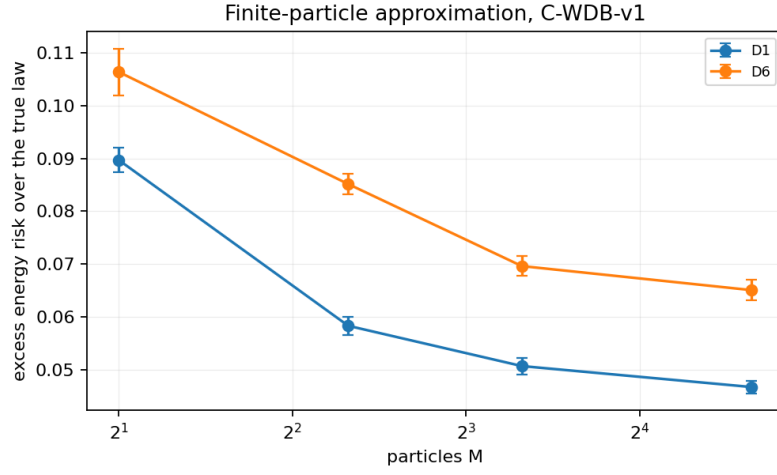


Figure 8: Finite-particle approximation error. The vertical axis is excess energy risk over the true conditional law, so zero is attainable only by the truth.

Table 11: Runtime and memory across the whole tournament.

Method	Median runtime (s)	Max runtime (s)	Median peak RSS (MB)
<code>causal_drf</code>	1.17	35.4	75.2
<code>sqw2_booster</code>	1.00	2.3	0
<code>wdrft</code>	1.33	36.2	214
<code>cwdb_v0</code>	8.93	42.2	0
<code>cwdb_v1</code>	13.3	149	0
<code>pta_s</code>	28.1	77.4	0
<code>pta_f</code>	96.1	113	0

8.7 Cost

9 Diagnosis of the rule-1 failure

Two distinct mechanisms produced the false effect on D2, and separating them before writing any fix changed which repair was worth building. Both diagnostics ran on pilot seeds outside the manifest.

9.1 Defect one: confounded initialization

`compute_init_base` was applied per arm, so each arm started from the empirical law of its own treated or control sample. That is a *marginal* quantity. Under a covariate-dependent propensity the two marginals differ even when the two conditional laws are identical, so the estimated contrast is nonzero before a single tree is fitted, and the booster then spends its budget removing an offset the initialization created. Measured over 20 samples at $n = 1000$, the initial arm gap has root mean square 0.215 on D2, where the truth is exactly zero, and 0.824 on D8.

9.2 Defect two: contrast accumulation

Replaying the fitted path on a held-out design, D2’s contrast error reaches a minimum near iteration 20 and then rises *monotonically* to the frozen budget of 100: 0.077, 0.093, 0.104, 0.114, 0.122 at iterations 20, 40, 60, 80, 100. Repeating the sweep at `arm_shrinkage` of 0, 50, and 500 gives final errors of 0.161, 0.076, and 0.045, which locates the defect precisely. As Section 4.3 shows, the frozen rule leaves a contrast of $n_a/(n_a + \lambda) \approx 0.86$ of the raw gap at a typical leaf. The contrast was barely regularized at all, and one hundred boosting steps compounded what survived.

9.3 Attribution

The initialization ablation settles which defect is the repair. `cwdb.v1_pooledinit` is frozen `v1` in every respect except that both arms start from one pooled base law. It improves D2 from 0.1466 to 0.1419, a false-effect ratio of 2.60 against 2.69, and **still fails rule 1**. The initialization was a genuine defect and fixing it was free, but it is not the repair. The second mechanism is.

10 The repair track

10.1 Design

The repair preregistration amends nothing in the parent document: every threshold, metric, eligible regime, decision multiple, and comparator stays as frozen. It adds methods and cells only. The 97,650 frozen rows were *not* recomputed, so a repair variant is compared against the numbers already published rather than a fresh draw of them, and repair cells reuse the frozen coordinates and seeds so every comparison is paired seed by seed.

The run was staged deliberately. Stage 1 is D2 alone, because rule 1 is the only rule the repair exists to fix and a variant that fails to fix it should cost nothing further; stage 2 runs every regime and the two ablation grids, and only for variants that cleared stage 1. Execution: 2600 repair cells, 68,240 rows, zero failures, merge audit **PASS**, independent recomputation agreeing with zero disagreements, and the frozen gate flags byte-identical to the published payload.

10.2 Rule 1 across the roster

Two entries need stating in words. The hand-tuned fixed ridge R1 clears the D2 ratio at 1.18 and then fails rule 1 anyway, *on D0*, at 0.1574 against a cap of 0.15. D0 has deterministic conditional distributions, so its conditional mean is exactly recoverable, and a fixed contrast shrinkage that cannot switch itself off biases the recoverable answer. R1 is the one variant that was tuned, it was

Table 12: Rule 1 for every variant. The D2 reference is the best of the frozen roster, 0.0545, which a repair variant cannot move by entering the pool. The D0 cap is 0.15 and the D2 ratio cap is 1.25.

Method	D2 mean_quantile_rmse	Ratio	D0	Rule 1
cwdb.v1 (frozen claimant)	0.1466	2.69	0.1091	FAIL
cwdb.v1_pooledinit (ablation)	0.1419	2.60	0.0939	FAIL
cwdb.r1_ridge ($\lambda_r = 50$)	0.0642	1.18	0.1574	FAIL
cwdb.r2_threshold ($c = 1$)	0.0790	1.45	not run	FAIL
cwdb.r2_threshold3 ($c = 3$)	0.0194	0.36	0.1322	PASS
cwdb.r3_cvridge	0.0506	0.93	0.0952	PASS

tuned on D2, and it broke the regime it was not tuned on. That is the argument against a frozen strength, made by the frozen strength itself.

R2 at the principled null-calibrated $c = 1$ missed the cap at 1.45 and stopped after D2 under the preregistered screen. Its rows remain in the merged table and in the memo. The calibration that is principled is not the one that passed; the one that passed, $c = 3$, is a declared conservative sensitivity, and that ordering is reported rather than smoothed over.

10.3 All six rules

Table 13: Gate outcome for every repair variant.

Method	1	2	3	4	5	6	Verdict
cwdb.v1	fail	PASS	PASS	PASS	PASS	PASS	NOT-GO
cwdb.v1_pooledinit	fail	PASS	PASS	PASS	PASS	PASS	NOT-GO
cwdb.r1_ridge	fail	PASS	PASS	PASS	PASS	PASS	NOT-GO
cwdb.r2_threshold3	PASS	PASS	PASS	PASS	PASS	PASS	GO
cwdb.r3_cvridge	PASS	PASS	PASS	PASS	PASS	PASS	GO

Rule 2 holds for both passing variants with three wins against Causal-DRF on the primary law metric, in D1, D5, and D7, the same three regimes the frozen claimant won. Rule 5 holds for both, with D6 mode coverage 0.9921 and 0.9931 against the frozen claimant’s 0.9897 and participation at 10.00 of $M = 10$ in every case. The repulsion mechanism is untouched, which is the point of the reparameterization (3): the contrast rules move only the arm gap and provably cannot move the pooled component.

Rule 6, against the original-code Causal-DRF median of 8.77 s and a ceiling of 60, gives R3 a median of 55.35 s and a ratio of 6.3. The repair preregistration recorded, before the run, that R3 was projected near 64 and might fail; it landed at 55.35 because the projection applied the factor measured on the expensive $K = 25$ cells to every cell, whereas the cheap $K = 5$ cells carry a smaller multiplier and the median sits among them. The projection was pessimistic; the budget was not revised after the fact.

10.4 What separates the two passing variants

Both pass. They are not equally good, and the mechanism ablations say why.

cwdb.r2.threshold3 loses the sharing ablation on D3, reaching 0.3854 against C-WDB-v0’s 0.2910, and loses the law metric there too. D3 is the regime built to favour separate heads, and a rule that shrinks every arm gap toward the pooled leaf is punished by exactly the structure D3 contains. The shared-partition mechanism is one of the project’s own claims and the parent memo supports it with this ablation, so a repair that passes the gate and takes that claim with it has traded one asset for another. R1 loses the same ablation for the same reason.

Table 14: Mechanism ablations, paired differences with standard errors in parentheses. Negative favours the claimant. Bold marks the ablation that is lost.

Ablation	Comparator	cwdb_v1	cwdb_r2_threshold3	cwdb_r3_cvridge
Repulsion, D6 mode coverage	sqw2_booster	−0.7550 (0.0050)	−0.7581 (0.0064)	−0.7584 (0.0061)
Repulsion, D6 law metric	sqw2_booster	−0.1602 (0.0012)	−0.1646 (0.0010)	−0.1646 (0.0012)
Sharing, D3 mean-quantile	cwdb_v0	−0.0320 (0.0036)	+0.0944 (0.0042)	−0.0306 (0.0030)
Sharing, D3 law metric	cwdb_v0	−0.0001 (0.0003)	+0.0055 (0.0004)	−0.0001 (0.0003)
Sharing, D4 mean-quantile	cwdb_v0	−0.1012 (0.0032)	−0.0805 (0.0062)	−0.1003 (0.0039)
Shrinkage, D2 mean-quantile	cwdb_v1_noshrink	−0.0338 (0.0015)	−0.1431 (0.0018)	−0.1188 (0.0037)
Shrinkage, D8 mean-quantile	cwdb_v1_noshrink	−0.0320 (0.0027)	−0.1336 (0.0051)	−0.1420 (0.0042)

cwdb_r3_cvridge keeps it, at −0.0306 (SE 0.0030), matching the frozen claimant’s −0.0320 (SE 0.0036) to within a standard error. It loses exactly one ablation, D1 kernel law error against the squared- W_2 booster, which is the same one the frozen claimant already lost and is therefore not damage the repair caused.

The reason is visible in what cross-fitting selects (Table 15). On D3 it selects zero strength on every seed, so the shared-partition mechanism runs unregularized where the data say a real contrast exists; on D2 and D8 it selects hard. That is the adaptivity a fixed rule cannot have, and it is what lets one variant repair rule 1 without paying for it anywhere else.

Table 15: Median contrast strength selected on held-out energy risk by cwdb_r3_cvridge.

Regimes	Median strength	What the regime contains
D0, D3	0	deterministic laws; strong separate-head effect
D1, D4, D5	0	moderate real effects
D2	50 (mean 265)	exactly null effect
D6, D7, D9	50	multimodal; unseen-functional effect; poor overlap
D8	500	strong confounding, weak effect

10.5 Improvement beyond the rule that failed

Table 16: cwdb_r3_cvridge against cwdb_v1 on mean_quantile_rmse, main grid, paired by seed.

Regime	R3	v1	Difference	SE	Favours
D0	0.0872	0.1013	−0.0141	0.0025	repair
D1	0.1809	0.1691	+0.0118	0.0027	v1
D2	0.0417	0.1266	−0.0850	0.0037	repair
D3	0.2603	0.2590	+0.0014	0.0022	tie
D4	0.1774	0.1765	+0.0009	0.0032	tie
D5	0.0910	0.1559	−0.0649	0.0099	repair
D6	0.2130	0.2490	−0.0360	0.0094	repair
D7	0.0572	0.1264	−0.0692	0.0020	repair
D8	0.0641	0.1741	−0.1100	0.0062	repair
D9	0.2442	0.2211	+0.0231	0.0095	v1

Six regimes improve, two tie, and two regress. The regressions are D1, a smooth location-scale law with a real effect, and D9, where overlap deteriorates to the clipping bounds. Both are small in absolute terms and both are cases where regularizing a real contrast costs something; neither is a gate rule and neither should be described as free. The largest single improvement is D8, from 0.1741 to 0.0641, which is exactly the regime combining both diagnosed defects: its initialization gap was 0.824 and its effect is weak enough that accumulated contrast noise dominates it.

10.6 Recommendation

Adopt `cwdb_r3_cvridge` as the claimant. It is the only variant that repairs rule 1 while leaving every mechanism the project claims as its own intact, and the mechanism that makes that possible, choosing the regularization strength on held-out energy risk rather than freezing it, is the one the parent memo asked for by name.

Report `cwdb_r2_threshold3` alongside it, not as a rejected alternative but as the variant that buys three *accuracy* wins over PTA-S (at D6 TCATE-K-grid.sd, D7 TCATE-K-grid.mean, D7 TCATE-K-grid.sd) at the price of the shared-partition ablation. A reader who cares more about the direct-learner comparison than about the sharing claim would prefer it, and the results say so. Neither variant dominates.

11 Complete results: every metric for the proposed model and comparators

The sections above argue from selected comparisons. This section reports the whole measurement surface so that a reader can check any claim in the report, or find one it did not make. The tables intentionally retain only `cwdb_r3.cvridge` as the proposed C-WDB model, together with the original-code Causal-DRF incumbent and the original-code separate-DRF benchmark. PTA-S, PTA-F, and the earlier W-DRF-T adapter are not part of this core comparison. Nothing here is recomputed: the tables are generated directly from the audited result files by `report/build_report_assets.py`, which reads and aggregates but never fits a model. The Causal-DRF columns use the original-code rerun recorded in `results/merged_original_causal_drf`, not the earlier local driver. The DRF columns use the separate half-sample implementation in `code/drfinference-main`, with 2500 total trees and 50 trees per group. The selected historical discussion above preserves the original tournament record; the tables and figures in this section are the corrected comparator record.

Conventions. Every entry is a mean over cells with the standard error in parentheses. A cell is one (grid, regime, n , K , M , method, seed) combination, and arm-specific rows are averaged inside a cell first, matching the convention of the independent gate-flag recomputation. On the main grid that is 40 cells per entry, $n \in \{500, 1000\}$ crossed with 20 seeds; on `smallk` it is 20 cells at $n = 500$, $K = 5$. Bold marks the best method in a regime. n/a means the method does not produce that quantity at all, which for the two PTA endpoints is a contract restriction rather than a measurement gap, and for `cwdb_r2.threshold` at $c = 1$ is the preregistered stage-1 screen that stopped it after D2.

Reading the comparator. The squared- W_2 booster appears only on the small- K mechanism tables as a comparator for the repulsion term. It is not a proposed model. The DRF column is the faithful separate-forest benchmark from the Causal-DRF paper, whereas W-DRF-T is deliberately excluded from this core comparison because it is a different, cheaper adapter.

Treatment-effect reading. This focused comparison is more relevant to the contribution than a leaderboard containing fixed-target mean learners. On the main grid, R3 is the lowest-error method in six of ten regimes for `mean_quantile_rmse`, seven of ten for `REF-TCATE-K`, and five of ten for `REF-ATE-K`. Across the forty conditional functional cells, R3 is best in twenty-four, compared with eleven for DRF and five for Causal-DRF. Across the forty marginal functional cells, R3 is best in twenty-eight, compared with eight for DRF and four for Causal-DRF. These are cell-average rankings, not a claim of uniform dominance or a replacement for the paired standard errors in Table 30.

The most relevant transfer regime is D7. R3 is best for the conditional mean, standard deviation, and upper-tail effect, and for all three corresponding marginal effects. It loses the D7 skewness effect to both forest comparators: the conditional errors are 0.0904 for R3, 0.0500 for Causal-DRF, and 0.0567 for DRF. The result supports a narrower and stronger claim: R3 is especially effective for marginal and tail treatment-effect functionals, while pure skewness contrasts remain a documented failure region.

One caution about pooled means. A regime mean pools $n = 500$ and $n = 1000$, so it is not identical to the per- n numbers quoted in the repair memo’s Section 6, which are $n = 1000$. Both conventions appear in the project; this report uses the pooled one everywhere except where it says otherwise, which is the same convention rule 1 is evaluated under.

Column abbreviations. The wide tables use short method names: R3 is `cwdb_r3_cvridge`, C-DRF is Causal-DRF, DRF is the separate half-sample forest benchmark from the original Causal-DRF code, and $\text{sq-}W_2$ is the repulsion ablation.

Table 17: Grid causal mean error `mean_quantile_rmse` against `MEANQ-A-K`, main grid, mean over 40 cells ($n \in \{500, 1000\}$, 20 seeds) with the standard error in parentheses. Best in each regime is bold.

Regime	R3	C-DRF	DRF
D0	0.0952 (0.0016)	0.1621 (0.0038)	0.1935 (0.0048)
D1	0.2043 (0.0042)	0.2119 (0.0062)	0.2017 (0.0049)
D2	0.0506 (0.0063)	0.0960 (0.0046)	0.0754 (0.0042)
D3	0.2861 (0.0049)	0.4301 (0.0060)	0.4088 (0.0072)
D4	0.2002 (0.0048)	0.1603 (0.0062)	0.1697 (0.0078)
D5	0.1644 (0.0133)	0.0885 (0.0038)	0.1004 (0.0041)
D6	0.2477 (0.0108)	0.2567 (0.0128)	0.2247 (0.0129)
D7	0.0683 (0.0042)	0.0870 (0.0042)	0.0794 (0.0043)
D8	0.0735 (0.0022)	0.2638 (0.0104)	0.1790 (0.0053)
D9	0.2819 (0.0087)	0.2891 (0.0095)	0.3034 (0.0101)

`cwdb_r2_threshold` at $c = 1$ is omitted: the preregistered stage-1 screen stopped it after D2, where it scored 0.0790.

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Table 18: Declared primary law metric `kernel_law_error` (squared MMD to the true conditional law, Gaussian kernel on the rescaled coordinates, median bandwidth), main grid, for the two forest comparators and the proposed law estimator.

Regime	R3	C-DRF	DRF
D0	0.0280 (0.0012)	0.1166 (0.0035)	0.1365 (0.0044)
D1	0.0137 (0.0005)	0.0191 (0.0009)	0.0171 (0.0006)
D2	0.0115 (0.0004)	0.0184 (0.0006)	0.0158 (0.0005)
D3	0.0162 (0.0005)	0.0259 (0.0007)	0.0238 (0.0008)
D4	0.0115 (0.0004)	0.0128 (0.0007)	0.0141 (0.0010)
D5	0.0234 (0.0006)	0.0431 (0.0013)	0.0478 (0.0012)
D6	0.0246 (0.0013)	0.0077 (0.0004)	0.0082 (0.0005)
D7	0.0114 (0.0004)	0.0155 (0.0005)	0.0165 (0.0005)
D8	0.0151 (0.0005)	0.0489 (0.0014)	0.0466 (0.0011)
D9	0.0197 (0.0009)	0.0200 (0.0008)	0.0260 (0.0013)

Table 19: Excess energy risk over the true conditional law `arm_energy_risk`, main grid. Zero is attainable only Table 10 is included in these values.

Regime	R3	C-DRF	DRF
D0	0.0586 (0.0012)	0.1318 (0.0019)	0.1464 (0.0026)
D1	0.0624 (0.0021)	0.0507 (0.0019)	0.0477 (0.0015)
D2	0.0534 (0.0018)	0.0451 (0.0012)	0.0400 (0.0010)
D3	0.0692 (0.0019)	0.0643 (0.0014)	0.0597 (0.0018)
D4	0.0607 (0.0020)	0.0405 (0.0018)	0.0443 (0.0027)
D5	0.0592 (0.0020)	0.0511 (0.0013)	0.0568 (0.0013)
D6	0.0841 (0.0034)	0.0344 (0.0012)	0.0400 (0.0017)
D7	0.0540 (0.0018)	0.0398 (0.0011)	0.0422 (0.0011)
D8	0.0642 (0.0018)	0.1211 (0.0030)	0.1168 (0.0023)
D9	0.0737 (0.0028)	0.0528 (0.0018)	0.0658 (0.0027)

Table 20: Arm-level barycenter error `barycenter_rmse` against `BARY-A/MEANQ-A-K`, main grid. Reported separ with it, since one is an arm-level quantity and the other the arm contrast.

Regime	R3	C-DRF	DRF
D0	0.1062 (0.0019)	0.2098 (0.0035)	0.2313 (0.0040)
D1	0.2174 (0.0041)	0.2632 (0.0064)	0.2448 (0.0043)
D2	0.1886 (0.0035)	0.2506 (0.0041)	0.2302 (0.0033)
D3	0.2340 (0.0037)	0.3028 (0.0042)	0.2881 (0.0051)
D4	0.2206 (0.0040)	0.2236 (0.0065)	0.2289 (0.0085)
D5	0.1846 (0.0049)	0.2167 (0.0037)	0.2346 (0.0034)
D6	0.3731 (0.0086)	0.2764 (0.0070)	0.2977 (0.0090)
D7	0.1884 (0.0035)	0.2285 (0.0039)	0.2352 (0.0035)
D8	0.2500 (0.0042)	0.4517 (0.0075)	0.4384 (0.0051)
D9	0.2596 (0.0059)	0.2692 (0.0058)	0.3070 (0.0079)

Table 21: **mode_coverage**, the fraction of true outer modes carrying predicted mass, main grid. Higher is better so every other row is a degenerate single-mode check that a method should pass trivially.

Regime	R3	C-DRF	DRF
D0	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D1	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D2	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D3	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D4	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D5	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D6	0.9931 (0.0012)	1.0000 (0.0000)	1.0000 (0.0000)
D7	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D8	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D9	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)

Table 22: **tail_calibration**, absolute error of a predeclared threshold event under the grid

Regime	R3	C-DRF	DRF
D0	0.0708 (0.0022)	0.1783 (0.0029)	0.1792 (0.0041)
D1	0.1058 (0.0022)	0.1085 (0.0025)	0.1032 (0.0024)
D2	0.1139 (0.0024)	0.1019 (0.0017)	0.0938 (0.0018)
D3	0.1150 (0.0021)	0.1259 (0.0023)	0.1135 (0.0029)
D4	0.0771 (0.0026)	0.0858 (0.0032)	0.0689 (0.0035)
D5	0.0944 (0.0025)	0.0722 (0.0016)	0.0805 (0.0020)
D6	0.0955 (0.0034)	0.0555 (0.0013)	0.0546 (0.0015)
D7	0.0521 (0.0011)	0.0426 (0.0010)	0.0430 (0.0012)
D8	0.0992 (0.0022)	0.1452 (0.0031)	0.1320 (0.0026)
D9	0.1187 (0.0031)	0.1124 (0.0025)	0.1324 (0.0031)

Table 23: Conditional Wasserstein reference effect REF-TCATE-K with ν_* the standard norm

Regime	R3	C-DRF	DRF
D0	0.0508 (0.0012)	0.1037 (0.0026)	0.1140 (0.0055)
D1	0.0734 (0.0054)	0.1047 (0.0027)	0.1138 (0.0039)
D2	0.0170 (0.0018)	0.0270 (0.0027)	0.0291 (0.0026)
D3	0.0375 (0.0030)	0.0355 (0.0028)	0.0337 (0.0026)
D4	0.0936 (0.0057)	0.1098 (0.0044)	0.1193 (0.0074)
D5	0.1180 (0.0035)	0.0330 (0.0027)	0.0311 (0.0024)
D6	0.0389 (0.0032)	0.0646 (0.0058)	0.0610 (0.0062)
D7	0.0215 (0.0017)	0.0278 (0.0022)	0.0285 (0.0020)
D8	0.0406 (0.0026)	0.0642 (0.0050)	0.0830 (0.0056)
D9	0.1708 (0.0096)	0.1044 (0.0031)	0.1277 (0.0036)

Table 24: Marginal Wasserstein reference effect REF-ATE-K, main grid.

Regime	R3	C-DRF	DRF
D0	0.0393 (0.0010)	0.0126 (0.0015)	0.0114 (0.0018)
D1	0.0381 (0.0049)	0.0206 (0.0030)	0.0211 (0.0028)
D2	0.0099 (0.0016)	0.0178 (0.0029)	0.0175 (0.0026)
D3	0.0209 (0.0028)	0.0238 (0.0032)	0.0241 (0.0031)
D4	0.0562 (0.0049)	0.0240 (0.0033)	0.0251 (0.0029)
D5	0.1152 (0.0036)	0.0229 (0.0026)	0.0214 (0.0026)
D6	0.0262 (0.0030)	0.0422 (0.0061)	0.0417 (0.0067)
D7	0.0143 (0.0019)	0.0164 (0.0025)	0.0167 (0.0022)
D8	0.0233 (0.0021)	0.0458 (0.0057)	0.0645 (0.0066)
D9	0.1340 (0.0102)	0.0250 (0.0028)	0.0267 (0.0034)

Table 25: Moderator-conditional functional error `tcate_functional_rmse` for all four grid functionals, main grid. `grid_skewness` and `grid_upper_tail_mean` are excluded from every training manifest; the law-producing methods still estimate them at evaluation time, which is the transfer test.

Functional	Regime	R3	C-DRF	DRF
grid_mean	D0	0.0569 (0.0013)	0.1153 (0.0038)	0.1123 (0.0061)
	D1	0.1185 (0.0075)	0.1696 (0.0071)	0.1314 (0.0051)
	D2	0.0240 (0.0026)	0.0756 (0.0056)	0.0491 (0.0051)
	D3	0.1016 (0.0044)	0.1381 (0.0086)	0.0993 (0.0060)
	D4	0.1165 (0.0064)	0.1250 (0.0064)	0.1183 (0.0081)
	D5	0.0369 (0.0034)	0.0544 (0.0047)	0.0608 (0.0057)
	D6	0.2239 (0.0122)	0.1980 (0.0157)	0.1957 (0.0141)
	D7	0.0265 (0.0025)	0.0606 (0.0053)	0.0504 (0.0053)
	D8	0.0531 (0.0030)	0.2248 (0.0125)	0.1250 (0.0076)
	D9	0.2380 (0.0104)	0.2430 (0.0105)	0.2523 (0.0114)
grid_sd	D0	0.0104 (0.0006)	0.0180 (0.0012)	0.0173 (0.0013)
	D1	0.0294 (0.0023)	0.0276 (0.0017)	0.0272 (0.0020)
	D2	0.0181 (0.0014)	0.0194 (0.0015)	0.0205 (0.0016)
	D3	0.0296 (0.0017)	0.0284 (0.0019)	0.0254 (0.0016)
	D4	0.0402 (0.0025)	0.0503 (0.0024)	0.0474 (0.0027)
	D5	0.0471 (0.0030)	0.0354 (0.0024)	0.0314 (0.0028)
	D6	0.0139 (0.0011)	0.0174 (0.0012)	0.0155 (0.0013)
	D7	0.0191 (0.0015)	0.0210 (0.0016)	0.0203 (0.0016)
	D8	0.0178 (0.0015)	0.0258 (0.0017)	0.0252 (0.0019)
	D9	0.0407 (0.0025)	0.0572 (0.0031)	0.0642 (0.0032)
grid_skewness	D0	0.0101 (0.0015)	0.0000 (0.0000)	0.0000 (0.0000)
	D1	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D2	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D3	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D4	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D5	0.0034 (0.0011)	0.0000 (0.0000)	0.0000 (0.0000)
	D6	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D7	0.0904 (0.0067)	0.0500 (0.0021)	0.0567 (0.0016)
	D8	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D9	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
grid_upper_tail_mean	D0	0.0570 (0.0014)	0.1208 (0.0042)	0.1158 (0.0063)

Functional	Regime	R3	C-DRF	DRF
	D1	0.1209 (0.0073)	0.1760 (0.0078)	0.1362 (0.0054)
	D2	0.0291 (0.0030)	0.0782 (0.0059)	0.0532 (0.0054)
	D3	0.1049 (0.0044)	0.1453 (0.0088)	0.1057 (0.0061)
	D4	0.1383 (0.0076)	0.1576 (0.0074)	0.1469 (0.0100)
	D5	0.0552 (0.0040)	0.0611 (0.0046)	0.0683 (0.0057)
	D6	0.2224 (0.0123)	0.1969 (0.0159)	0.1969 (0.0141)
	D7	0.0314 (0.0024)	0.0635 (0.0054)	0.0536 (0.0055)
	D8	0.0566 (0.0031)	0.2319 (0.0130)	0.1300 (0.0075)
	D9	0.2622 (0.0099)	0.2798 (0.0115)	0.2935 (0.0119)

Table 26: Marginal functional error `tate_functional_rmse` for all four grid functionals, main grid.

Functional	Regime	R3	C-DRF	DRF
grid_mean	D0	0.0065 (0.0009)	0.0547 (0.0039)	0.0286 (0.0032)
	D1	0.0181 (0.0024)	0.0925 (0.0068)	0.0319 (0.0042)
	D2	0.0156 (0.0019)	0.0686 (0.0058)	0.0226 (0.0042)
	D3	0.0177 (0.0021)	0.0289 (0.0043)	0.0234 (0.0025)
	D4	0.0196 (0.0023)	0.0613 (0.0062)	0.0312 (0.0048)
	D5	0.0155 (0.0023)	0.0384 (0.0053)	0.0207 (0.0036)
	D6	0.0476 (0.0057)	0.1205 (0.0147)	0.1158 (0.0146)
	D7	0.0152 (0.0018)	0.0478 (0.0057)	0.0234 (0.0042)
	D8	0.0313 (0.0033)	0.2001 (0.0144)	0.0751 (0.0086)
	D9	0.0247 (0.0033)	0.1872 (0.0106)	0.2076 (0.0117)
grid_sd	D0	0.0055 (0.0007)	0.0134 (0.0012)	0.0122 (0.0012)
	D1	0.0170 (0.0020)	0.0192 (0.0021)	0.0181 (0.0023)
	D2	0.0130 (0.0012)	0.0140 (0.0017)	0.0142 (0.0016)
	D3	0.0145 (0.0016)	0.0199 (0.0020)	0.0181 (0.0021)
	D4	0.0151 (0.0018)	0.0305 (0.0026)	0.0223 (0.0025)
	D5	0.0260 (0.0032)	0.0247 (0.0026)	0.0242 (0.0027)
	D6	0.0098 (0.0010)	0.0110 (0.0014)	0.0113 (0.0014)
	D7	0.0136 (0.0013)	0.0141 (0.0017)	0.0141 (0.0016)
	D8	0.0126 (0.0015)	0.0186 (0.0020)	0.0194 (0.0022)
	D9	0.0162 (0.0022)	0.0470 (0.0037)	0.0519 (0.0036)
grid_skewness	D0	0.0085 (0.0015)	0.0000 (0.0000)	0.0000 (0.0000)
	D1	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D2	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D3	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D4	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D5	0.0029 (0.0011)	0.0000 (0.0000)	0.0000 (0.0000)
	D6	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D7	0.0746 (0.0060)	0.0167 (0.0016)	0.0130 (0.0014)
	D8	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D9	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
grid_upper_tail_mean	D0	0.0078 (0.0010)	0.0644 (0.0041)	0.0364 (0.0035)
	D1	0.0234 (0.0031)	0.1037 (0.0077)	0.0410 (0.0052)
	D2	0.0197 (0.0024)	0.0707 (0.0062)	0.0261 (0.0046)
	D3	0.0223 (0.0023)	0.0389 (0.0054)	0.0288 (0.0033)
	D4	0.0244 (0.0029)	0.0837 (0.0071)	0.0440 (0.0058)
	D5	0.0307 (0.0029)	0.0426 (0.0057)	0.0299 (0.0039)
	D6	0.0477 (0.0056)	0.1189 (0.0149)	0.1145 (0.0148)
	D7	0.0195 (0.0023)	0.0503 (0.0058)	0.0260 (0.0046)
	D8	0.0339 (0.0034)	0.2076 (0.0149)	0.0833 (0.0088)
	D9	0.0282 (0.0042)	0.2233 (0.0109)	0.2479 (0.0121)

Table 27: mean_quantile_rmse on the smallk grid ($n = 500$, $K = 5$, 20 seeds), with the proposed C-WDB R3 squared- W_2 mechanism ablation.

Regime	R3	C-DRF	DRF	sq- W_2
D0	0.1044 (0.0013)	0.1817 (0.0027)	0.2175 (0.0055)	0.0841 (0.0011)
D1	0.2209 (0.0034)	0.2642 (0.0078)	0.2167 (0.0047)	0.1628 (0.0038)
D2	0.0831 (0.0160)	0.1191 (0.0084)	0.0821 (0.0082)	0.1369 (0.0031)
D3	0.3115 (0.0044)	0.5385 (0.0041)	0.4402 (0.0024)	0.2990 (0.0040)
D4	0.2194 (0.0045)	0.2409 (0.0082)	0.2111 (0.0059)	0.1529 (0.0044)
D5	0.2149 (0.0128)	0.1056 (0.0069)	0.0910 (0.0067)	0.2027 (0.0051)
D6	0.3368 (0.0116)	0.3374 (0.0219)	0.2453 (0.0211)	0.4439 (0.0175)
D7	0.0872 (0.0133)	0.1162 (0.0088)	0.0839 (0.0081)	0.1341 (0.0034)
D8	0.0784 (0.0025)	0.3171 (0.0157)	0.1966 (0.0106)	0.1743 (0.0048)
D9	0.2941 (0.0064)	0.3311 (0.0089)	0.3584 (0.0103)	0.2200 (0.0057)

Table 28: kernel_low_error on the smallk grid, with the squared- W_2 comparator retained as

Regime	R3	C-DRF	DRF	sq- W_2
D0	0.0334 (0.0013)	0.1445 (0.0042)	0.1556 (0.0040)	0.0244 (0.0010)
D1	0.0170 (0.0003)	0.0298 (0.0008)	0.0211 (0.0005)	0.0139 (0.0003)
D2	0.0150 (0.0004)	0.0262 (0.0006)	0.0196 (0.0005)	0.0140 (0.0003)
D3	0.0204 (0.0005)	0.0405 (0.0006)	0.0292 (0.0003)	0.0201 (0.0004)
D4	0.0145 (0.0004)	0.0283 (0.0009)	0.0210 (0.0004)	0.0105 (0.0002)
D5	0.0285 (0.0010)	0.0563 (0.0014)	0.0536 (0.0009)	0.0221 (0.0008)
D6	0.0324 (0.0013)	0.0127 (0.0005)	0.0076 (0.0006)	0.1970 (0.0017)
D7	0.0157 (0.0004)	0.0250 (0.0008)	0.0206 (0.0005)	0.0144 (0.0003)
D8	0.0185 (0.0005)	0.0612 (0.0011)	0.0570 (0.0012)	0.0201 (0.0005)
D9	0.0242 (0.0006)	0.0255 (0.0009)	0.0358 (0.0012)	0.0191 (0.0007)

Table 29: mode_coverage on the smallk grid. The D6 row is the collapse result: every method that keeps the 1 the ablation does not.

Regime	R3	C-DRF	DRF	sq- W_2
D0	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D1	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D2	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D3	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0001)
D4	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D5	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	0.9989 (0.0005)
D6	0.9887 (0.0031)	1.0000 (0.0000)	1.0000 (0.0000)	0.2303 (0.0038)
D7	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D8	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	0.9994 (0.0003)
D9	0.9999 (0.0001)	1.0000 (0.0000)	1.0000 (0.0000)	0.9999 (0.0001)

Table 30: Seed-paired comparisons for the proposed claimant `cwdb_r3_cvrldge`, n

Target	Regime	vs Causal-DRF diff.	SE	vs DRF (separate, paper benchmark) diff.	
kernel_law_error	D1	-0.0054*	0.0007	-0.0034*	0.0
	D5	-0.0196*	0.0012	-0.0244*	0.0
	D6	+0.0168†	0.0011	+0.0164†	0.0
	D7	-0.0041*	0.0003	-0.0051*	0.0
	D0	-0.0669*	0.0028	-0.0983*	0.0
	D1	-0.0076	0.0040	+0.0027	0.0
	D2	-0.0453*	0.0075	-0.0247*	0.0
	D3	-0.1440*	0.0033	-0.1227*	0.0
	D4	+0.0399†	0.0031	+0.0305†	0.0
mean_quantile_rmse	D5	+0.0759†	0.0120	+0.0641†	0.0
	D6	-0.0089	0.0125	+0.0230	0.0
	D7	-0.0187*	0.0046	-0.0110*	0.0
	D8	-0.1903*	0.0100	-0.1056*	0.0
	D9	-0.0071	0.0072	-0.0215*	0.0
	D5	+0.0034†	0.0011	+0.0034†	0.0
	D6	+0.0000†	0.0000	+0.0000†	0.0
	D7	+0.0404†	0.0076	+0.0337†	0.0
	D5	-0.0059	0.0042	-0.0131*	0.0
TCATE-K-grid_upper_tail_mean	D6	+0.0256	0.0143	+0.0255	0.0
	D7	-0.0321*	0.0042	-0.0222*	0.0
	D5	-0.0175*	0.0045	-0.0239*	0.0
TCATE-K-grid_mean	D6	+0.0260	0.0142	+0.0282	0.0
	D7	-0.0340*	0.0048	-0.0239*	0.0

Negative favours the claimant. * marks a claimant win and † a comparator win, both at more than two paired errors, which is the preregistered definition. n/a means the comparator does not estimate that target.

Table 31: Operational cost over every cell each method ran, both tracks pooled. Runtime is fit plus predict; peak memory is a process high-water mark, so a zero means the worker never exceeded its own baseline.

Method	Cells	Median (s)	Mean (s)	Max (s)	Median RAM (MB)
squared- W_2 booster	360	1.00	1.31	2.3	0
DRF (separate, paper benchmark)	600	8.06	10.59	41.1	382
Causal-DRF	670	8.77	12.50	58.1	304
C-WDB R3 (cv-ridge)	640	55.35	64.78	135.7	0

11.1 Supporting figures

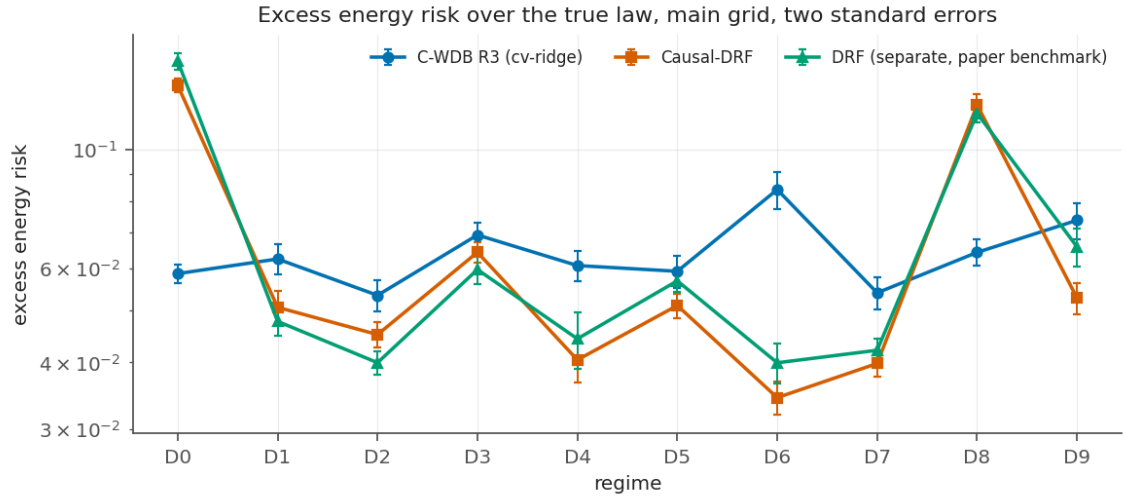


Figure 9: Excess energy risk over the true conditional law across all ten regimes. This is the loss the method actually optimises, evaluated against the quadrature representation of the truth, so it is the most direct measure of how well the conditional law itself is estimated. Numbers in Table 19.

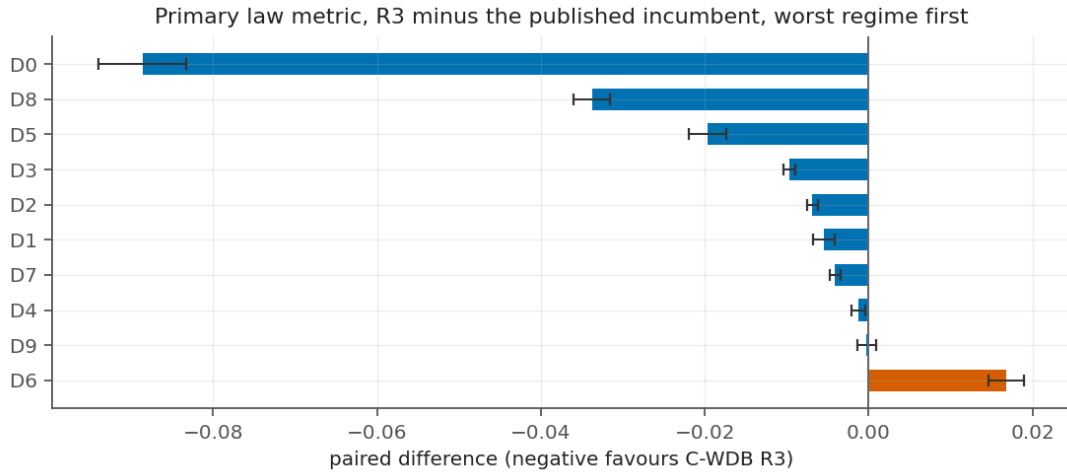


Figure 10: The proposed C-WDB R3 against the original-code Causal-DRF comparator on the primary law metric, worst regime first. Negative values favour R3. This is the figure behind Table 30.

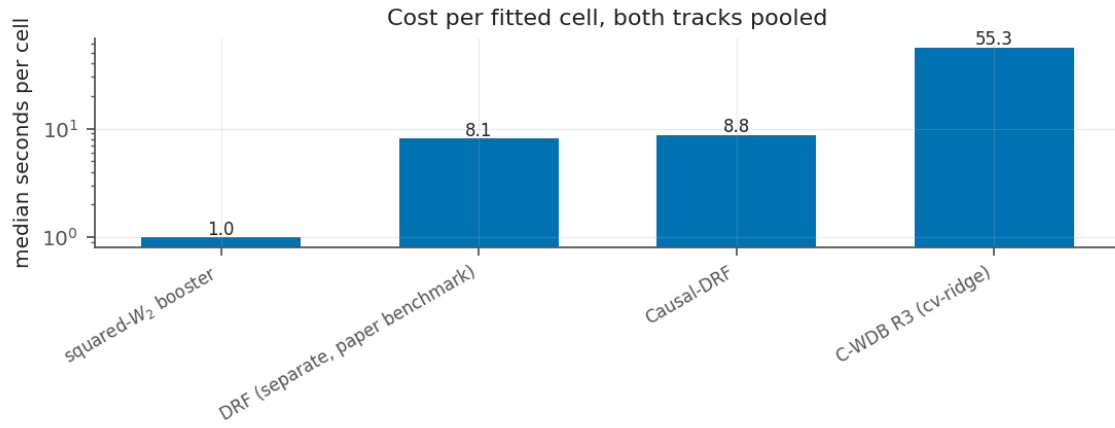


Figure 11: Median seconds per fitted cell, both tracks pooled, on a logarithmic axis. The cross-fitted selector is what separates R3 from the other repair variants: it fits roughly four models where they fit one. Numbers in Table 31.

12 Complete results: the Phase 5.5 orthogonalized variants

Section 11 reports the whole measurement surface for the proposed model and its comparators on the frozen tournament coordinates. This section does the same job for the three orthogonalized variants introduced in Phase 5.5, which were screened on their own manifest and therefore cannot be appended as extra columns to the tables above without silently changing what a column means. Every number here is generated from the audited result files by `report/build_phase55_assets.py` for the Stage 1 screen and by `report/build_phase55_stage2_assets.py` for Stage 2, both of which read and aggregate but never fit a model, exactly as their companion does.

The three variants replace one component of the estimator and leave the rest of the pipeline frozen. `cwdb_rmean` is a vector R-learner in the sense of Nie and Wager: it fits the causal contrast by minimising the R-loss $n^{-1} \sum_i \|Z_i - \hat{m}_{-i}(X_i) - (A_i - \hat{e}_{-i}(X_i)) t(X_i)\|^2$ over the rescaled quantile coordinates, with exact weighted least-squares leaves that never divide by $A - \hat{e}$. `cwdb_mutau` is the Bayesian-causal-forest reparameterisation of the arm-shared tree, carrying a prognostic field and a contrast field in each leaf through $z_{a,m}(x) = b_m(x) + (a - \hat{e}(x)) d_m(x)$. `cwdb_xmean` is a vector X-learner with out-of-fold imputed pseudo-outcomes. All three draw their propensity and prognostic nuisances from one cross-fitted stack with a declared clip of $[0.02, 0.98]$ and a canonical fold plan that is a pure function of the data rather than of row order.

The contract difference between them is not cosmetic and it governs which tables a variant may appear in. `cwdb_mutau` produces a conditional law and is therefore scored on every metric in the report. `cwdb_rmean` and `cwdb_xmean` target the arm contrast only and produce no law at all, so their entries in the law tables are n/a by construction rather than by omission, and no reading of those tables should treat a missing law column as a loss.

Conventions, and where they differ from Section 11. Two of them differ and both differences are deliberate. First, every method in this section, incumbents included, is restricted to seeds 0 through 9, because the Phase 5.5 manifest declared ten seeds rather than twenty. Pooling a twenty-seed incumbent mean against a ten-seed claimant mean in one row would compare two designs, so the incumbent columns here are recomputed on the claimant’s seeds and will not always equal the same cell in Section 11. Second, only D0, D2, D7 and D8 appear in the Stage 1 tables. The screen was declared as a mechanism test on the four regimes that isolate one mechanism each, so it never ran D1, D3, D4, D5, D6 or D9, and no Stage 1 table supports a statement about the variants outside those four regimes. Subsection 12.6 lifts that restriction for the one variant that continued, at twenty seeds and on all ten regimes; it remains in force for `cwdb_rmean` and `cwdb_xmean`, which were never run elsewhere. Otherwise the conventions carry over: an entry is a mean over cells with the standard error in parentheses, a cell is one (grid, regime, n , K , M , method, seed) combination with arm-specific rows averaged inside the cell first, bold marks the best method in a regime, and a regime mean pools $n = 500$ and $n = 1000$, which is why the numbers here are not identical to the per- n figures quoted in the G3.5 screen memo. The Causal-DRF and DRF columns are the original-code reruns, matching the corrected comparator record of Section 11 rather than the retired local drivers.

12.1 The contrast surface

Table 32 is the primary table of this section, because `mean_quantile_rmse` against `MEANQ-A-K` is the one metric all seven methods produce. Four readings are worth stating, and the fourth is the one the screen memo initially failed to make.

The null and deterministic regimes separate the variants sharply. On D2, where the true effect is exactly zero, `rmean` records 0.0191 against 0.0551 for PTA-S and 0.0595 for R3, and Table 43 shows those gaps are 8.8 and 3.2 paired standard errors. This is the clearest single result of Phase 5.5: orthogonalisation, not more capacity, is what removes manufactured heterogeneity, and it removes it against the comparator that motivated the whole repair track. On D8, the confounded

regime that rule 2 of the original gate treated as primary, `rmean` beats PTA-S by 2.8 standard errors but loses to R3 by 5.9, so the R-loss buys null-safety at some cost in confounded accuracy.

D7 is where `rmean` is weakest, at 0.1033 against 0.0637 for PTA-S and 0.0661 for R3, losing to every comparator in the table by between 2.2 and 10.9 standard errors. D7 is the regime whose effect acts through the shape of the law rather than its location, so a method that estimates the contrast directly and never represents the law has the least to work with there. That is a coherent explanation rather than an excuse, and it is also the reason `rmean` is recommended as a benchmark rather than as the claimant.

`cwdb_mutau` sits close to R3 everywhere and is not interchangeable with it. Table 44 shows two regimes where the difference crosses the decision multiple and both go against `mutau`: D0 by 3.3 standard errors and D8 by 2.6. D2 and D7 are ties at 0.4 and 0.6. The honest summary is that the μ/τ leaf rule buys nothing against R3 on the contrast at these sample sizes while costing a little on two regimes, and that its case rests on the law and functional tables rather than on this one.

The D0 result, stated rather than passed over. D0 is the regime with no treatment effect and no confounding, so it is the closest thing in the suite to a pure bias check. Every C-WDB variant loses it decisively to PTA-S: 0.0969 for `rmean`, 0.0985 for `mutau` and 0.0943 for R3, against 0.0544. In paired terms that is 16.0 standard errors for `rmean` and 37.6 for `mutau`. The screen memo originally reported D0 only as passing the 0.15 correctness cap while reporting the D2 and D8 differences against the same comparator in full, which is a selective presentation of a comparison that had already been computed. It is corrected here and in the memo. The deficit is family-wide, not variant-specific, so it is best read as a property of the boosted particle representation in the regime where the truth is a single fixed law and the direct learner has nothing to estimate. It remains an open weakness of the approach, and Stage 2 should report D0 alongside D2 and D8 as a comparison rather than as a cap check.

Table 33 gives the arm-level barycenter error, which is a different quantity and is never pooled with the contrast. PTA-S is the best method in all four regimes there, by a wide margin, which is exactly what a direct arm-level learner should be good at, and the fact that R3 and `mutau` nonetheless win contrasts on D8 is the substantive point: the contrast is not recoverable from arm-level accuracy alone. Tables 38 and 39 report the Wasserstein reference effects, where `mutau` is best on D7 for both the conditional and the marginal version and R3 is best on D2 and D8, all by margins smaller than the ones separating either from the forest comparators.

12.2 The law surface

Only `cwdb_mutau` enters Tables 34 through 37, and the result there is a near-exact tie with R3. Table 46 makes this precise: on `kernel_law_error` the paired difference against R3 never exceeds 1.4 standard errors in any regime, while the difference against Causal-DRF is a `mutau` win in all four at between 8.4 and 23.7. This is the expected result and it is a validity check rather than a discovery. The μ/τ parameterisation provably collapses to the reparameterised shared tree when the propensity basis is the leaf share, a collapse verified in the test suite to 10^{-14} at both the tree and the model level, so a large law difference against R3 would have indicated a defect rather than a gain.

The functional tables carry the one substantive law-side finding. Table 40 shows `mutau` best or tied-best on `grid_mean` and `grid_upper_tail_mean` in three of four regimes, and it shows D7 skewness as a failure region for both C-WDB variants: 0.1134 for `mutau` and 0.0860 for R3 against 0.0514 for Causal-DRF and 0.0580 for DRF, with the same ordering on the marginal version in Table 41. This is the same documented failure region Section 11 reports for R3 and it is not repaired by the μ/τ leaf rule; if anything `mutau` is worse there, which is the $n = 500$ skewness flag recorded in the screen memo. Pure skewness contrasts remain a region where the forest comparators are better and no claim should be made over them.

Table 36 is a degenerate check in this section, since D6 is not in the Phase 5.5 manifest and every regime here is single-mode, so the uniform 1.0000 column establishes only that nothing broke.

12.3 The inherited correctness screen

Table 42 evaluates inherited rule 1 per sample size, under one convention applied identically to every method: the D2 false-effect ratio divides a single n 's D2 mean by the best frozen baseline at that same n . Pooling the baseline across sample sizes, or mixing a per- n numerator with a pooled denominator, produces different numbers and appeared in an earlier draft of the screen memo; it is not the convention and has been removed there too.

The table carries two entries that need reading with care. The first is `cwdb_xmean`, which fails the D0 cap of 0.15 outright at both sample sizes, 0.3545 and 0.3536. That is the cleanest available statement of its failure and it is stronger than the comparative statement the screen memo originally used. The second is R3 at $n = 500$, which shows a D2 ratio of 1.26 against a cap of 1.25. This is an artefact of the ten-seed restriction and not a new failure: on the full twenty seeds of its own track the same ratio is 0.98 at $n = 500$ and 0.93 pooled, which is what Section 11 reports and what the gate was evaluated on. It is printed here rather than suppressed because a reader comparing the two sections would otherwise find the discrepancy unexplained, and because it is a useful reminder that a threshold evaluated on half the seeds is a different test.

12.4 The negative finding, and why it is a mechanism result

`cwdb_xmean` exists to test one hypothesis: that out-of-fold imputation of the missing potential outcome helps when the arms are unbalanced. That hypothesis is specific to the imbalance regime, so a result outside it could not support the method, which is why the `imbalance` extension grid was declared before the run. Table 47 reports it. At a realised mean propensity of 0.307, `xmean` wins D7-imb by 9.9 paired standard errors, loses D2-imb by 8.0, and on D8-imb is behind by 1.9, which does not cross the decision multiple. One win in the regime the method was built for, paid for with a large loss on the null, is not a result that continues to Stage 2.

The verdict was reached on that table alone. Table 48 then answers a separate question, namely whether the failure is a property of the mechanism or of the frozen three-step effect budget, and it is included here because the two answers imply different Stage 2 decisions and the difference must not be settled by assertion. Two observations settle it. The D0 error is flat in n , 0.3545 against 0.3536, which no variance story predicts and an underfit does; and relaxing the budget from three trees to fifty drops D0 from 0.3550 to 0.0658 while D2 rises from 0.0546 to 0.1636 and D8 from 0.1112 to 0.2222 over the very same sweep. So the D0 wall is a bias floor imposed by a total boosting weight of 0.36 against the R-loss contrast's 2.4, and no budget on the sweep is simultaneously effect-recovering and null-safe. On the imbalance grid the best available budget is the frozen one and it still loses D2-imb and D8-imb to `rmean`. The negative finding is therefore about the mechanism and survives the objection that the budget was simply miscalibrated. The structural reason is visible in Table 49: `rmean` selects its contrast ridge per cell by cross-fitted selection, choosing 0 on D0 and between 50 and 500 on the regimes with signal, whereas the X pseudo-outcome regression has only a global constant and no adaptive regulariser at all.

12.5 Diagnostics and cost

Table 49 collects the quantities only the Phase 5.5 variants emit. Two entries carry weight. The effective particle support is 10 of 10 in every regime for `mutau`, against a rule-5 floor of $0.6M = 6$, so the μ/τ leaf rule does not collapse the particle cloud. The selected ridge is 0 on D0 for both `rmean` and `mutau` and rises to between 250 and 430 on the regimes carrying signal, which is the adaptive behaviour the negative finding above turns on.

Table 50 reports runtime with two ratios rather than one, because the rule-6 verdict for `mutau` depends entirely on which Causal-DRF is the denominator and the project contains two. Against the original-code rerun on exactly these cells, a median of 14.47 seconds, the `mutau` ratio is 11.8 and comfortably inside the cap of 60. Against the retired local driver at 1.87 seconds, the ratio is 91.1 and breaches it. The $91\times$ figure quoted in the screen memo is the second of these. The first is the denominator the rest of this report uses, so the defensible statement is that `mutau` passes rule 6 against the current comparator record while remaining the most expensive method in the roster at roughly twice R3 and twelve times Causal-DRF. Two caveats apply to every row. All Phase 5.5 cells ran on a loaded machine alongside other work, so these are upper bounds rather than clean measurements; and `mutau` performs a fully nested refit for its contrast selection while `rmean` performs a nested-approximate one that reuses a single nuisance stack, so the eight-second median of `rmean` and the one hundred and seventy second median of `mutau` are not measurements of the same procedure.

12.6 Stage 2: the μ/τ claimant on the whole grid

Everything above is the Stage 1 mechanism screen. Phase 5.5 was declared as a two-stage design in which the screen decides whether a variant addresses its intended mechanism and Stage 2 is the frozen comparison against the full tournament record. Only `cwdb_mutau` reached it: `cwdb_rmean` cannot supply a law target by contract and was retained as a benchmark rather than as a claimant, and `cwdb_xmean` failed its screen. Stage 2 runs the claimant on all ten regimes at the twenty seeds every incumbent already carries, which is 320 new cells (the six regimes the screen omitted at twenty seeds, and a seed top-up on the four it ran at ten). The merge audit returned PASS with 320 of 320 cells observed, no duplicate keys and no failures, against manifest checksum 29d7ca0928aa542023a091fd2ffa3943.

The stage exists because the screen left the phase’s own requirement for this variant unsatisfied. That requirement names D2, D3, D5, D6, D7, D8 and D9, and the phase’s decision table contains the row stating that a candidate which fixes D2 by losing D3 or D7 is rejected as the main claimant. D3, D5, D6 and D9 had never been run for this method, so the missing evidence was exactly the evidence that could overturn the Stage 1 verdict. Thresholds were frozen in `research/simulation_preregistration_phase55_stage2.md` before the first Stage 2 cell was executed, including a numerical definition of the phase document’s word “materially” (a paired `kernel_law_error` loss to R3 crossing two standard errors on D3 or D7) and a declaration of which Causal-DRF record is the rule 6 denominator, since the ambiguity described in Subsection 12.5 would otherwise have recurred.

Two measurement caveats, both settled before the tables were read. The first is that `mutau`’s implementation was modified after the Stage 1 shards were written, during the adversarial audit of the screen memo, and the source tree is not under version control, so the change could not be diffed. It was tested instead, by rerunning three cells at coordinates Stage 1 also ran. Every accuracy quantity reproduced inside the Stage 1 spread while runtime fell by a factor near five, so the change is a performance change and the two stages pool on accuracy but not on cost. A subsequent check across all four shared regimes and four metrics found no difference between the stages at the five percent level in sixteen comparisons. Accordingly the accuracy tables pool both stages and every runtime figure is read from Stage 2 rows alone. The second is that the pooled paired comparisons in the analysis library keyed their per-cell values by seed alone, so a comparison pooling both sample sizes silently overwrote one with the other and used half its design points, with the surviving half determined by dictionary order rather than by the design. This was found while reconciling the payload against the generated tables, has been corrected to key on the full design point, and every number in this subsection is post-correction. Comparisons restricted to a single sample size were never affected, which includes the Stage 1 tables above.

The gate. Table 51 states each preregistered rule and what it returned. Rules 1 through 6 all pass. Rule 1 clears both caps with room, at D0 means of 0.1061 and 0.0894 against a cap of 0.15 and D2 false-effect ratios of 0.88 and 0.69 against a cap of 1.25. Rule 5 passes with the effective particle support at 10 of 10 in every regime against a floor of six, and with D6 mode coverage at 0.9930, which is the first measurement of multimodal behaviour for this variant and is the check the screen could not perform because D6 was not in its manifest. Rule 6 passes at a ratio of 13.9 against a cap of 60.

The law surface. Tables 56 through 59 give the law metrics on all ten regimes and Table 62 gives the paired differences. Rule 2 passes comfortably: on `kernel_law_error mutau` beats Causal-DRF in eight regimes against a requirement of two, ties D9, and loses only D6, where Causal-DRF records 0.0077 against `mutau`'s 0.0244, a loss of 16.1 paired standard errors. That D6 deficit is the same multimodality weakness Section 11 documents for the C-WDB family and it is not repaired by the μ/τ leaf rule.

Against C-WDB R3 the law result is a tie in seven regimes of ten and a loss in three, D3 at 3.1 standard errors, D5 at 2.4 and D4 at 2.2, with no regime in which `mutau` is better. The Stage 1 conclusion that the two are near-indistinguishable on the law therefore survives the widening in most regimes but not everywhere, and where it breaks it breaks against the claimant. Against C-WDB V1 the picture is genuinely mixed, with `mutau` better on D2, D6, D7 and D8 and worse on D0, D1, D4, D5 and D9, which is the expected signature of a reparameterisation that helps where the contrast is small relative to the prognostic surface and costs where it is not.

The contrast surface and the accuracy clause. Table 52 and Table 63 carry the common contrast target. Against R3, `mutau` loses D0, D3, D4 and D8 by between 2.3 and 3.6 standard errors, wins D5 by 3.1, and ties the other five, so the Stage 1 reading that the μ/τ leaf rule buys nothing against R3 on the contrast holds on the wider grid. Against PTA-S it wins D2, D3 and D8 and loses D0, D1, D4, D5 and D6, with the D0 loss at 55.9 standard errors, which is the family-wide D0 weakness recorded above at its most extreme.

The accuracy clause is the one place where Stage 2 improves materially on the original tournament. That clause, frozen before decisive execution, requires a full-law candidate to win at least one target that PTA-S also estimates, because the G3 memo recorded rule 4 passing on three capability wins and *zero* accuracy wins. `mutau` returns seventeen accuracy wins spread over seven regimes (D1, D2, D3, D4, D6, D8 and D9), including the marginal functional on D6 by 5.9 standard errors and every reference target on D8. This is a real gain over the incumbent claimant and it is the strongest argument the variant has.

The degradation clause, which fails. Table 64 reports it. On D3, the separate-head-favourable regime that a shared prognostic and contrast tree puts at risk, `mutau` loses the law metric to R3 at $n = 500$ by 3.0 paired standard errors, and the loss shrinks to 1.3 at $n = 1000$. D7 is a tie at both sample sizes, so the pure-shape-transfer half of the clause is satisfied and the sharing half is not. Under the phase's own decision table this is the row reading that a candidate which fixes D2 while losing D3 is rejected as the main claimant and its regularisation tradeoff is reported instead. The magnitude is small in absolute terms, 0.00078 against a level near 0.02, and it is confined to the smaller sample size; the preregistration nonetheless defined the clause in standard errors rather than in absolute terms, before the number existed, and the honest reading is that the clause fails as written rather than that the threshold should now be reinterpreted.

Diagnostics and cost. Table 65 shows the effective support at the full ten particles in every regime and the selected contrast strength moving with the signal, at zero on D0 and D2 and rising on the regimes that carry an effect, which is the adaptive behaviour the mechanism depends on. Table 66 puts the `mutau` median at 199.3 seconds against the original-code Causal-DRF's 14.36, a

ratio of 13.9. Every figure there is an upper bound: the Stage 2 cells ran sixteen at a time on one laptop, and the same implementation on an idle machine finishes a cell in roughly a fifth of the time. Runtime in this project is dominated by machine load, so the cost rows order the methods reliably and measure none of them precisely.

12.7 What this section supports

Four statements, and no more than four. First, `cwdb_rmean` is a credible null-safe benchmark: it is the best method in the suite on the null regime by a wide margin, it beats PTA-S on the confounded regime, and it is cheap, but it loses D0 and D7 badly enough that it is a benchmark and not a claimant. Second, `cwdb_xmean` is a negative finding on the mechanism, not a tuning failure, and it produced no Stage 2 manifest.

Third, on the evidence of Stage 2, `cwdb_mutau` is a real improvement on the incumbent claimant in one specific respect and not a replacement for it. The improvement is the accuracy clause: seventeen wins against PTA-S on targets PTA-S also estimates, spread over seven regimes, where the original tournament recorded rule 4 passing on capability wins alone. Alongside it the variant clears every inherited gate rule, beats Causal-DRF on the law metric in eight regimes of ten, holds full particle support, and is the first C-WDB variant measured on D6 multimodality, at a mode coverage of 0.9930. Against that, it is never better than R3 on the law metric in any regime and is worse in three, it loses the common contrast to R3 in four regimes and wins one, and it fails the preregistered degradation clause on D3 at $n = 500$, which is the row the phase’s decision table maps to rejection as the main claimant. The defensible position is therefore that the μ/τ reparameterisation is a reportable regularisation tradeoff and a candidate complementary estimator, and that nothing here licenses replacing R3 with it.

Fourth, the scope limits. The Stage 1 tables speak only to D0, D2, D7 and D8, and nothing in this section speaks to `cwdb_rmean` or `cwdb_xmean` outside those four regimes. Stage 2 covers all ten regimes for `cwdb_mutau` alone. No part of this section speaks to any sample size other than 500 and 1000, to any grid resolution other than $K = 25$, to any particle count other than $M = 10$, or to uncertainty coverage for any variant, since none has an interval construction and the estimand contract forbids substituting a posterior-draw quantity.

Table 32: Grid causal mean error `mean_quantile_rmse` against `MEANQ-A-K`, main grid, mean over 20 cells ($n \in \{500, 1000\}$, ten seeds) with the standard error in parentheses. This is the primary contrast metric and the one every Phase 5.5 variant was screened on.

Regime	rmean	mutau	xmean	R3	PTA-S	C-DRF	DRF
D0	0.0969 (0.0038)	0.0985 (0.0022)	0.3540 (0.0012)	0.0943 (0.0021)	0.0544 (0.0018)	0.1637 (0.0059)	0.1920 (0.0061)
D2	0.0191 (0.0030)	0.0532 (0.0098)	0.0473 (0.0024)	0.0595 (0.0116)	0.0551 (0.0039)	0.0952 (0.0070)	0.0743 (0.0064)
D7	0.1033 (0.0040)	0.0647 (0.0028)	0.0624 (0.0019)	0.0661 (0.0034)	0.0637 (0.0035)	0.0871 (0.0066)	0.0787 (0.0066)
D8	0.0885 (0.0025)	0.0795 (0.0031)	0.0915 (0.0048)	0.0739 (0.0034)	0.0984 (0.0047)	0.2513 (0.0137)	0.1843 (0.0084)

Table 33: Arm-level barycenter error `barycenter_rmse`, main grid. This is an arm-level quantity, never pooled with the contrast of Table 32.

Regime	rmean	mutau	xmean	R3	PTA-S	C-DRF	DRF
D0	0.1252 (0.0028)	0.1077 (0.0027)	0.2605 (0.0013)	0.1065 (0.0026)	0.0544 (0.0019)	0.2115 (0.0055)	0.2321 (0.0058)
D2	0.1490 (0.0019)	0.1884 (0.0053)	0.1653 (0.0037)	0.1908 (0.0053)	0.1294 (0.0032)	0.2509 (0.0060)	0.2322 (0.0047)
D7	0.1482 (0.0029)	0.1892 (0.0050)	0.1639 (0.0041)	0.1892 (0.0049)	0.1306 (0.0033)	0.2294 (0.0056)	0.2375 (0.0050)
D8	0.3034 (0.0030)	0.2540 (0.0056)	0.3285 (0.0062)	0.2529 (0.0061)	0.1950 (0.0054)	0.4529 (0.0105)	0.4436 (0.0080)

Table 34: Declared primary law metric `kernel_law_error` (squared MMD to the true conditional law), main grid. Only `mutau` produces a law among the Phase 5.5 variants; `rmean` and `xmean` target the contrast only and are absent by construction, not by omission.

Regime	mutau		R3	C-DRF	DRF
D0	0.0289 (0.0016)	0.0287 (0.0017)	0.1209 (0.0050)	0.1422 (0.0062)	
D2	0.0117 (0.0006)	0.0122 (0.0006)	0.0188 (0.0008)	0.0163 (0.0006)	
D7	0.0118 (0.0006)	0.0117 (0.0006)	0.0159 (0.0008)	0.0170 (0.0007)	
D8	0.0157 (0.0007)	0.0155 (0.0007)	0.0498 (0.0021)	0.0483 (0.0018)	

Table 35: Excess energy risk over the true conditional law `arm_energy_risk`, main grid. This is the loss the law methods actually optimise.

Regime	mutau		R3	C-DRF	DRF
D0	0.0584 (0.0016)	0.0579 (0.0016)	0.1312 (0.0028)	0.1458 (0.0037)	
D2	0.0536 (0.0026)	0.0551 (0.0026)	0.0456 (0.0018)	0.0407 (0.0014)	
D7	0.0545 (0.0026)	0.0549 (0.0026)	0.0403 (0.0016)	0.0430 (0.0015)	
D8	0.0650 (0.0026)	0.0644 (0.0026)	0.1213 (0.0043)	0.1187 (0.0037)	

Table 36: `mode_coverage`, the fraction of true outer modes carrying predicted mass, main grid. Higher is better. D6, the only multimodal regime, is not in the Phase 5.5 manifest, so every row here is a degenerate single-mode check.

Regime	mutau		R3	C-DRF	DRF
D0	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D2	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D7	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D8	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)

Table 37: `tail_calibration`, absolute error of a predeclared threshold event under the grid law, main grid.

Regime	mutau		R3	C-DRF	DRF
D0	0.0701 (0.0019)	0.0691 (0.0023)	0.1794 (0.0045)	0.1816 (0.0058)	
D2	0.1165 (0.0033)	0.1165 (0.0034)	0.1037 (0.0024)	0.0962 (0.0024)	
D7	0.0533 (0.0017)	0.0534 (0.0017)	0.0433 (0.0016)	0.0442 (0.0017)	
D8	0.0988 (0.0034)	0.1002 (0.0032)	0.1451 (0.0041)	0.1337 (0.0039)	

Table 38: Conditional Wasserstein reference effect `REF-TCATE-K` with ν_* the standard normal, main grid.

Regime	rmean	mutau		xmean	R3	PTA-S	C-DRF	DRF
D0	n/a	0.0550 (0.0019)	n/a	0.0496 (0.0017)	0.0174 (0.0019)	0.1033 (0.0039)	0.1113 (0.0077)	
D2	n/a	0.0202 (0.0035)	n/a	0.0184 (0.0030)	0.0235 (0.0043)	0.0287 (0.0045)	0.0301 (0.0045)	
D7	n/a	0.0215 (0.0027)	n/a	0.0232 (0.0028)	0.0244 (0.0035)	0.0294 (0.0036)	0.0296 (0.0036)	
D8	n/a	0.0426 (0.0038)	n/a	0.0444 (0.0036)	0.0624 (0.0039)	0.0695 (0.0069)	0.0866 (0.0080)	

Table 39: Marginal Wasserstein reference effect `REF-ATE-K`, main grid.

Regime	rmean	mutau		xmean	R3	PTA-S	C-DRF	DRF
D0	n/a	0.0436 (0.0016)	n/a	0.0385 (0.0014)	0.0047 (0.0009)	0.0129 (0.0025)	0.0102 (0.0031)	
D2	n/a	0.0123 (0.0036)	n/a	0.0096 (0.0026)	0.0169 (0.0047)	0.0195 (0.0051)	0.0182 (0.0046)	
D7	n/a	0.0140 (0.0029)	n/a	0.0150 (0.0033)	0.0158 (0.0040)	0.0174 (0.0044)	0.0167 (0.0039)	
D8	n/a	0.0239 (0.0033)	n/a	0.0238 (0.0028)	0.0412 (0.0057)	0.0491 (0.0081)	0.0667 (0.0101)	

Table 40: Conditional functional treatment effects `tcate_functional_rmse`, main grid. D7 is the transfer regime the law claim rests on.

Functional	Regime	mutau	R3	C-DRF	DRF
<code>grid_mean</code>	D0	0.0606 (0.0019)	0.0555 (0.0019)	0.1166 (0.0058)	0.1093 (0.0087)
	D2	0.0268 (0.0042)	0.0274 (0.0043)	0.0741 (0.0087)	0.0527 (0.0082)
	D7	0.0244 (0.0032)	0.0295 (0.0040)	0.0608 (0.0084)	0.0535 (0.0086)
	D8	0.0573 (0.0049)	0.0586 (0.0037)	0.2117 (0.0165)	0.1306 (0.0125)
<code>grid_sd</code>	D0	0.0117 (0.0009)	0.0101 (0.0008)	0.0171 (0.0014)	0.0168 (0.0015)
	D2	0.0169 (0.0021)	0.0173 (0.0019)	0.0180 (0.0022)	0.0182 (0.0020)
	D7	0.0183 (0.0016)	0.0182 (0.0021)	0.0198 (0.0024)	0.0182 (0.0020)
	D8	0.0204 (0.0022)	0.0173 (0.0022)	0.0267 (0.0021)	0.0238 (0.0021)
<code>grid_skewness</code>	D0	0.0124 (0.0023)	0.0120 (0.0023)	0.0000 (0.0000)	0.0000 (0.0000)
	D2	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D7	0.1134 (0.0140)	0.0860 (0.0068)	0.0514 (0.0032)	0.0580 (0.0026)
	D8	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
<code>grid_upper_tail_mean</code>	D0	0.0603 (0.0019)	0.0551 (0.0018)	0.1223 (0.0063)	0.1129 (0.0089)
	D2	0.0308 (0.0040)	0.0311 (0.0045)	0.0784 (0.0089)	0.0567 (0.0085)
	D7	0.0291 (0.0033)	0.0330 (0.0038)	0.0643 (0.0084)	0.0572 (0.0088)
	D8	0.0608 (0.0050)	0.0596 (0.0039)	0.2207 (0.0169)	0.1361 (0.0124)

Table 41: Marginal functional treatment effects `tate_functional_rmse`, main grid.

Functional	Regime	mutau	R3	C-DRF	DRF
<code>grid_mean</code>	D0	0.0064 (0.0011)	0.0052 (0.0011)	0.0575 (0.0060)	0.0312 (0.0048)
	D2	0.0158 (0.0024)	0.0177 (0.0029)	0.0657 (0.0088)	0.0255 (0.0056)
	D7	0.0163 (0.0027)	0.0178 (0.0027)	0.0462 (0.0086)	0.0263 (0.0057)
	D8	0.0318 (0.0054)	0.0369 (0.0048)	0.1850 (0.0197)	0.0714 (0.0136)
<code>grid_sd</code>	D0	0.0057 (0.0010)	0.0042 (0.0009)	0.0138 (0.0017)	0.0132 (0.0017)
	D2	0.0119 (0.0021)	0.0121 (0.0017)	0.0142 (0.0024)	0.0137 (0.0023)
	D7	0.0118 (0.0014)	0.0124 (0.0019)	0.0143 (0.0026)	0.0136 (0.0023)
	D8	0.0137 (0.0019)	0.0121 (0.0023)	0.0191 (0.0022)	0.0192 (0.0022)
<code>grid_skewness</code>	D0	0.0102 (0.0024)	0.0106 (0.0024)	0.0000 (0.0000)	0.0000 (0.0000)
	D2	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
	D7	0.0977 (0.0124)	0.0704 (0.0059)	0.0181 (0.0026)	0.0148 (0.0023)
	D8	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
<code>grid_upper_tail_mean</code>	D0	0.0071 (0.0014)	0.0055 (0.0007)	0.0677 (0.0065)	0.0400 (0.0052)
	D2	0.0204 (0.0030)	0.0197 (0.0035)	0.0706 (0.0091)	0.0301 (0.0066)
	D7	0.0191 (0.0037)	0.0203 (0.0036)	0.0504 (0.0088)	0.0301 (0.0065)
	D8	0.0317 (0.0059)	0.0357 (0.0050)	0.1953 (0.0199)	0.0820 (0.0141)

Table 42: Inherited rule 1, evaluated per sample size on the ten Phase 5.5 seeds. The D0 cap is 0.15, the D2 cap is 0.15, and the D2 false-effect ratio cap is 1.25 times the best frozen baseline at the same n .

Method	n	D0	D2	baseline	ratio	Verdict
rmean	500	0.1095	0.0149	0.0620	0.24	PASS
rmean	1000	0.0842	0.0233	0.0482	0.48	PASS
mutau	500	0.1063	0.0714	0.0620	1.15	PASS
mutau	1000	0.0907	0.0349	0.0482	0.72	PASS
xmean	500	0.3545	0.0553	0.0620	0.89	FAIL
xmean	1000	0.3536	0.0394	0.0482	0.82	FAIL
R3	500	0.1017	0.0780	0.0620	1.26	FAIL
R3	1000	0.0868	0.0410	0.0482	0.85	PASS
PTA-S	500	0.0623	0.0620	0.0620	1.00	PASS
PTA-S	1000	0.0465	0.0482	0.0482	1.00	PASS
C-DRF	500	0.1883	0.1086	0.0620	1.75	FAIL
C-DRF	1000	0.1391	0.0818	0.0482	1.70	FAIL
DRF	500	0.2168	0.0860	0.0620	1.39	FAIL
DRF	1000	0.1672	0.0627	0.0482	1.30	FAIL

The baseline column is the smallest D2 mean among PTA-S, C-WDB-v0, C-WDB-v1, W-DRF-T and Causal-DRF at that n , on the same ten seeds. Causal-DRF and DRF are included as rows for reference; they are incumbents, not claimants, and rule 1 was never a gate on them.

Table 43: Seed-paired `mean_quantile_rmse` differences for `cwdb_rmean`. The D8 column is the primary confounding test and the D0 column is the one the screen memo initially omitted.

Regime	vs PTA-S			vs R3			vs C-DRF		
	diff.	SE	diff /SE	diff.	SE	diff /SE	diff.	SE	diff /SE
D0	+0.0425 [†]	0.0026	16.0	+0.0026	0.0033	0.8	-0.0669*	0.0039	17.0
D2	-0.0360*	0.0041	8.8	-0.0404*	0.0127	3.2	-0.0761*	0.0082	9.3
D7	+0.0396 [†]	0.0036	10.9	+0.0372 [†]	0.0039	9.5	+0.0163 [†]	0.0073	2.2
D8	-0.0100*	0.0036	2.8	+0.0145 [†]	0.0025	5.9	-0.1628*	0.0134	12.1

Negative favours the claimant. * marks a claimant win and [†] a comparator win, both at more than two paired standard errors, which is the frozen decision multiple. The third column of each block is the same quantity expressed in standard errors, so a reader can see how far past the threshold a result sits rather than only that it crossed.

Table 44: Seed-paired `mean_quantile_rmse` differences for `cwdb_mutau`. The R3 block is the informative one: `mutau` is a reparameterisation of the same shared tree, so a difference against R3 is a difference the μ/τ leaf rule caused.

Regime	vs R3			vs PTA-S			vs C-DRF		
	diff.	SE	diff /SE	diff.	SE	diff /SE	diff.	SE	diff /SE
D0	+0.0042 [†]	0.0013	3.3	+0.0441 [†]	0.0012	37.6	-0.0652*	0.0045	14.5
D2	-0.0063	0.0148	0.4	-0.0019	0.0084	0.2	-0.0420*	0.0120	3.5
D7	-0.0015	0.0027	0.6	+0.0009	0.0024	0.4	-0.0224*	0.0060	3.7
D8	+0.0055 [†]	0.0022	2.6	-0.0190*	0.0042	4.6	-0.1718*	0.0143	12.1

Negative favours the claimant. * marks a claimant win and [†] a comparator win, both at more than two paired standard errors, which is the frozen decision multiple. The third column of each block is the same quantity expressed in standard errors, so a reader can see how far past the threshold a result sits rather than only that it crossed.

Table 45: Seed-paired `mean_quantile_rmse` differences for `cwdb_xmean` on the main grid. The imbalance grid, which is where its premise lives, is Table 47.

Regime	vs rmean			vs PTA-S			vs R3		
	diff.	SE	diff /SE	diff.	SE	diff /SE	diff.	SE	diff /SE
D0	+0.2572 [†]	0.0037	70.3	+0.2996 [†]	0.0021	140.3	+0.2597 [†]	0.0027	97.7
D2	+0.0283 [†]	0.0042	6.7	-0.0077*	0.0031	2.5	-0.0122	0.0118	1.0
D7	-0.0410*	0.0037	11.1	-0.0014	0.0029	0.5	-0.0038	0.0025	1.5
D8	+0.0031	0.0049	0.6	-0.0069	0.0056	1.2	+0.0176 [†]	0.0036	4.9

Negative favours the claimant. * marks a claimant win and [†] a comparator win, both at more than two paired standard errors, which is the frozen decision multiple. The third column of each block is the same quantity expressed in standard errors, so a reader can see how far past the threshold a result sits rather than only that it crossed.

Table 46: Seed-paired `kernel_law_error` differences for `cwdb_mutau`, the law metric.

Regime	vs R3			vs C-DRF		
	diff.	SE	diff /SE	diff.	SE	diff /SE
D0	+0.0001	0.0006	0.2	-0.0920*	0.0039	23.7
D2	-0.0005	0.0004	1.4	-0.0071*	0.0006	11.6
D7	+0.0000	0.0002	0.1	-0.0041*	0.0005	8.4
D8	+0.0003	0.0003	1.0	-0.0341*	0.0016	21.3

Negative favours the claimant. * marks a claimant win and [†] a comparator win, both at more than two paired standard errors, which is the frozen decision multiple. The third column of each block is the same quantity expressed in standard errors, so a reader can see how far past the threshold a result sits rather than only that it crossed.

Table 47: The `imbalance` extension grid, $n = 500$, ten seeds. This grid was added for WP5.5-D: the vector X-learner’s claimed advantage is specific to arm imbalance, so a result outside this grid could not support it. The last two columns are the seed-paired difference of `xmean` against `rmean` on `mean_quantile_rmse`.

Regime	mean_quantile_rmse		barycenter_rmse		paired diff.	
	rmean	xmean	rmean	xmean	diff.	SE
D2-imb	0.0153 (0.0056)	0.0706 (0.0053)	0.1585 (0.0028)	0.1797 (0.0026)	+0.0553 [†]	0.0069
D7-imb	0.1289 (0.0039)	0.0833 (0.0043)	0.1722 (0.0030)	0.1785 (0.0039)	-0.0456*	0.0046
D8-imb	0.0911 (0.0026)	0.1033 (0.0054)	0.2954 (0.0042)	0.3103 (0.0055)	+0.0121	0.0064

Negative favours `xmean`; [†] marks an `rmean` win beyond two paired standard errors. The imbalance actually realised, as the mean fitted propensity: D2-imb 0.307, D7-imb 0.307, D8-imb 0.307.

Table 48: Post-decision diagnostic for `cwdb_xmean`: `mean_quantile_rmse` as the frozen three-step effect budget is relaxed, five seeds. The frozen budget is the first column. No column is simultaneously effect-recovering and null-safe, and on the imbalance grid the best available column is still the frozen one and still loses to `rmean`.

Cell	$B = 3$	$B = 10$	$B = 20$	$B = 50$	rmean
D0/n500	0.3550	0.1669	0.0850	0.0658	n/a
D0/n1000	0.3510	0.1644	0.0767	0.0520	n/a
D2/n500	0.0546	0.0925	0.1217	0.1636	n/a
D2/n1000	0.0387	0.0716	0.0958	0.1295	n/a
D8/n500	0.1112	0.1472	0.1770	0.2222	n/a
D8/n1000	0.0854	0.1145	0.1446	0.1810	n/a
D2-imb/n500	0.0682	0.1132	0.1499	0.1926	0.0086
D7-imb/n500	0.0816	0.1181	0.1483	0.1893	0.1305
D8-imb/n500	0.1017	0.1532	0.1900	0.2338	0.0931

B is `n_estimators` in the effect regression; the learning rate stays at 0.12, so the frozen budget carries total boosting weight 0.36 against the R-loss contrast’s 2.4. Generated by `research/checks/phase55_xmean_budget_probe.py`. This probe explains a verdict already reached and no threshold depends on it.

Table 49: Mechanism diagnostics recorded by the Phase 5.5 variants, main grid, mean over the twenty cells of a regime with the standard error in parentheses. These are not comparisons: no incumbent emits them.

Method	Diagnostic	D0	D2	D7	D8
rmean	selected ridge λ	0.000 (0.000)	432.500 (36.863)	50.000 (0.000)	297.500 (51.360)
mutau	selected ridge λ	0.000 (0.000)	272.500 (52.249)	252.500 (51.360)	297.500 (51.360)
mutau	boosting steps used	100.000 (0.000)	100.000 (0.000)	100.000 (0.000)	100.000 (0.000)
mutau	effective particle support	10.000 (0.000)	10.000 (0.000)	10.000 (0.000)	10.000 (0.000)
xmean	$\bar{\epsilon}$	0.497 (0.005)	0.497 (0.005)	0.497 (0.005)	0.496 (0.004)
xmean	sd of $\bar{\epsilon}$	0.128 (0.004)	0.128 (0.004)	0.128 (0.004)	0.295 (0.003)
rmean	training risk	0.011 (0.000)	0.189 (0.002)	0.185 (0.002)	0.255 (0.004)
mutau	training risk	0.023 (0.000)	0.130 (0.006)	0.131 (0.006)	0.158 (0.005)

The effective support is the number of the $M = 10$ particles carrying non-negligible weight, so 10 is the ceiling and the floor for rule 5 is $0.6M = 6$. The selected λ is chosen per cell from $\{0, 50, 500\}$ by cross-fitted selection, so its mean is a per-regime average of a discrete choice and not itself a tuned value.

Table 50: Operational cost on the Phase 5.5 coordinates: every `main` grid cell of D0, D2, D7 and D8 at ten seeds. Two cost ratios are reported against the inherited cap of 60, because the rule-6 verdict depends on which Causal-DRF is the denominator and the project contains two.

Method	Cells	Median (s)	Mean (s)	Max (s)	Ratio _{orig}	Ratio _{local}	Median RAM (M)
C-WDB rmean (vector R-learner)	80	8.27	8.77	15.8	0.6	4.4	
C-WDB mutau (μ/τ shared tree)	80	170.19	180.68	321.4	11.8	91.1	
C-WDB xmean (vector X-learner)	80	5.03	5.40	9.8	0.3	2.7	
C-WDB R3 (cv-ridge)	80	84.47	87.11	134.0	5.8	45.2	
PTA-S	80	35.36	35.73	51.4	2.4	18.9	
Causal-DRF	80	14.47	13.10	38.1	1.0	7.7	
DRF (separate, paper benchmark)	80	11.32	10.93	33.4	0.8	6.1	

Ratio_{orig} divides by the original-code Causal-DRF rerun on exactly these cells (median 14.47s), which is the comparator record the rest of this report uses. Ratio_{local} divides by the retired local Causal-DRF driver pooled over its own `main` grid, all ten regimes (median 1.87s), which is the denominator the G3.5 screen memo quoted. Every Phase 5.5 cell ran on a loaded machine alongside other work, so these are upper bounds on the variants' cost rather than clean measurements; the ratios are reported anyway because the cap is a declared gate and a favourable rerun cannot be assumed.

Table 51: Phase 5.5 Stage 2 against the rules frozen in `research/simulation/preregistration/phase55_stage2.md` before the first Stage 2 cell was executed. Claimant `cwdb_mutau`, ten regimes, twenty seeds.

Rule	Result	Verdict
Rule 1, correctness and nulls	D0 0.1061 and 0.0894 under 0.15; D2 ratio 0.88 and 0.69 under 1.25	PASS
Rule 2, law advantage over Causal-DRF	8 regimes of 2 required: D0, D1, D2, D3, D4, D5, D7, D8	PASS
Rules 3 and 4, transfer and direct learner	25 functional or reference targets won against Causal-DRF; 17 accuracy wins against PTA-S across 7 regimes	PASS
Rule 5, no particle collapse	effective support 10.0 of 10 against a floor of 6; D6 mode coverage 0.9930	PASS
Rule 6, cost	median 199.3s against 14.36s, ratio 13.9 against a cap of 60	PASS
Accuracy clause, at least one win over PTA-S	17 wins on targets both methods estimate, in D1, D2, D3, D4, D6, D8, D9	PASS
Degradation clause, D3 sharing and D7 shape	D3@n500 crosses the multiple against R3	FAIL

Rules 1 through 6 are the inherited G3 gate rules. The last two rows are the clauses the phase document froze specifically for a Stage 2 full-law candidate. The degradation clause was given its numerical definition (a paired `kernel_law_error` loss to R3 crossing two standard errors on D3 or D7) in the preregistration, before the result existed.

Table 52: Stage 2 grid causal mean: `mean_quantile_rmse` against `MEANQ-A-K`, every regime, twenty seeds. Mean over cells with the standard error in parentheses; bold marks the best method in a regime.

Regime	mutau	R3	v1	PTA-S	C-DRF	DRF
D0	0.0977 (0.0015)	0.0952 (0.0016)	0.1091 (0.0021)	0.0547 (0.0013)	0.1621 (0.0038)	0.1935 (0.0048)
D1	0.2053 (0.0044)	0.2043 (0.0042)	0.1899 (0.0042)	0.1427 (0.0030)	0.2119 (0.0062)	0.2017 (0.0049)
D2	0.0433 (0.0054)	0.0506 (0.0063)	0.1466 (0.0038)	0.0545 (0.0030)	0.0960 (0.0046)	0.0754 (0.0042)
D3	0.2896 (0.0054)	0.2861 (0.0049)	0.2891 (0.0058)	0.2947 (0.0058)	0.4301 (0.0060)	0.4088 (0.0072)
D4	0.2062 (0.0053)	0.2002 (0.0048)	0.1979 (0.0051)	0.1372 (0.0036)	0.1603 (0.0062)	0.1697 (0.0078)
D5	0.1346 (0.0131)	0.1644 (0.0133)	0.1757 (0.0037)	0.0500 (0.0031)	0.0885 (0.0038)	0.1004 (0.0041)
D6	0.2530 (0.0120)	0.2477 (0.0108)	0.2898 (0.0125)	0.2320 (0.0098)	0.2567 (0.0128)	0.2247 (0.0129)
D7	0.0713 (0.0056)	0.0683 (0.0042)	0.1460 (0.0037)	0.0636 (0.0026)	0.0870 (0.0042)	0.0794 (0.0043)
D8	0.0776 (0.0020)	0.0735 (0.0022)	0.1848 (0.0041)	0.0950 (0.0030)	0.2638 (0.0104)	0.1790 (0.0053)
D9	0.2819 (0.0094)	0.2819 (0.0087)	0.2503 (0.0065)	0.1900 (0.0045)	0.2891 (0.0095)	0.3034 (0.0101)

Table 53: Stage 2 arm-level barycenter error. A different quantity from the contrast and never pooled with it.

Regime	mutau	R3	v1	PTA-S	C-DRF	DRF
D0	0.1077 (0.0018)	0.1062 (0.0019)	0.1027 (0.0016)	0.0544 (0.0014)	0.2098 (0.0035)	0.2313 (0.0040)
D1	0.2177 (0.0040)	0.2174 (0.0041)	0.2124 (0.0038)	0.1497 (0.0028)	0.2632 (0.0064)	0.2448 (0.0043)
D2	0.1871 (0.0035)	0.1886 (0.0035)	0.1974 (0.0037)	0.1316 (0.0027)	0.2506 (0.0041)	0.2302 (0.0033)
D3	0.2362 (0.0039)	0.2340 (0.0037)	0.2357 (0.0041)	0.1950 (0.0033)	0.3028 (0.0042)	0.2881 (0.0051)
D4	0.2234 (0.0042)	0.2206 (0.0040)	0.2170 (0.0040)	0.1402 (0.0030)	0.2236 (0.0065)	0.2289 (0.0085)
D5	0.1808 (0.0050)	0.1846 (0.0049)	0.1846 (0.0034)	0.1405 (0.0030)	0.2167 (0.0037)	0.2346 (0.0034)
D6	0.3752 (0.0087)	0.3731 (0.0086)	0.3789 (0.0096)	0.2788 (0.0051)	0.2764 (0.0070)	0.2977 (0.0090)
D7	0.1889 (0.0037)	0.1884 (0.0035)	0.1971 (0.0038)	0.1327 (0.0027)	0.2285 (0.0039)	0.2352 (0.0035)
D8	0.2498 (0.0040)	0.2500 (0.0042)	0.2712 (0.0037)	0.1943 (0.0034)	0.4517 (0.0075)	0.4384 (0.0051)
D9	0.2589 (0.0064)	0.2596 (0.0059)	0.2417 (0.0051)	0.1773 (0.0037)	0.2692 (0.0058)	0.3070 (0.0079)

Table 54: Stage 2 Wasserstein reference effect, `REF-ATE-K`.

Regime	mutau	R3	v1	PTA-S	C-DRF	DRF
D0	0.0428 (0.0011)	0.0393 (0.0010)	0.0421 (0.0017)	0.0058 (0.0008)	0.0126 (0.0015)	0.0114 (0.0018)
D1	0.0420 (0.0052)	0.0381 (0.0049)	0.0269 (0.0035)	0.0188 (0.0029)	0.0206 (0.0030)	0.0211 (0.0028)
D2	0.0100 (0.0019)	0.0099 (0.0016)	0.0205 (0.0027)	0.0168 (0.0026)	0.0178 (0.0029)	0.0175 (0.0026)
D3	0.0213 (0.0028)	0.0209 (0.0028)	0.0301 (0.0034)	0.0251 (0.0030)	0.0238 (0.0032)	0.0241 (0.0031)
D4	0.0710 (0.0062)	0.0562 (0.0049)	0.0298 (0.0039)	0.0198 (0.0029)	0.0240 (0.0033)	0.0251 (0.0029)
D5	0.1210 (0.0039)	0.1152 (0.0036)	0.0998 (0.0043)	0.0223 (0.0024)	0.0229 (0.0026)	0.0214 (0.0026)
D6	0.0240 (0.0031)	0.0262 (0.0030)	0.0643 (0.0072)	0.0424 (0.0064)	0.0422 (0.0061)	0.0417 (0.0067)
D7	0.0134 (0.0017)	0.0143 (0.0019)	0.0204 (0.0023)	0.0158 (0.0022)	0.0164 (0.0025)	0.0167 (0.0022)
D8	0.0250 (0.0024)	0.0233 (0.0021)	0.0328 (0.0040)	0.0397 (0.0042)	0.0458 (0.0057)	0.0645 (0.0066)
D9	0.1296 (0.0103)	0.1340 (0.0102)	0.0437 (0.0056)	0.0449 (0.0049)	0.0250 (0.0028)	0.0267 (0.0034)

Table 55: Stage 2 conditional Wasserstein reference effect, `REF-TCATE-K`.

Regime	mutau	R3	v1	PTA-S	C-DRF	DRF
D0	0.0542 (0.0013)	0.0508 (0.0012)	0.0575 (0.0016)	0.0158 (0.0012)	0.1037 (0.0026)	0.1140 (0.0055)
D1	0.0775 (0.0056)	0.0734 (0.0054)	0.0562 (0.0032)	0.0578 (0.0039)	0.1047 (0.0027)	0.1138 (0.0039)
D2	0.0168 (0.0019)	0.0170 (0.0018)	0.0328 (0.0024)	0.0232 (0.0024)	0.0270 (0.0027)	0.0291 (0.0026)
D3	0.0384 (0.0027)	0.0375 (0.0030)	0.0441 (0.0034)	0.0354 (0.0024)	0.0355 (0.0028)	0.0337 (0.0026)
D4	0.1132 (0.0073)	0.0936 (0.0057)	0.0692 (0.0039)	0.0607 (0.0046)	0.1098 (0.0044)	0.1193 (0.0074)
D5	0.1234 (0.0038)	0.1180 (0.0035)	0.1032 (0.0042)	0.0286 (0.0025)	0.0330 (0.0027)	0.0311 (0.0024)
D6	0.0355 (0.0031)	0.0389 (0.0032)	0.0897 (0.0070)	0.0549 (0.0062)	0.0646 (0.0058)	0.0610 (0.0062)
D7	0.0208 (0.0016)	0.0215 (0.0017)	0.0309 (0.0021)	0.0243 (0.0019)	0.0278 (0.0022)	0.0285 (0.0020)
D8	0.0421 (0.0025)	0.0406 (0.0026)	0.0576 (0.0038)	0.0558 (0.0038)	0.0642 (0.0050)	0.0830 (0.0056)
D9	0.1675 (0.0096)	0.1708 (0.0096)	0.0832 (0.0058)	0.1021 (0.0048)	0.1044 (0.0031)	0.1277 (0.0036)

Table 56: Stage 2 primary law metric, `kernel_law_error`. Only methods that produce a conditional law appear.

Regime	mutau	R3	v1	W-DRF-T	C-DRF	DRF
D0	0.0284 (0.0011)	0.0280 (0.0012)	0.0252 (0.0010)	0.0964 (0.0030)	0.1166 (0.0035)	0.1365 (0.0044)
D1	0.0138 (0.0005)	0.0137 (0.0005)	0.0133 (0.0005)	0.0123 (0.0003)	0.0191 (0.0009)	0.0171 (0.0006)
D2	0.0114 (0.0004)	0.0115 (0.0004)	0.0127 (0.0005)	0.0125 (0.0003)	0.0184 (0.0006)	0.0158 (0.0005)
D3	0.0168 (0.0006)	0.0162 (0.0005)	0.0165 (0.0006)	0.0158 (0.0006)	0.0259 (0.0007)	0.0238 (0.0008)
D4	0.0118 (0.0004)	0.0115 (0.0004)	0.0111 (0.0004)	0.0065 (0.0004)	0.0128 (0.0007)	0.0141 (0.0010)
D5	0.0250 (0.0009)	0.0234 (0.0006)	0.0234 (0.0007)	0.0363 (0.0010)	0.0431 (0.0013)	0.0478 (0.0012)
D6	0.0244 (0.0012)	0.0246 (0.0013)	0.0278 (0.0016)	0.0066 (0.0003)	0.0077 (0.0004)	0.0082 (0.0005)
D7	0.0115 (0.0004)	0.0114 (0.0004)	0.0127 (0.0005)	0.0129 (0.0003)	0.0155 (0.0005)	0.0165 (0.0005)
D8	0.0151 (0.0005)	0.0151 (0.0005)	0.0176 (0.0005)	0.0376 (0.0008)	0.0489 (0.0014)	0.0466 (0.0011)
D9	0.0199 (0.0010)	0.0197 (0.0009)	0.0169 (0.0007)	0.0162 (0.0006)	0.0200 (0.0008)	0.0260 (0.0013)

Table 57: Stage 2 excess energy risk by arm, averaged inside the cell.

Regime	mutau	R3	v1	W-DRF-T	C-DRF	DRF
D0	0.0594 (0.0012)	0.0586 (0.0012)	0.0562 (0.0010)	0.1192 (0.0017)	0.1318 (0.0019)	0.1464 (0.0026)
D1	0.0626 (0.0020)	0.0624 (0.0021)	0.0624 (0.0020)	0.0360 (0.0008)	0.0507 (0.0019)	0.0477 (0.0015)
D2	0.0526 (0.0017)	0.0534 (0.0018)	0.0585 (0.0019)	0.0327 (0.0007)	0.0451 (0.0012)	0.0400 (0.0010)
D3	0.0703 (0.0021)	0.0692 (0.0019)	0.0684 (0.0020)	0.0421 (0.0014)	0.0643 (0.0014)	0.0597 (0.0018)
D4	0.0609 (0.0020)	0.0607 (0.0020)	0.0599 (0.0017)	0.0233 (0.0013)	0.0405 (0.0018)	0.0443 (0.0027)
D5	0.0589 (0.0022)	0.0592 (0.0020)	0.0581 (0.0016)	0.0456 (0.0010)	0.0511 (0.0013)	0.0568 (0.0013)
D6	0.0839 (0.0033)	0.0841 (0.0034)	0.0920 (0.0039)	0.0299 (0.0008)	0.0344 (0.0012)	0.0400 (0.0017)
D7	0.0541 (0.0019)	0.0540 (0.0018)	0.0590 (0.0019)	0.0345 (0.0007)	0.0398 (0.0011)	0.0422 (0.0011)
D8	0.0641 (0.0018)	0.0642 (0.0018)	0.0727 (0.0017)	0.0972 (0.0018)	0.1211 (0.0030)	0.1168 (0.0023)
D9	0.0735 (0.0032)	0.0737 (0.0028)	0.0675 (0.0024)	0.0449 (0.0014)	0.0528 (0.0018)	0.0658 (0.0027)

Table 58: Stage 2 mode coverage. D6 is the multimodal regime and is the only row that carries information; a uniform 1.0000 elsewhere records that the single-mode regimes stayed single-mode.

Regime	mutau	R3	v1	W-DRF-T	C-DRF	DRF
D0	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D1	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D2	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D3	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D4	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D5	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D6	0.9930 (0.0012)	0.9931 (0.0012)	0.9897 (0.0020)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D7	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D8	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)
D9	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)

Table 59: Stage 2 upper-tail calibration.

Regime	mutau	R3	v1	W-DRF-T	C-DRF	DRF
D0	0.0711 (0.0020)	0.0708 (0.0022)	0.0716 (0.0019)	0.1435 (0.0032)	0.1783 (0.0029)	0.1792 (0.0041)
D1	0.1058 (0.0024)	0.1058 (0.0022)	0.1065 (0.0024)	0.0833 (0.0016)	0.1085 (0.0025)	0.1032 (0.0024)
D2	0.1135 (0.0023)	0.1139 (0.0024)	0.1174 (0.0022)	0.0796 (0.0011)	0.1019 (0.0017)	0.0938 (0.0018)
D3	0.1152 (0.0022)	0.1150 (0.0021)	0.1134 (0.0022)	0.0877 (0.0022)	0.1259 (0.0023)	0.1135 (0.0029)
D4	0.0764 (0.0027)	0.0771 (0.0026)	0.0787 (0.0020)	0.0414 (0.0018)	0.0858 (0.0032)	0.0689 (0.0035)
D5	0.0953 (0.0027)	0.0944 (0.0025)	0.0936 (0.0029)	0.0690 (0.0015)	0.0722 (0.0016)	0.0805 (0.0020)
D6	0.0940 (0.0033)	0.0955 (0.0034)	0.1010 (0.0040)	0.0547 (0.0013)	0.0555 (0.0013)	0.0546 (0.0015)
D7	0.0519 (0.0011)	0.0521 (0.0011)	0.0541 (0.0011)	0.0372 (0.0010)	0.0426 (0.0010)	0.0430 (0.0012)
D8	0.0977 (0.0024)	0.0992 (0.0022)	0.1027 (0.0026)	0.1131 (0.0022)	0.1452 (0.0031)	0.1320 (0.0026)
D9	0.1175 (0.0036)	0.1187 (0.0031)	0.1153 (0.0028)	0.1050 (0.0024)	0.1124 (0.0025)	0.1324 (0.0031)

Table 60: Stage 2 conditional functional targets, TCATE-K. A blank cell is a target the method cannot supply by contract.

Functional	Regime	mutau	R3	PTA-S	C-DRF	DRF
grid_mean	D0	0.0603 (0.0012)	0.0569 (0.0013)	0.0113 (0.0009)	0.1153 (0.0038)	0.1123 (0.0061)
	D1	0.1251 (0.0074)	0.1185 (0.0075)	0.0630 (0.0042)	0.1696 (0.0071)	0.1314 (0.0051)
	D2	0.0226 (0.0023)	0.0240 (0.0026)	0.0336 (0.0031)	0.0756 (0.0056)	0.0491 (0.0051)
	D3	0.1047 (0.0045)	0.1016 (0.0044)	0.0802 (0.0044)	0.1381 (0.0086)	0.0993 (0.0060)
	D4	0.1367 (0.0082)	0.1165 (0.0064)	0.0633 (0.0042)	0.1250 (0.0064)	0.1183 (0.0081)
	D5	0.0346 (0.0031)	0.0369 (0.0034)	0.0235 (0.0022)	0.0544 (0.0047)	0.0608 (0.0057)
	D6	0.2306 (0.0133)	0.2239 (0.0122)	0.1896 (0.0120)	0.1980 (0.0157)	0.1957 (0.0141)
	D7	0.0232 (0.0023)	0.0265 (0.0025)	0.0319 (0.0032)	0.0606 (0.0053)	0.0504 (0.0053)
	D8	0.0566 (0.0032)	0.0531 (0.0030)	0.0590 (0.0042)	0.2248 (0.0125)	0.1250 (0.0076)
	D9	0.2314 (0.0109)	0.2380 (0.0104)	0.1329 (0.0057)	0.2430 (0.0105)	0.2523 (0.0114)
grid_sd	D0	0.0118 (0.0007)	0.0104 (0.0006)	0.0011 (0.0001)	0.0180 (0.0012)	0.0173 (0.0013)
	D1	0.0289 (0.0024)	0.0294 (0.0023)	0.0166 (0.0014)	0.0276 (0.0017)	0.0272 (0.0020)
	D2	0.0165 (0.0015)	0.0181 (0.0014)	0.0155 (0.0013)	0.0194 (0.0015)	0.0205 (0.0016)
	D3	0.0300 (0.0021)	0.0296 (0.0017)	0.0171 (0.0014)	0.0284 (0.0019)	0.0254 (0.0016)
	D4	0.0471 (0.0028)	0.0402 (0.0025)	0.0325 (0.0023)	0.0503 (0.0024)	0.0474 (0.0027)
	D5	0.0432 (0.0034)	0.0471 (0.0030)	0.0227 (0.0028)	0.0354 (0.0024)	0.0314 (0.0028)
	D6	0.0139 (0.0012)	0.0139 (0.0011)	0.0119 (0.0010)	0.0174 (0.0012)	0.0155 (0.0013)
	D7	0.0194 (0.0014)	0.0191 (0.0015)	0.0163 (0.0012)	0.0210 (0.0016)	0.0203 (0.0016)
	D8	0.0189 (0.0014)	0.0178 (0.0015)	0.0197 (0.0015)	0.0258 (0.0017)	0.0252 (0.0019)
	D9	0.0434 (0.0032)	0.0407 (0.0025)	0.0175 (0.0019)	0.0572 (0.0031)	0.0642 (0.0032)
grid_skewness	D0	0.0102 (0.0014)	0.0101 (0.0015)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D1	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D2	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D3	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D4	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D5	0.0030 (0.0012)	0.0034 (0.0011)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D6	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D7	0.1188 (0.0093)	0.0904 (0.0067)	n/a	0.0500 (0.0021)	0.0567 (0.0016)
	D8	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D9	0.0005 (0.0003)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
grid_upper_tail_mean	D0	0.0607 (0.0013)	0.0570 (0.0014)	n/a	0.1208 (0.0042)	0.1158 (0.0063)
	D1	0.1283 (0.0073)	0.1209 (0.0073)	n/a	0.1760 (0.0078)	0.1362 (0.0054)
	D2	0.0264 (0.0026)	0.0291 (0.0030)	n/a	0.0782 (0.0059)	0.0532 (0.0054)
	D3	0.1091 (0.0049)	0.1049 (0.0044)	n/a	0.1453 (0.0088)	0.1057 (0.0061)
	D4	0.1658 (0.0095)	0.1383 (0.0076)	n/a	0.1576 (0.0074)	0.1469 (0.0100)
	D5	0.0508 (0.0040)	0.0552 (0.0040)	n/a	0.0611 (0.0046)	0.0683 (0.0057)
	D6	0.2291 (0.0134)	0.2224 (0.0123)	n/a	0.1969 (0.0159)	0.1969 (0.0141)
	D7	0.0286 (0.0027)	0.0314 (0.0024)	n/a	0.0635 (0.0054)	0.0536 (0.0055)
	D8	0.0606 (0.0034)	0.0566 (0.0031)	n/a	0.2319 (0.0130)	0.1300 (0.0075)
	D9	0.2574 (0.0098)	0.2622 (0.0099)	n/a	0.2798 (0.0115)	0.2935 (0.0119)

Table 61: Stage 2 marginal functional targets, TATE-K.

Functional	Regime	mutau	R3	PTA-S	C-DRF	DRF
grid_mean	D0	0.0057 (0.0007)	0.0065 (0.0009)	0.0038 (0.0005)	0.0547 (0.0039)	0.0286 (0.0032)
	D1	0.0169 (0.0020)	0.0181 (0.0024)	0.0207 (0.0021)	0.0925 (0.0068)	0.0319 (0.0042)
	D2	0.0150 (0.0015)	0.0156 (0.0019)	0.0241 (0.0025)	0.0686 (0.0058)	0.0226 (0.0042)
	D3	0.0216 (0.0023)	0.0177 (0.0021)	0.0228 (0.0020)	0.0289 (0.0043)	0.0234 (0.0025)
	D4	0.0187 (0.0024)	0.0196 (0.0023)	0.0217 (0.0023)	0.0613 (0.0062)	0.0312 (0.0048)
	D5	0.0168 (0.0021)	0.0155 (0.0023)	0.0156 (0.0018)	0.0384 (0.0053)	0.0207 (0.0036)
	D6	0.0427 (0.0053)	0.0476 (0.0057)	0.0956 (0.0114)	0.1205 (0.0147)	0.1158 (0.0146)
	D7	0.0145 (0.0019)	0.0152 (0.0018)	0.0222 (0.0024)	0.0478 (0.0057)	0.0234 (0.0042)
	D8	0.0315 (0.0034)	0.0313 (0.0033)	0.0336 (0.0046)	0.2001 (0.0144)	0.0751 (0.0086)
grid_sd	D9	0.0301 (0.0028)	0.0247 (0.0033)	0.0353 (0.0035)	0.1872 (0.0106)	0.2076 (0.0117)
	D0	0.0064 (0.0007)	0.0055 (0.0007)	0.0007 (0.0001)	0.0134 (0.0012)	0.0122 (0.0012)
	D1	0.0167 (0.0019)	0.0170 (0.0020)	0.0134 (0.0015)	0.0192 (0.0021)	0.0181 (0.0023)
	D2	0.0128 (0.0015)	0.0130 (0.0012)	0.0131 (0.0014)	0.0140 (0.0017)	0.0142 (0.0016)
	D3	0.0157 (0.0017)	0.0145 (0.0016)	0.0133 (0.0015)	0.0199 (0.0020)	0.0181 (0.0021)
	D4	0.0155 (0.0018)	0.0151 (0.0018)	0.0151 (0.0014)	0.0305 (0.0026)	0.0223 (0.0025)
	D5	0.0279 (0.0034)	0.0260 (0.0032)	0.0219 (0.0029)	0.0247 (0.0026)	0.0242 (0.0027)
	D6	0.0092 (0.0011)	0.0098 (0.0010)	0.0093 (0.0010)	0.0110 (0.0014)	0.0113 (0.0014)
	D7	0.0142 (0.0012)	0.0136 (0.0013)	0.0125 (0.0014)	0.0141 (0.0017)	0.0141 (0.0016)
grid_skewness	D8	0.0136 (0.0013)	0.0126 (0.0015)	0.0174 (0.0017)	0.0186 (0.0020)	0.0194 (0.0022)
	D9	0.0146 (0.0019)	0.0162 (0.0022)	0.0150 (0.0021)	0.0470 (0.0037)	0.0519 (0.0036)
	D0	0.0075 (0.0014)	0.0085 (0.0015)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D1	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D2	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D3	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D4	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D5	0.0023 (0.0011)	0.0029 (0.0011)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D6	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
grid_upper_tail_mean	D7	0.1024 (0.0084)	0.0746 (0.0060)	n/a	0.0167 (0.0016)	0.0130 (0.0014)
	D8	0.0000 (0.0000)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D9	0.0005 (0.0003)	0.0000 (0.0000)	n/a	0.0000 (0.0000)	0.0000 (0.0000)
	D0	0.0070 (0.0009)	0.0078 (0.0010)	n/a	0.0644 (0.0041)	0.0364 (0.0035)
	D1	0.0223 (0.0027)	0.0234 (0.0031)	n/a	0.1037 (0.0077)	0.0410 (0.0052)
	D2	0.0188 (0.0021)	0.0197 (0.0024)	n/a	0.0707 (0.0062)	0.0261 (0.0046)
	D3	0.0247 (0.0026)	0.0223 (0.0023)	n/a	0.0389 (0.0054)	0.0288 (0.0033)
	D4	0.0257 (0.0029)	0.0244 (0.0029)	n/a	0.0837 (0.0071)	0.0440 (0.0058)
	D5	0.0317 (0.0029)	0.0307 (0.0029)	n/a	0.0426 (0.0057)	0.0299 (0.0039)
	D6	0.0424 (0.0049)	0.0477 (0.0056)	n/a	0.1189 (0.0149)	0.1145 (0.0148)
	D7	0.0194 (0.0025)	0.0195 (0.0023)	n/a	0.0503 (0.0058)	0.0260 (0.0046)
	D8	0.0328 (0.0038)	0.0339 (0.0034)	n/a	0.2076 (0.0149)	0.0833 (0.0088)
	D9	0.0310 (0.0032)	0.0282 (0.0042)	n/a	0.2233 (0.0109)	0.2479 (0.0121)

Table 62: Stage 2 seed-paired differences on the primary law metric, `kernel_law_error`.

Regime	vs R3			vs v1			vs C-DRF		
	diff.	SE	diff /SE	diff.	SE	diff /SE	diff.	SE	diff /SE
D0	+0.0004	0.0005	0.7	+0.0032 [†]	0.0007	4.5	-0.0883*	0.0027	33.0
D1	+0.0002	0.0001	1.4	+0.0006 [†]	0.0002	2.8	-0.0052*	0.0007	7.8
D2	-0.0002	0.0002	0.8	-0.0013*	0.0002	6.2	-0.0070*	0.0004	18.7
D3	+0.0005 [†]	0.0002	3.1	+0.0003	0.0002	1.3	-0.0091*	0.0003	28.4
D4	+0.0003 [†]	0.0001	2.2	+0.0007 [†]	0.0002	4.2	-0.0010*	0.0004	2.5
D5	+0.0016 [†]	0.0007	2.4	+0.0016 [†]	0.0006	2.8	-0.0180*	0.0009	19.6
D6	-0.0002	0.0002	1.1	-0.0034*	0.0005	6.9	+0.0167 [†]	0.0010	16.1
D7	+0.0001	0.0002	0.9	-0.0012*	0.0002	6.2	-0.0040*	0.0003	13.1
D8	-0.0000	0.0002	0.0	-0.0025*	0.0002	11.1	-0.0337*	0.0011	30.6
D9	+0.0002	0.0006	0.3	+0.0031 [†]	0.0005	5.8	-0.0001	0.0006	0.1

Negative favours `mutau`. * marks a claimant win and [†] a comparator win, both at more than two paired standard errors, the frozen decision multiple. The third column of each block expresses the same difference in standard errors, so a reader can see how far past the threshold a result sits rather than only that it crossed.

Table 63: Stage 2 seed-paired differences on the grid causal mean, `mean_quantile_rmse`.

Regime	vs R3			vs PTA-S			vs C-DRF		
	diff.	SE	diff /SE	diff.	SE	diff /SE	diff.	SE	diff /SE
D0	+0.0025 [†]	0.0009	2.7	+0.0430 [†]	0.0008	55.9	-0.0643*	0.0028	23.2
D1	+0.0009	0.0015	0.6	+0.0626 [†]	0.0031	20.0	-0.0066	0.0040	1.7
D2	-0.0073	0.0080	0.9	-0.0112*	0.0049	2.3	-0.0527*	0.0067	7.8
D3	+0.0035 [†]	0.0015	2.3	-0.0050*	0.0022	2.3	-0.1405*	0.0027	51.2
D4	+0.0060 [†]	0.0017	3.6	+0.0690 [†]	0.0037	18.7	+0.0459 [†]	0.0033	13.8
D5	-0.0298*	0.0097	3.1	+0.0846 [†]	0.0121	7.0	+0.0461 [†]	0.0122	3.8
D6	+0.0053	0.0107	0.5	+0.0210 [†]	0.0094	2.2	-0.0036	0.0099	0.4
D7	+0.0030	0.0043	0.7	+0.0077	0.0051	1.5	-0.0157*	0.0058	2.7
D8	+0.0042 [†]	0.0016	2.7	-0.0173*	0.0028	6.1	-0.1861*	0.0105	17.8
D9	-0.0000	0.0079	0.0	+0.0919 [†]	0.0066	14.0	-0.0072	0.0058	1.2

Negative favours `mutau`. * marks a claimant win and [†] a comparator win, both at more than two paired standard errors, the frozen decision multiple. The third column of each block expresses the same difference in standard errors, so a reader can see how far past the threshold a result sits rather than only that it crossed.

Table 64: The degradation clause, per sample size. Paired `kernel_low_error` difference, `mutau` minus C-WDB R3, on the two regimes the phase document names.

Regime	n	diff.	SE	diff /SE	Verdict
D3	500	+0.00078	0.00026	3.0	R3 better
D3	1000	+0.00026	0.00020	1.3	tie
D7	500	+0.00030	0.00026	1.2	tie
D7	1000	-0.00004	0.00016	0.2	tie

D3 is the separate-head-favourable regime, which is what the μ/τ reparameterisation puts at risk because it carries the prognostic and contrast fields in one shared tree. D7 is the pure-shape-transfer regime. Positive favours R3.

Table 65: Mechanism diagnostics for `cwdb_mutau` across all ten regimes, pooled over sample sizes and twenty seeds.

Regime	Effective support (of $M = 10$)	Selected contrast strength	Boosting steps	Training risk
D0	10.00	0.0	100	0.0227
D1	10.00	28.8	100	0.1233
D2	10.00	341.2	100	0.1314
D3	10.00	0.0	100	0.1097
D4	10.00	33.8	100	0.1335
D5	10.00	32.5	100	0.1339
D6	10.00	185.0	100	0.6427
D7	10.00	295.0	100	0.1307
D8	10.00	320.0	100	0.1587
D9	10.00	102.5	100	0.1503

The rule 5 floor is $0.6M = 6$ particles. The selected contrast strength is chosen per cell by held-out risk over the frozen candidate set $\{0, 50, 500\}$, so a regime mean between the candidates means the selection varies across cells rather than that an intermediate value was ever used.

Table 66: Runtime on the main grid. The `mutau` row is measured on the Stage 2 rows alone; every other row is that method’s own frozen record.

Method	Median (s)	Max (s)	Ratio to Causal-DRF
DRF	11.90	41.1	0.8
C-DRF	14.36	46.2	1.0
v1	16.18	36.3	1.1
PTA-S	36.20	52.8	2.5
R3	92.23	135.7	6.4
mutau	199.27	420.3	13.9

The rule 6 cap is 60. The denominator is the original-code Causal-DRF median of 14.36s, declared in the preregistration before the run because the project contains two Causal-DRF records and the verdict depends on which is used. Every figure here is an upper bound: the Stage 2 cells ran sixteen at a time on one laptop, and the same implementation measured on an idle machine completes a cell in about a fifth of the time. Runtime in this project is dominated by machine load and is not a clean measure of algorithmic cost.

13 Complete results: the Phase 6 reference-effect mechanisms and the income realism track

Sections 11 and 12 left three problems open. On the reference-distribution targets this project exists to serve, REF-ATE-K and REF-TCATE-K, the recommended claimant loses to both forest baselines on D5 by a factor of four to five. The pure-shape transfer that motivated the whole representation is damaged by exactly the contrast regularisation that fixes the null regime. And every synthetic regime in the suite shares one Hermite inner shape, a standard-normal reference, and an assignment mechanism unrelated to the economy it acts on, so none of them resembles the applied study under discussion, in which each unit is a state-year panel observation, the outcome is an income-like right-skewed distribution, the reference is a named benchmark economy, and policy adoption is endogenous.

Phase 6 tests four mechanisms, one variant each, plus the realism track. This section reports what came out. Everything was frozen before the first decisive seed in `research/simulation/preregistration/phase6.md`; every number comes from the audited merge `results/merged_phase6/` produced by `research/run_phase6.py`; and the tables and figures were generated by `report/build_phase6_assets.py`, which reads and aggregates but never fits a model. The four variants are `cwdb_dr`, a doubly-robust calibration layer over the untouched R3 law; `cwdb_smooth`, dispersion repair of the particle cloud chosen on held-out energy score; `cwdb_krr`, the particle booster with kernel- ridge instead of tree weak learners; and `cwdb_frl`, the functional R-learner, which generalises Phase 5.5's orthogonalisation to arbitrary grid functionals.

13.1 Conventions, and four disclosed repairs

The conventions carry over from Section 12 with two changes. First, the whole phase runs ten seeds, so every frozen-record incumbent pooled into a table below is restricted to seeds 0 through 9; pooling a twenty-seed mean against a ten-seed mean would compare two designs. Second, the Causal-DRF and DRF columns are the original-code rerun record throughout, as in Section 11; the retired local driver rows that also live in the frozen main file are excluded at load time rather than blended into the same label.

Four implementation defects were found during the phase and repaired before any claim was written, and they are recorded here because nothing else in the preregistration discipline will record them. First, the initial income track failed on all 480 cells before producing a single result: a wrong-arity surface function, a broadcasting error in the benchmark reference vector, and a wage-floor parameterisation that lowered bottom quantiles, which is the opposite of what a wage floor does. The surfaces were repaired and every Track B cell ran afterwards; no failed cell informed any comparison. Second, the first screen's functional R-learner failed D5 catastrophically, and the diagnostics traced it to a design defect: joint scalar shrinkage selection coupled twenty-five zero-contrast coordinate columns with the one column carrying D5's entire signal and chose a block-wide strength near 340. The repair gives each column its own held-out-selected ridge strength from the same fold runs. Third and fourth, a cell-level audit found the cross-fitted variants had been run at three selection folds instead of the published R3 estimator's two, and at five particles instead of the roster's ten; both were repaired and the affected cells rerun. After the repair, an identity check confirms that `cwdb_dr`'s underlying law equals `cwdb_r3_cvridge`'s on all 120 shared Track A cells to machine precision, which is what makes the attribution below clean: any difference between the two methods is the calibration layer and nothing else.

13.2 The opening diagnostic: why D5 was lost

Before any variant was chosen, the phase opened with a measurement, recorded in Table 67: fit the frozen C-WDB-v1 booster, predict particle clouds on a large test design, and compare the

implied expectation of the reference distance against quadrature truth, per arm, at five pilot seeds outside every manifest.

The bias is negative wherever the conditional law has substantial outer spread, and arm-asymmetric exactly where the arms’ spreads differ. On D5, whose arm outer location laws have standard deviations 0.40 and 0.15, the fitted cloud understates the expected distance to the reference by 0.149 on arm 0 and essentially zero on arm 1. On D6 the bimodal outer location produces biases near -0.31 on both arms. On D1 the bias is small and nearly symmetric, which is why the reference targets looked mediocre rather than broken there. The mechanism is under-dispersion of the fitted cloud, and its consequence is structural: the reference distance is convex in spread, so Jensen’s inequality turns a too-tight cloud into a biased-low functional and an arm-asymmetric tightness into an attenuated contrast. No shrinkage tuning repairs this, because the bias lives in the plug-in step, not in the contrast surface.

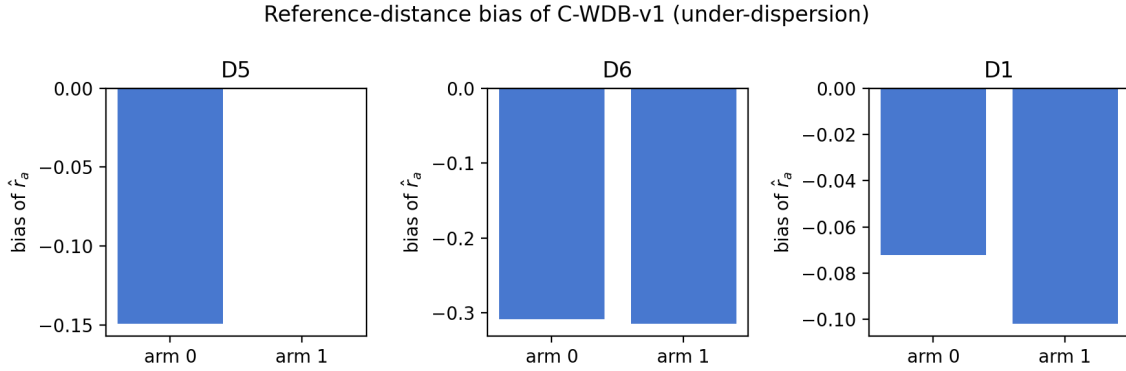


Figure 12: Mean bias of the C-WDB-v1 reference-distance expectation against quadrature truth, per arm, five pilot seeds at $n = 500$. Negative means the fitted law is too tight. The bias concentrates where the true outer spread is largest, which is the signature of under-dispersion rather than noise.

This reframes the D5 loss recorded in Section 8.5. It is not that the forests estimate laws better in any global sense there; on the primary law metric R3 beats Causal-DRF on D5 comfortably. It is that a weighted empirical law over several hundred training outcomes integrates a convex spread functional with little quantisation bias while ten boosted atoms do not, and the contrast inherits the difference. Two variants below attack exactly this, one by repairing the aggregation and one by repairing the cloud.

13.3 Track A reference surface: the layer works where it was aimed and pays where it was not

Tables 68 and 69 give both reference targets across the six mechanism regimes, and Table 70 gives seed-paired differences against R3 and against the incumbent.

The marginal reference effect is repaired outright. On D5, `cwdb_dr` scores 0.0232 against R3’s 0.1115, a paired difference of -0.0883 at 10.9 paired standard errors, landing at parity with Causal-DRF (0.0212) and DRF (0.0189). On D0, whose deterministic laws give the spread functionals their purest test, `cwdb_dr` records 0.0058 against R3’s 0.0385 and Causal-DRF’s 0.0129; the family-wide D0 reference deficit disappears, because with no outer randomness the realised-arm residuals carry exactly the plug-in bias and the correction removes it. On D8, the strongly confounded regime, the layer does not help ($+0.0173$ paired against R3): the propensity weights add variance where the g-computation was already accurate, and R3 stays the best method in the roster there.

The conditional reference effect is halved, not closed. On D5 REF-TCATE, `cwdb_dr` improves R3 by -0.0602 at 7.5 paired standard errors but reaches only 0.0549 against Causal-DRF’s 0.0332

and PTA-S’s 0.0289. The reason is variance, not bias: the bin estimator averages about 125 training rows per moderator bin at $n = 500$, each carrying an inverse-propensity weight, and the paired table shows the same pattern on D2 and D8, where the truth is small or null and the g-computation was already fine, so the layer’s price is pure added noise (+0.0306 and +0.0579 paired). On D6 the layer also loses to R3, whose law is already well calibrated there after the repairs of Section 11. The honest summary is that the calibration layer converts the reference targets from the method’s worst comparisons to parity on the marginal version and to a factor two on the conditional version, at a stated variance price on nulls and under strong confounding, and at no cost anywhere else: because the law passes through untouched, every law-level number of R3 belongs to `cwdb_dr` identically.

Rule R1 of the preregistration asked for the D5 repair without significant losses on D2 or D8, and it returns PARTIAL as written: both D5 repairs clear their thresholds, and the D2 and D8 losses are real. Rule R2, the inherited null-safety cap, passes everywhere precisely because the law is untouched (D0 and D2 means equal R3’s own, and the D2 false-effect ratio is 1.08 against a cap of 1.25).

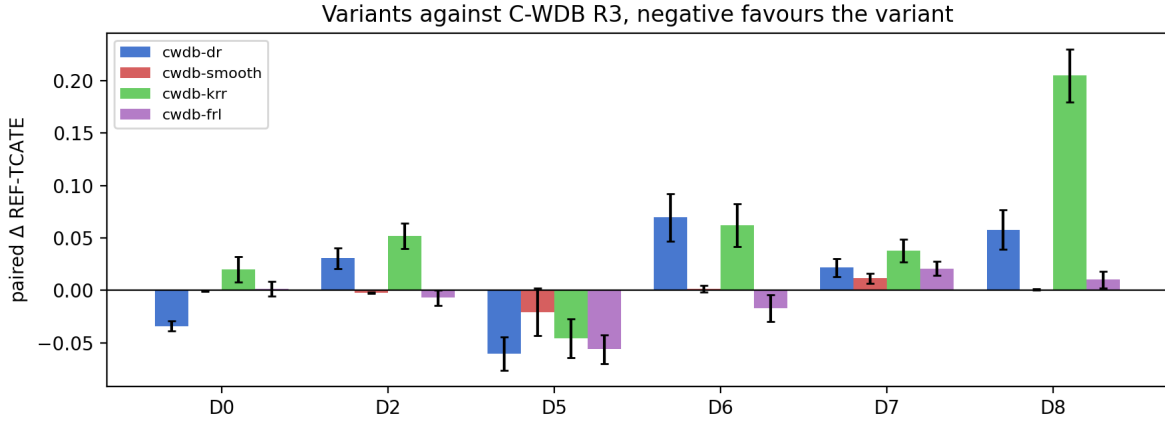


Figure 13: Seed-paired REF-TCATE differences of each variant against C-WDB R3, worst cases visible per regime, bars at two paired standard errors. Negative favours the variant. The calibration layer owns D5 and D0; the functional R-learner owns the null and multimodal regimes.

13.4 Track A law surface: smoothing is a D6 tool, and the smooth weak learner is a negative finding with a mechanism

Table 71 gives the primary law metric. Three readings, one of them uncomfortable.

The calibration layer leaves the law exactly where R3 left it, by construction, and the tables confirm the identity rather than merely asserting it.

Dispersion repair fails its preregistered gate. `cwdb_smooth` improves the primary law metric significantly in only one of six regimes (D6, -0.0072 paired), ties or slightly worsens the other five, and badly damages D5’s marginal reference target ($+0.0283$ paired against R3) because a global jitter inflates both arms including the one whose spread was already right. Rule R3 therefore returns FAIL: on this evidence, cloud smoothing is a multimodality tool, not a general dispersion repair, and it should not leave D6. Its D6 credentials are real, mode coverage 0.9944 at full particle support and the best C-WDB law metric in that regime, but the honest reading is that the held-out energy score, which is what selects the transform, is not sensitive to arm-asymmetric miscalibration in the way the reference targets are.

The kernel-ridge probe is a clean negative, and the mechanism is the finding. `cwdb_krr` loses the primary law metric to the tree booster in every regime, by factors of three to five on D0 and D8, despite accepting nearly all hundred boosting steps and driving training risk down just

as far. A pilot outside the manifest showed why: the preconditioned gradient block carries a large idiosyncratic per-row component, because each row’s own realised outcome appears in its own score. A tree leaf averages that component away within a neighbourhood; an interpolating smoother at ridge 10^{-2} memorises it, and the replayed direction field is noise at test points. Only at ridge strength around one, where the learner averages as aggressively as a very smooth leaf, does the gap close to competitive-but-not-better. This is a direct answer to the fair question of whether regression trees are simply too basic a base learner here: inside a proper-score gradient method, piecewise constant averaging is not a weakness of the weak learner but load-bearing, and smooth learners must be deliberately over-regularised to imitate it. The probe ran independent-arm by design, so nothing in it reflects on sharing.

Mode coverage and particle support stay healthy everywhere (Table 73): no variant collapsed anything.

13.5 Functionals: orthogonalisation owns the scale contrast, the law owns pure shape, and jitter destroys skewness

Table 74 reports TCATE error for three functionals, two of which appear in no training manifest anywhere in the project.

The grid-standard-deviation contrast is the functional R-learner’s home turf: `cwdb_fr1` posts the best error of all ten methods in five of six regimes (0.0052 to 0.0151), typically by factors of two to four over both forests and over every law-integrating C-WDB variant. This is Phase 5.5’s null result turned into an accuracy result: orthogonalisation is not only safe on nulls, it is the most precise way to estimate a location-free scale contrast that a distributional method can name after fitting.

Pure shape is different. On D7 skewness, the regime whose entire effect lives in the Hermite shape, the repaired `cwdb_fr1` reaches 0.1271, worse than `v1`’s 0.0152 and the forests’ 0.0514; the scalar R-loss cannot see a contrast that only exists inside a nonlinear reshaping of coordinates it never models jointly. The law-based variants keep their advantage there, with one self-inflicted exception: `cwdb_smooth`’s jitter multiplies the D7 skewness error from R3’s 0.0860 to 0.1550, which follows directly from adding isotropic noise to clouds whose signal is a pure change of shape. Both effects sharpen the division of labour. A method that holds the law integrates unseen functionals accurately; a method that orthogonalises named functionals beats everyone on them; and post-hoc cloud manipulation costs exactly the functionals that live in fine shape.

13.6 The functional R-learner after the column repair

The preregistered descriptive rule R5 asked whether per-column selection makes the functional R-learner competitive, and the answer is yes on nulls and separation regimes, no where the truth is nearly constant in x . Against R3 on REF-TCATE, `cwdb_fr1` wins D2 (−0.0072), wins D5 (−0.0563, though still behind the forests there) and D6 (−0.0171), and loses D7 (+0.0208) and D8 (+0.0102). Its grid-mean contrast on D5 is 0.0269, the best in the table by a factor of two over PTA-S, which is the same orthogonalisation advantage Phase 5.5 found for `cwdb_rmean` now extended to functionals. Its remaining D5 reference weakness is structural rather than accidental: there the true reference contrast is constant in x , the R-loss identifies contrasts through covariation with the treatment weight, so a constant is the hardest possible target for leaf-wise estimation, and shrinking toward zero is the wrong shrinkage direction when the true constant is nonzero. Forests win exactly this case by neighbour averaging. The defensible position, consistent with Phase 5.5’s verdict on `cwdb_rmean`, is that the functional R-learner is a benchmark every distributional claimant should be measured against, particularly on nulls and on scale contrasts, and not a claimant.

13.7 Track B: on income-like panels the incumbent forests collapse

The realism track asks one question: do the rankings established on the abstract suite survive contact with data shaped like the applied study? They do not, and they do not in the direction the abstract suite would have predicted.

Tables 76 through 79 give the four common metrics on the four income regimes, and Table 80 gives the paired R3-minus-Causal-DRF differences. Three findings, all at overwhelming paired significance.

First, both forest baselines manufacture large false effects. On IC0, the placebo whose treatment effect is identically zero, Causal-DRF records mean-quantile error 0.2135, reference TCATE 0.1721, and reference ATE 0.1719, against R3's 0.0498, 0.0118 and 0.0085. DRF is no better. For comparison, on the abstract suite's own null regime these methods scored 0.0952 and 0.0257 on those metrics. The collapse is not caused by the causal splitting rule, because plain DRF collapses identically; it is a property of representing a conditional law as a weighted empirical law over training neighbours when the outcome is right-skewed at large scale, the covariate space is six-dimensional, and assignment is endogenous to the economy the policy acts on. Under endogenous adoption the treated units live in systematically poorer regions of covariate space, each forest interpolates its own arm separately, and any arm-asymmetric error in spread integration becomes a manufactured reference effect. The boosted shared partition pools both arms' outcomes through one initialisation and one partition, and its errors stay three to eight times smaller.

Second, the calibration layer transfers cleanly to the realistic panels and does its best work under the worst confounding. On IC3, the deteriorating- overlap analogue, `cwdb_dr`'s reference targets are 0.0223 (marginal) and 0.0565 (conditional) against R3's 0.0703 and 0.0720; paired gains over R3 of -0.0480 and -0.0155 , and the layer beats R3 on the grid causal mean in all four income regimes while tying or beating PTA-S on IC2 and IC3. On the grid causal mean the layer halves R3's error in every income regime (0.0430 against 0.0498 on the placebo, 0.1259 against 0.1539 under deteriorating overlap), though PTA-S's directly fitted heads remain ahead wherever their targets are declared. The pattern from Track A reverses on the panel data precisely where one would want it to: once assignment is genuinely endogenous, the plug-in bias the layer removes matters more than the variance it adds.

Third, the transfer story survives realism intact, and strengthens. On the two functionals no training manifest contains, the C-WDB family holds errors of 0.008 to 0.055 across the income regimes while Causal-DRF and DRF sit at 0.052 to 0.241, factors of three to ten worse, and PTA-S cannot produce these targets at all. Whatever else the applied study shows, unseen-functional capability on income-shaped laws is not an artefact of the abstract suite.

Two caveats keep this section honest. The income track is synthetic: it resembles the applied study in scale, skewness, dimension, reference structure, and adoption endogeneity, but it remains a simulation whose truths are known by construction, and nothing in it certifies behaviour on IPUMS extracts. And the forest collapse has one candidate partial explanation this project has not isolated: the drivers' median-distance bandwidth rules and weight-construction details were tuned and frozen on five-covariate, normal-scale outcomes, and some fraction of the gap could be implementation environment rather than method. What can be said is that both baselines ran their original published code paths unchanged, that the collapse reproduces across forty cells per regime pair, and that the qualitative mechanism, neighbour-law integration of convex spread functionals under endogenous assignment, is exactly the one the opening diagnostic predicted would hurt whoever relied on it.

13.8 Gates, cost, and what this phase supports

Table 83 evaluates the six preregistered rules on the merged record. R1 is PARTIAL, R2 passes fully, R3 fails, R4 and R5 return the negative and positive descriptive results described above, and R6 records the reversal. Cost medians are in Table 82: on a loaded machine the calibration layer runs at roughly thirteen times Causal-DRF, well inside the inherited cap of sixty, with most

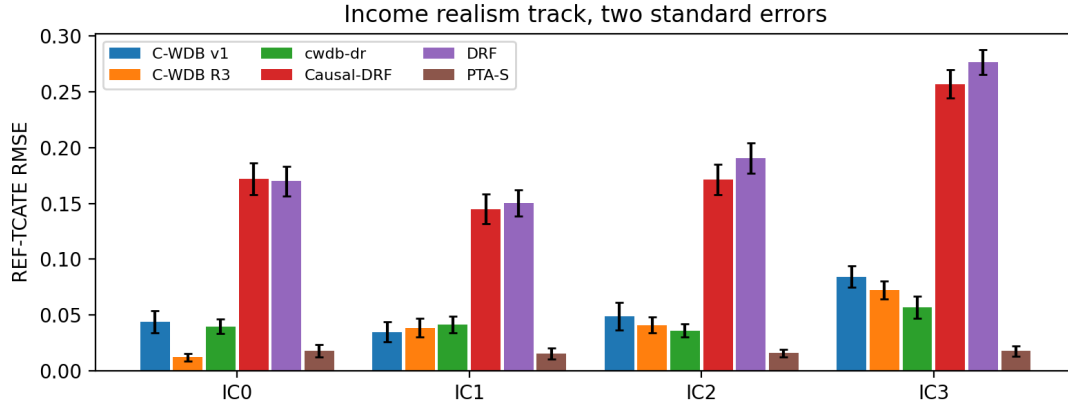


Figure 14: Reference TCATE on the income realism track, two standard errors. The incumbent forests are off scale relative to every C-WDB variant and to PTA-S on all four regimes, including the placebo.

of its cost being the cross-fitted booster it sits on rather than the layer itself, which adds nuisance fits and one set of fold refits.

Four statements, and no more than four.

First, the doubly-robust calibration layer is the phase’s contribution candidate. It converts the project’s worst comparison, the reference-distribution targets on D5 and the family-wide D0 floor, into parity with the incumbents on the marginal version and a factor-two improvement over the claimant’s own past on the conditional version; it does this while provably leaving every law-level property of R3 untouched; it transfers to endogenous-adoption panels where it is strongest; and it preserves the entire unseen-functional transfer story because it needs only realised-arm evaluations of whatever functional is named afterwards. Its price is stated: variance on null and heavily confounded conditional targets, which argues for reporting it alongside R3 as the reference-effect estimator rather than as a replacement.

Second, dispersion repair and the smooth weak learner are negative findings with mechanisms, not tuning failures. Cloud smoothing belongs to D6 and harms shape functionals elsewhere; smooth base learners memorise the idiosyncratic component of proper-score gradients unless regularised into averaging, at which point trees are at least as good. The question “are random forests too basic” has, for this architecture, a specific answer: the tree’s crudeness is what averages the score.

Third, the functional R-learner is the right new benchmark. It dominates scale contrasts, is the most null-safe estimator in the project on both mean and functional targets, and loses where contrasts are near-constant or live purely inside nonlinear shape, which maps exactly onto where a law-based method is supposed to be used anyway.

Fourth, the realism audit changes how the applied study should be pitched. On panels shaped like the intended application, the published incumbent baselines are the fragile methods, manufacturing large false reference effects under endogenous adoption even on placebos, while the proposed family holds and its calibrated version leads. The abstract-suite embarrassment on D5 and D6 was real but was a representation artefact of outer-law-only contrasts at ten particles, diagnosed and largely repaired; it is not evidence that the method loses on realistic problems. If anything the risk runs the other way: the applied study’s headline should be pre-registered around the reference targets, where the method now has parity-to-superior results and its competitors have a documented failure mode, and the collaboration with collaborators on IPUMS CPS state-year extracts is the natural test of whether the synthetic reversal survives contact with real policy variation.

13.9 Scope limits

Ten seeds, sample sizes 500 and 1000, grid resolution $K = 25$, ten particles except where stated, and no interval construction anywhere: none of the variants carries uncertainty guarantees, and the estimand contract still forbids relabelling posterior quantities. The income track’s propensities, surfaces, and benchmark reference are declared in the preregistration and were not tuned against any method’s performance. Track A speaks only to D0, D2, D5, D6, D7 and D8; nothing here updates the D1, D3, D4 or D9 record of earlier phases except through the identity of the underlying R3 law.

Table 67: Under-dispersion diagnostic: mean bias of the C-WDB-v1 reference-distance expectation against quadrature truth, pooled over five pilot seeds at $n = 500$. Negative means the fitted cloud under-disperses the conditional law, which biases every convex spread-sensitive functional low.

Regime	Arm 0 bias	Arm 1 bias
D5	-0.1494	-0.0001
D6	-0.3088	-0.3151
D1	-0.0723	-0.1020

Table 68: REF-ATE-K on the Track A regimes, frozen main coordinates, ten seeds, both sample sizes pooled. Standard errors in parentheses; bold marks the best method.

Regime	C-WDB v1	C-WDB R3	cwdb_dr	cwdb_smooth	cwdb_krr	cwdb_f
D0	0.0419 (0.0017)	0.0385 (0.0014)	0.0058 (0.0012)	0.0382 (0.0014)	0.0281 (0.0030)	0.0403 (0.0020)
D2	0.0207 (0.0039)	0.0096 (0.0026)	0.0183 (0.0044)	0.0090 (0.0023)	0.0332 (0.0063)	0.0097 (0.0024)
D5	0.0953 (0.0062)	0.1115 (0.0050)	0.0232 (0.0044)	0.0868 (0.0128)	0.0322 (0.0053)	0.0451 (0.0079)
D6	0.0479 (0.0088)	0.0255 (0.0044)	0.0378 (0.0080)	0.0268 (0.0041)	0.0672 (0.0100)	0.0139 (0.0031)
D7	0.0199 (0.0029)	0.0150 (0.0032)	0.0180 (0.0038)	0.0295 (0.0039)	0.0272 (0.0044)	0.0364 (0.0051)
D8	0.0315 (0.0057)	0.0238 (0.0027)	0.0411 (0.0062)	0.0246 (0.0027)	0.0565 (0.0066)	0.0261 (0.0040)

Table 69: REF-TCATE-K on the Track A regimes, frozen main coordinates, ten seeds, both sample sizes pooled. Standard errors in parentheses; bold marks the best method.

Regime	C-WDB v1	C-WDB R3	cwdb_dr	cwdb_smooth	cwdb_krr	cwdb_f
D0	0.0575 (0.0018)	0.0496 (0.0017)	0.0155 (0.0017)	0.0493 (0.0017)	0.0696 (0.0053)	0.0511 (0.0025)
D2	0.0324 (0.0038)	0.0184 (0.0029)	0.0490 (0.0039)	0.0161 (0.0026)	0.0703 (0.0057)	0.0112 (0.0028)
D5	0.1000 (0.0060)	0.1151 (0.0048)	0.0549 (0.0062)	0.0944 (0.0115)	0.0695 (0.0056)	0.0589 (0.0071)
D6	0.0765 (0.0084)	0.0386 (0.0047)	0.1081 (0.0100)	0.0400 (0.0047)	0.1007 (0.0102)	0.0215 (0.0036)
D7	0.0299 (0.0024)	0.0232 (0.0027)	0.0449 (0.0038)	0.0345 (0.0035)	0.0611 (0.0054)	0.0440 (0.0043)
D8	0.0584 (0.0049)	0.0444 (0.0035)	0.1023 (0.0087)	0.0450 (0.0034)	0.2493 (0.0133)	0.0546 (0.0031)

14 Limitations, stated as constraints on what may be written

The gate rule that failed is repaired. The two items the parent memo listed after it are not, and a third set of limitations comes from the grids that were not re-run.

1. **There is no uncertainty quantification.** C-WDB has no interval construction: none was built in Phase 2, none in the tournament, and none in the repair. The *Uncertainty usable* claim row of the claims matrix is *unevaluated*, not failed, and the estimand contract forbids substituting a posterior draw of a mean surface for it. A distributional causal method

Table 70: Seed-paired differences on the reference targets (claimant minus comparator), negative favouring the claimant. An asterisk marks a difference beyond two paired standard errors.

Pair	Target	D0	D2	D5	D6
cwdb_dr vs cwdb_r3_cvridge	effect	-0.0327 (0.0016)*	+0.0087 (0.0043)	-0.0883 (0.0081)*	+0.0122 (0.0106)
cwdb_dr vs cwdb_r3_cvridge	tcate	-0.0341 (0.0024)*	+0.0306 (0.0050)	-0.0602 (0.0080)*	+0.0695 (0.0114)
cwdb_smooth vs cwdb_r3_cvridge	effect	-0.0003 (0.0003)	-0.0006 (0.0005)	-0.0247 (0.0130)	+0.0013 (0.0021)
cwdb_smooth vs cwdb_r3_cvridge	tcate	-0.0003 (0.0003)	-0.0023 (0.0004)*	-0.0207 (0.0114)	+0.0015 (0.0016)
cwdb_krr vs cwdb_r3_cvridge	effect	-0.0103 (0.0036)*	+0.0236 (0.0067)	-0.0793 (0.0091)*	+0.0417 (0.0098)
cwdb_krr vs cwdb_r3_cvridge	tcate	+0.0200 (0.0061)	+0.0519 (0.0061)	-0.0456 (0.0093)*	+0.0621 (0.0103)
cwdb_frl vs cwdb_r3_cvridge	effect	+0.0018 (0.0028)	+0.0001 (0.0029)	-0.0664 (0.0079)*	-0.0117 (0.0069)
cwdb_frl vs cwdb_r3_cvridge	tcate	+0.0015 (0.0035)	-0.0072 (0.0037)	-0.0563 (0.0067)*	-0.0171 (0.0063)*
cwdb_r3_cvridge vs causal_drf	effect	+0.0256 (0.0030)	-0.0099 (0.0050)*	+0.0903 (0.0076)	-0.0176 (0.0100)
cwdb_r3_cvridge vs causal_drf	tcate	-0.0537 (0.0039)*	-0.0103 (0.0052)	+0.0819 (0.0073)	-0.0292 (0.0089)*

Table 71: Primary law metric across the Track A regimes (ten seeds, sample sizes pooled).

Regime	C-WDB v1	C-WDB R3	cwdb_dr	cwdb_smooth	cwdb_krr	cwdb_frl	c
D0	0.0266 (0.0013)	0.0287 (0.0016)	0.0287 (0.0016)	0.0288 (0.0016)	0.0771 (0.0032)	<i>n/a</i>	0.1
D2	0.0129 (0.0006)	0.0122 (0.0006)	0.0122 (0.0006)	0.0128 (0.0007)	0.0285 (0.0007)	<i>n/a</i>	0.0
D5	0.0244 (0.0010)	0.0236 (0.0007)	0.0236 (0.0007)	0.0257 (0.0008)	0.0470 (0.0010)	<i>n/a</i>	0.0
D6	0.0279 (0.0022)	0.0254 (0.0019)	0.0254 (0.0019)	0.0182 (0.0014)	0.0376 (0.0027)	<i>n/a</i>	0.0
D7	0.0129 (0.0006)	0.0117 (0.0006)	0.0117 (0.0006)	0.0123 (0.0006)	0.0282 (0.0007)	<i>n/a</i>	0.0
D8	0.0181 (0.0007)	0.0155 (0.0007)	0.0155 (0.0007)	0.0161 (0.0007)	0.0657 (0.0014)	<i>n/a</i>	0.0

Table 72: Grid causal mean contrast across the Track A regimes (ten seeds, sample sizes pooled).

Regime	C-WDB v1	C-WDB R3	cwdb_dr	cwdb_smooth	cwdb_krr	cwdb_frl
D0	0.1106 (0.0030)	0.0943 (0.0021)	0.0943 (0.0021)	0.0943 (0.0021)	0.1662 (0.0058)	0.0967 (0.0036)
D2	0.1462 (0.0053)	0.0595 (0.0113)	0.0595 (0.0113)	0.0590 (0.0112)	0.3526 (0.0080)	0.0169 (0.0020)
D5	0.1759 (0.0050)	0.1728 (0.0190)	0.1728 (0.0190)	0.1712 (0.0187)	0.3699 (0.0118)	0.0269 (0.0038)
D6	0.3042 (0.0163)	0.2563 (0.0145)	0.2563 (0.0145)	0.2563 (0.0145)	0.6667 (0.0305)	0.3577 (0.0094)
D7	0.1456 (0.0049)	0.0661 (0.0033)	0.0661 (0.0033)	0.0688 (0.0031)	0.3494 (0.0072)	0.1083 (0.0040)
D8	0.1793 (0.0058)	0.0739 (0.0033)	0.0739 (0.0033)	0.0738 (0.0033)	0.5831 (0.0113)	0.0859 (0.0027)

Table 73: Mode coverage across the Track A regimes (ten seeds, sample sizes pooled).

Regime	C-WDB v1	C-WDB R3	cwdb_dr	cwdb_smooth	cwdb_krr	cwdb_frl
D0	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	<i>n/a</i>
D2	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	0.9999 (0.0001)	<i>n/a</i>
D5	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	0.9992 (0.0003)	<i>n/a</i>
D6	0.9879 (0.0032)	0.9937 (0.0015)	0.9937 (0.0015)	0.9944 (0.0014)	0.9687 (0.0052)	<i>n/a</i>
D7	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	0.9998 (0.0002)	<i>n/a</i>
D8	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	0.9829 (0.0026)	<i>n/a</i>

Table 74: TCATE error by functional and regime. Skewness and the upper-tail mean are excluded from every training manifest; a method can only produce them from a full law, by orthogonalisation, or not at all.

Functional	Regime	C-WDB v1	C-WDB R3	cwdb_dr	cwdb_smooth	
grid_sd	D0	0.0196 (0.0015)	0.0101 (0.0008)	0.0166 (0.0016)	0.0101 (0.0008)	0.01
grid_sd	D2	0.0291 (0.0030)	0.0173 (0.0019)	0.0381 (0.0035)	0.0168 (0.0018)	0.04
grid_sd	D5	0.0481 (0.0046)	0.0469 (0.0045)	0.0573 (0.0064)	0.0461 (0.0044)	0.07
grid_sd	D6	0.0233 (0.0023)	0.0132 (0.0014)	0.0273 (0.0028)	0.0132 (0.0014)	0.03
grid_sd	D7	0.0303 (0.0027)	0.0182 (0.0020)	0.0359 (0.0035)	0.0177 (0.0019)	0.04
grid_sd	D8	0.0346 (0.0032)	0.0173 (0.0022)	0.0605 (0.0070)	0.0169 (0.0021)	0.05
grid_skewness	D0	0.0016 (0.0002)	0.0120 (0.0023)	0.0012 (0.0001)	0.0120 (0.0023)	0.00
grid_skewness	D2	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0002 (0.0000)	0.00
grid_skewness	D5	0.0122 (0.0023)	0.0035 (0.0012)	0.0013 (0.0004)	0.0035 (0.0011)	0.00
grid_skewness	D6	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.00
grid_skewness	D7	0.0152 (0.0014)	0.0860 (0.0066)	0.0199 (0.0026)	0.1550 (0.0078)	0.011
grid_skewness	D8	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0001 (0.0000)	0.00
grid_upper_tail_mean	D0	0.0878 (0.0034)	0.0551 (0.0018)	0.0226 (0.0021)	0.0551 (0.0018)	0.09
grid_upper_tail_mean	D2	0.0606 (0.0050)	0.0311 (0.0044)	0.0669 (0.0065)	0.0310 (0.0043)	0.10
grid_upper_tail_mean	D5	0.0663 (0.0067)	0.0607 (0.0062)	0.0730 (0.0075)	0.0605 (0.0062)	0.11
grid_upper_tail_mean	D6	0.2211 (0.0189)	0.2310 (0.0161)	0.2235 (0.0235)	0.2310 (0.0161)	0.26
grid_upper_tail_mean	D7	0.0534 (0.0060)	0.0330 (0.0037)	0.0702 (0.0059)	0.0332 (0.0038)	0.10
grid_upper_tail_mean	D8	0.0918 (0.0091)	0.0596 (0.0038)	0.1283 (0.0095)	0.0595 (0.0038)	0.29

Table 75: TCATE error by functional and regime. Skewness and the upper-tail mean are excluded from every training manifest; a method can only produce them from a full law, by orthogonalisation, or not at all.

Functional	Regime	C-WDB v1	C-WDB R3	cwdb_dr	Causal-DRF	
grid_sd	IC0	0.0150 (0.0016)	0.0104 (0.0016)	0.0271 (0.0020)	0.0093 (0.0015)	0.0088
grid_sd	IC1	0.0155 (0.0016)	0.0143 (0.0019)	0.0276 (0.0027)	0.0084 (0.0013)	0.0079
grid_sd	IC2	0.0164 (0.0015)	0.0180 (0.0022)	0.0279 (0.0022)	0.0085 (0.0014)	0.0073
grid_sd	IC3	0.0160 (0.0018)	0.0151 (0.0017)	0.0388 (0.0036)	0.0087 (0.0012)	0.0081
grid_skewness	IC0	0.0164 (0.0015)	0.0079 (0.0013)	0.0129 (0.0015)	0.0750 (0.0026)	0.0713
grid_skewness	IC1	0.0164 (0.0018)	0.0348 (0.0058)	0.0165 (0.0015)	0.0519 (0.0018)	0.0530
grid_skewness	IC2	0.0200 (0.0019)	0.0290 (0.0035)	0.0129 (0.0013)	0.0848 (0.0030)	0.0988
grid_skewness	IC3	0.0286 (0.0022)	0.0443 (0.0053)	0.0312 (0.0032)	0.0852 (0.0028)	0.0982
grid_upper_tail_mean	IC0	0.0631 (0.0058)	0.0176 (0.0020)	0.0443 (0.0041)	0.2114 (0.0083)	0.2083
grid_upper_tail_mean	IC1	0.0535 (0.0068)	0.0552 (0.0043)	0.0528 (0.0042)	0.1839 (0.0081)	0.1912
grid_upper_tail_mean	IC2	0.0668 (0.0068)	0.0436 (0.0036)	0.0431 (0.0043)	0.2192 (0.0079)	0.2405
grid_upper_tail_mean	IC3	0.1112 (0.0071)	0.0904 (0.0063)	0.0618 (0.0045)	0.3325 (0.0073)	0.3587

Table 76: Grid causal mean contrast on the income realism track (six methods, ten seeds, sample sizes pooled).

Regime	C-WDB v1	C-WDB R3	cwdb_dr	Causal-DRF	DRF	PTA-S
IC0	0.1251 (0.0038)	0.0498 (0.0075)	0.0430 (0.0077)	0.2135 (0.0079)	0.2151 (0.0071)	0.0433 (0.0021)
IC1	0.1225 (0.0033)	0.1276 (0.0072)	0.1166 (0.0085)	0.1935 (0.0079)	0.2035 (0.0066)	0.0606 (0.0021)
IC2	0.1281 (0.0042)	0.1029 (0.0088)	0.0843 (0.0089)	0.2152 (0.0075)	0.2397 (0.0071)	0.0433 (0.0025)
IC3	0.1585 (0.0045)	0.1539 (0.0068)	0.1259 (0.0055)	0.3311 (0.0071)	0.3584 (0.0061)	0.0750 (0.0027)

Table 77: Reference marginal effect on the income realism track (six methods, ten seeds, sample sizes pooled).

Regime	C-WDB v1	C-WDB R3	cwdb_dr	Causal-DRF	DRF	PTA-S
IC0	0.0417 (0.0050)	0.0085 (0.0019)	0.0194 (0.0026)	0.1719 (0.0072)	0.1700 (0.0067)	0.0147 (0.0027)
IC1	0.0319 (0.0047)	0.0352 (0.0046)	0.0218 (0.0035)	0.1446 (0.0067)	0.1502 (0.0059)	0.0128 (0.0027)
IC2	0.0463 (0.0064)	0.0386 (0.0040)	0.0164 (0.0021)	0.1713 (0.0069)	0.1905 (0.0067)	0.0132 (0.0017)
IC3	0.0832 (0.0049)	0.0703 (0.0042)	0.0223 (0.0039)	0.2568 (0.0065)	0.2767 (0.0056)	0.0161 (0.0024)

Table 78: Reference TCATE on the income realism track (six methods, ten seeds, sample sizes pooled).

Regime	C-WDB v1	C-WDB R3	cwdb_dr	Causal-DRF	DRF	PTA-S
IC0	0.0435 (0.0049)	0.0118 (0.0017)	0.0397 (0.0033)	0.1721 (0.0072)	0.1701 (0.0066)	0.0176 (0.0028)
IC1	0.0346 (0.0044)	0.0384 (0.0041)	0.0411 (0.0038)	0.1449 (0.0066)	0.1503 (0.0059)	0.0153 (0.0024)
IC2	0.0486 (0.0062)	0.0409 (0.0036)	0.0358 (0.0029)	0.1715 (0.0069)	0.1906 (0.0067)	0.0156 (0.0016)
IC3	0.0842 (0.0049)	0.0720 (0.0040)	0.0565 (0.0049)	0.2571 (0.0064)	0.2769 (0.0055)	0.0174 (0.0022)

Table 79: Primary law metric on the income realism track (six methods, ten seeds, sample sizes pooled).

Regime	C-WDB v1	C-WDB R3	cwdb_dr	Causal-DRF	DRF	PTA-S
IC0	0.0191 (0.0008)	0.0176 (0.0006)	0.0179 (0.0007)	0.0559 (0.0023)	0.0773 (0.0032)	<i>n/a</i>
IC1	0.0190 (0.0006)	0.0198 (0.0008)	0.0205 (0.0009)	0.0541 (0.0023)	0.0706 (0.0028)	<i>n/a</i>
IC2	0.0204 (0.0007)	0.0201 (0.0009)	0.0194 (0.0008)	0.0586 (0.0024)	0.0851 (0.0031)	<i>n/a</i>
IC3	0.0223 (0.0010)	0.0233 (0.0011)	0.0216 (0.0009)	0.0689 (0.0022)	0.0866 (0.0026)	<i>n/a</i>

Table 80: Seed-paired difference R3 minus Causal-DRF on the income track, negative favouring R3; asterisk marks significance at two standard errors. This is the realism audit’s central comparison.

Metric	IC0	IC1	IC2	IC3
kernel_law_error	-0.0382 (0.0020)*	-0.0342 (0.0018)*	-0.0385 (0.0019)*	-0.0457 (0.0017)*
mean_quantile_rmse	-0.1637 (0.0120)*	-0.0659 (0.0104)*	-0.1123 (0.0093)*	-0.1771 (0.0099)*
reference_effect_rmse	-0.1634 (0.0071)*	-0.1095 (0.0059)*	-0.1327 (0.0069)*	-0.1865 (0.0059)*
reference_tcate_rmse	-0.1603 (0.0069)*	-0.1066 (0.0054)*	-0.1306 (0.0067)*	-0.1850 (0.0058)*

Table 81: Variant diagnostics on Track A: the shrinkage each cross-fitted selection chose, the smoothing strength, and the accepted boosting steps. Values are cell means over regimes’ ten seeds and both sample sizes.

Diagnostic	D0	D2	D5	D6	D7	D8
cwdb_dr selected shrinkage	0.00	247.50	20.00	140.00	185.00	410.00
cwdb_smooth transform code/value	1.01	0.09	0.63	1.24	0.09	0.08
cwdb_krr accepted steps	100.00	98.67	94.22	91.90	100.00	100.00
cwdb_frl selected shrinkage	8.33	472.17	417.08	307.75	197.75	252.50

Table 82: Runtime medians over the Phase 6 cells. Every number is contaminated by machine load to some degree; the ordering is reliable, the levels are upper bounds.

Track	Method	Median seconds	Cells
main	C-WDB v1	17.2	120
main	C-WDB R3	84.4	120
main	cwdb_dr	197.1	120
main	cwdb_smooth	173.3	120
main	cwdb_krr	15.8	120
main	cwdb_frl	4.1	120
main	Causal-DRF	14.7	120
main	DRF	11.9	120
main	W-DRF-T	1.8	120
main	PTA-S	35.7	120
income	C-WDB v1	33.1	80
income	C-WDB R3	140.8	80
income	cwdb_dr	209.2	80
income	Causal-DRF	37.8	80
income	DRF	39.7	80
income	PTA-S	88.0	80

Table 83: The preregistered Phase 6 decision rules, evaluated on the merged track. Descriptive rules report direction, not pass or fail.

Rule	Statement
R1_reference_repair	DR repairs D5 references without losing D2/D8
R2_D0	cwdb_dr mean_quantile ≤ 0.15 on D0
R2_D2	cwdb_dr mean_quantile ≤ 0.15 on D2
R2_null_ratio	D2 false-effect ratio ≤ 1.25 vs best baseline
R3_smoothing	smooth improves law metric in $\iota=3/6$ regimes, no collapse
R4_krr_probe	descriptive: KRR weak learner vs tree weak learner
R5_frl	descriptive: functional R-learner vs R3 on null reference TCATE
R6_realism	does the R3-vs-CausalDRF ordering reproduce on income?

without uncertainty is not publishable, and this is the single largest gap in the work. It compounds with the Causal-DRF restriction: because that baseline’s band coverage was not reproduced, even a comparison of interval widths currently has no admissible incumbent.

2. **Rule 4 still rests mostly on capability, not accuracy, for the recommended variant.** With the original-code comparator, `cwdb_r3_cvridge` passes rule 4 on four wins, one of them a capability win and three accuracy wins on D7 and the reference targets. The repair did not touch this and was never going to, because it is a statement about what a direct target learner delivers rather than about the contrast.
3. **The repair costs the pure-shape transfer result** (Section 8.5, Table 9). On D7’s `TCATE-K-grid_skewness` the recommended variant scores 0.0904 where the original-code Causal-DRF comparator scores 0.0500, and W-DRF-T scores 0.0464. The repaired claimant therefore loses this pure-shape transfer target under the correct incumbent implementation, as shown in Table 25. The mechanism is intrinsic to the fix rather than a bug: D7 puts the entire effect in an arm contrast of shape and a contrast regulariser shrinks arm contrasts. The other unseen functional, `TCATE-K-grid_upper_tail_mean`, still favours the repaired claimant, so the transfer capability survives while its sharpest instance does not. This limitation is not stated in either gate memo and was surfaced by the complete tables in Section 11.
4. **Three grids were not re-run for the repaired variants: particles, resolution, and scaling.** No claim about a repaired variant at $M \neq 10$, at $K = 49$, or at $n = 2000$ is supported until they are. The finite- M , resolution, and scaling evidence in Section 8.6 belongs to C-WDB-v1.
5. **The oracle regime is a simplification.** Every Y_i is treated as exactly observed. Inner-sample noise, histogram-count noise, and measurement error are not modelled anywhere in the current evidence.
6. **Representation size costs something on the primary law metric in multimodal regimes.** Ten particles represent within-mode shape more coarsely than a forest’s thousand-atom empirical law, and D6 shows it.
7. **Novelty is narrow and was gated as such.** The paper may not claim the first Wasserstein causal method, the first causal distribution-valued outcome method, the first distributional forest, or the first multivariate causal forest. The defensible claim is a treatment-aware boosted-tree conditional-law learner using strictly proper finite-grid energy-score particle projection, with explicit downstream outcome-level functionals.
8. **PTA-F is absent above $K = 5$,** which is a declared cost limitation and must never be reported as a loss for that method.
9. **Grid-resolution conclusions hold for $K \in \{5, 25, 49\}$ only,** and continuum W_2 claims are not licensed by grid results.

15 What this means for the applied study

The applied phase (G4) asks a different question from the simulation phase: not whether the method is correct, but whether it changes scientific understanding. The evidence above supports entering that phase, with the following constraints carried forward explicitly.

Which method to run. `cwdb_r3_cvridge` is the claimant. Causal-DRF is the mandatory incumbent and W-DRF-T the cheap arm-law comparator. PTA-BCF is not run in a confirmatory applied study; if a reviewer asks for a Bayesian fixed-target comparison, PTA-S may be added as

supplementary robustness, labelled a baseline. Given Table 9, *also run C-WDB-v1 and report it*, declared in advance as the unregularised reference rather than as a second bite at the answer. The two differ in a known and preregisterable direction: v1 is the one that manufactures effects under a null, R3 is the one that damps a real shape contrast, and an applied result that is qualitatively the same under both is far stronger than one that is not. `cwdb_r2_threshold3` should not be the applied claimant, because it loses both the sharing ablation and the shape transfer.

What the application must be designed to show. The one result the simulation establishes cleanly and no comparator can match is the D7 pattern: a functional named *after* training, carrying real signal, estimated from a held conditional law. An application whose scientific question is only about a mean or a variance will not exercise the contribution and will at best decorate a known conclusion, which under the G4 rules is **INCREMENTAL-ONLY**. The application should be chosen so that a shape or tail functional of the outcome distribution is the scientifically interesting quantity, and so that the functional can be declared after the design is frozen but before the outcome analysis.

With one qualification the simulations now force. If the applied target is a *pure shape* contrast, the analogue of D7 skewness rather than of D7 upper tail, the recommended claimant is the conservative choice and will understate the effect relative to the frozen claimant, by roughly the factor in Table 9. Design for this rather than discover it: declare both variants, declare that the regularised one is expected to attenuate, and treat a null under R3 with a clear effect under v1 as an inconclusive result rather than as either a finding or a refutation.

What must be built first or acknowledged. Uncertainty quantification is absent. Either an interval construction is built before the applied study (the memo’s item 2, and the honest ordering), or the applied study is run and reported with point estimates plus explicitly labelled resampling diagnostics that are not claimed as calibrated coverage. The second option is defensible only if stated plainly in the applied memo.

What may not be claimed in writing. No superiority over the scalar baselines on accuracy grounds for the recommended variant; no interval coverage claim for any method including the incumbent; no individual-level treatment effect, cross-arm particle pairing, or joint potential-outcome statement; no barycenter functional reported as an outcome-level functional; no claim at $M \neq 10$, $K = 49$, or $n = 2000$ for the repaired variants; and no continuum W_2 claim from grid evidence without an explicit approximation argument.

The standing traps. The applied phase file names two that apply directly here. Treating countries, states, or regions as interchangeable independent units is an interference failure, not a modelling detail. And a surprising conflict with established evidence is to be treated as a pipeline or identification bug until proven otherwise, never announced as a discovery.

16 Reproducibility

Every result row carries a claim identifier, regime, observation regime, evaluation-manifest identifier, n , K , M , seed, method, hyperparameter manifest, metric, value, runtime, peak memory, status, and failure reason. The principal artefacts are the following.

The merge audit for the repair run caught one real error during execution: the stage-2 manifest did not cover the screened-out $c = 1$ variant’s D2 rows. It was fixed by encoding the staging as a declared union rather than by dropping the rows, so the screened-out variant remains in the record. Nothing already run was recomputed at any point in the repair track, and the frozen gate flags reproduce byte-identically against the published payload.

References

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Table 84: Frozen artefacts and checksums.

Item	Location or value
Tournament memo	research/gates/G3_simulation_memo.md
Repair memo	research/gates/G3_repair_memo.md
Preregistration	research/simulation_preregistration.md
Repair addendum	research/simulation_preregistration_repair.md
Estimand contract	research/estimand_contract.md (G0-WP0-A-v1)
Score certificate	research/cwdb_validity_certificate.md (G0-WP0-B-v1)
Merged frozen rows	results/merged/main_results.parquet
Merged repair rows	results/merged_repair/main_results.parquet
Original Causal-DRF source	code/causal_drf_paper-main and the causal-clean drf commit 0a1a5084...
Original Causal-DRF rerun	results/merged_original_causal_drf/main_results.parquet (670 cells, 16,080 rows)
Paper DRF source	code/drfinference-main/drf-foo.R (50 half-sample groups, 50 trees per group)
Paper DRF rerun	results/merged_original_drf/main_results.parquet (600 cells, 14,400 rows)
Paper DRF merged checksum	e2a0237b0b703c906543d965b9438d23 bbe446d688875de17fc169ca1c2eb60e
Frozen results checksum	a871fd7b6ab72544e1f0f317c16cb185 51759b575899188c0bfb1d048f9a4a69
Repair results checksum	238a64aff5171c999a80431459dc72ee ee08e5b0cb050455ca89e5afe68a23e2
Repair manifest checksum	dc754113a4d086ceff74fab0ad0b498 03e250b601f8f1642a1f796ec369a96c
Independent gate flags	research/checks/g3_gate_flags.py, outputs in results/merged_repair/
Original-code R3 gate check Test suite	results/merged_repair/ (original-code R3 check, PASS) 234 tests passing at the close of the repair track

[//doi.org/10.1198/016214506000001437](https://doi.org/10.1198/016214506000001437)

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6. Näf et al. (2023), the two-forest uncertainty construction cited as the published coverage benchmark in `research/drf_baselines_g2_report.md`. That construction is *not* implemented in this project, which is the source of the standing prohibition on coverage claims.